

A 14.08-to-135.69Token/s ReRAM-on-Logic Stacked Outlier-Free Large-Language-Model Accelerator with Block-Clustered Weight-Compression and Adaptive Parallel-Speculative-Decoding

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Outline

- **Introduction**
- **Overall Architecture**
- **Key Features**
 - Local Rotation Unit (LRU) with Decomposed FWHT
 - ReRAM-Stacked PNM (RS-PNM) with Blockwise VQ
 - Adaptive Parallel Speculative Decoding (APSD)
 - Workload-Decoupled Out-of-Order Scheduler (WDOS)
- **Experiment Results**
- **Summary**

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■ Overall Architecture

■ Key Features

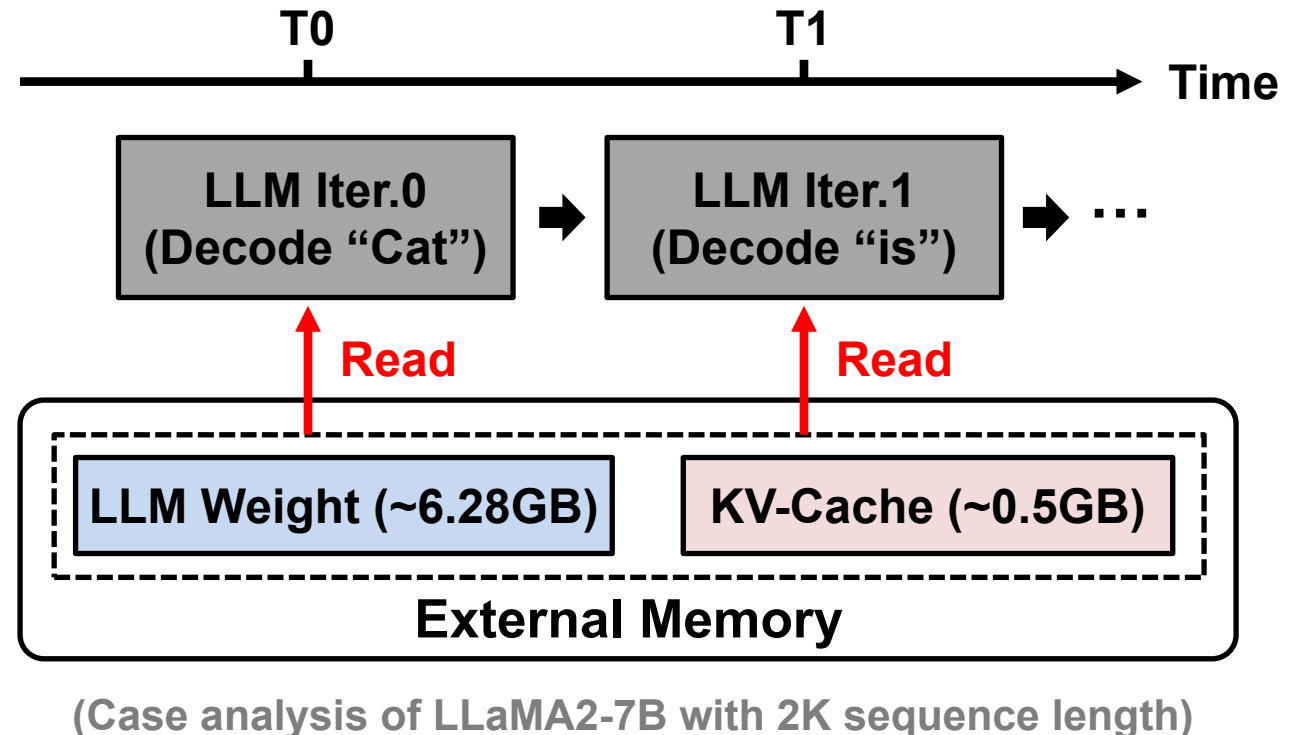
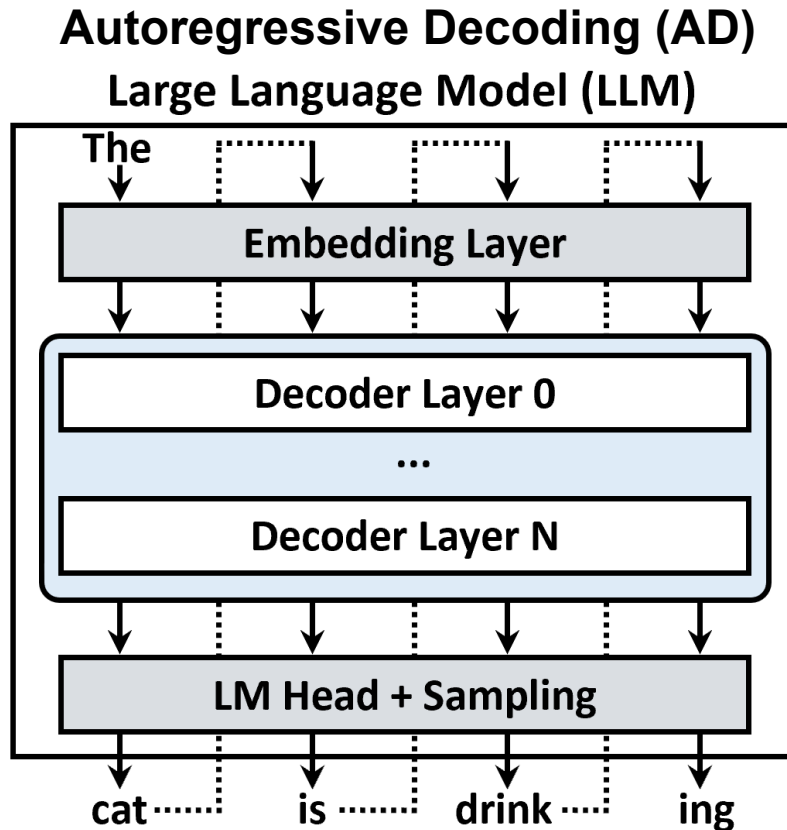
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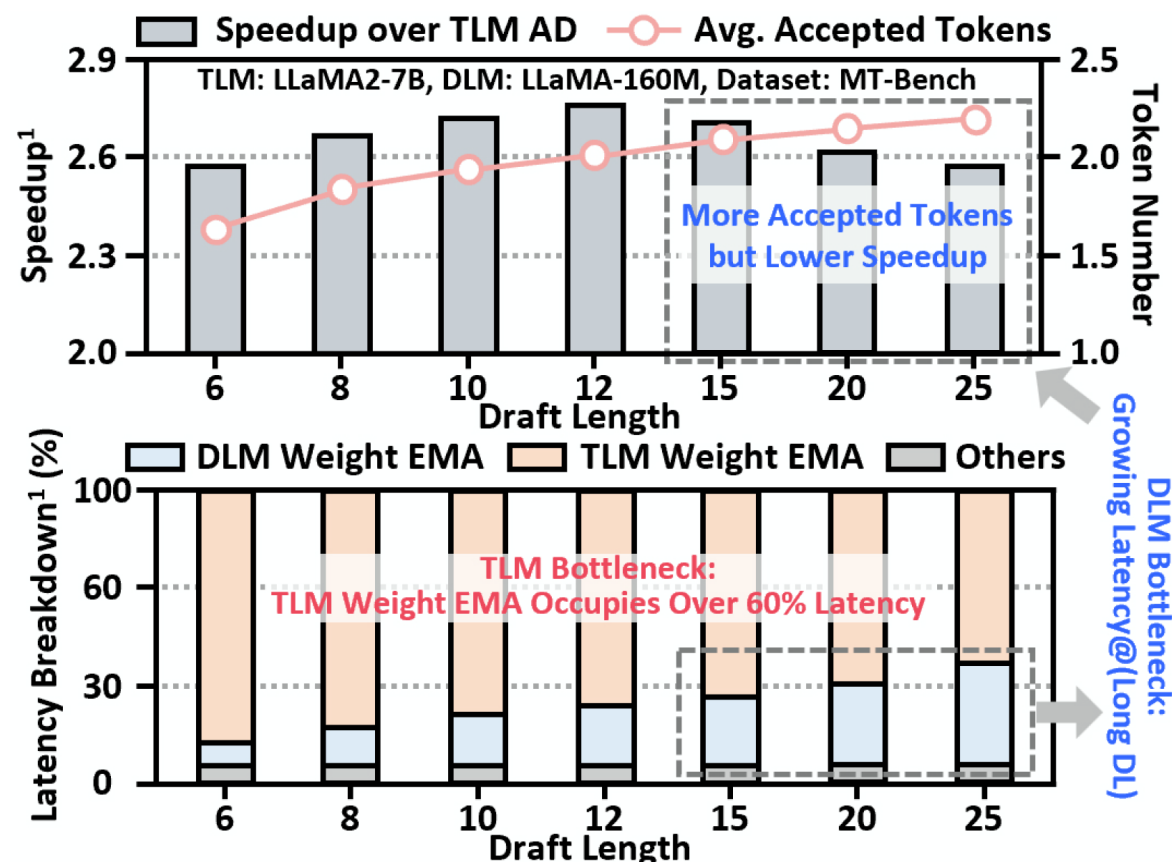
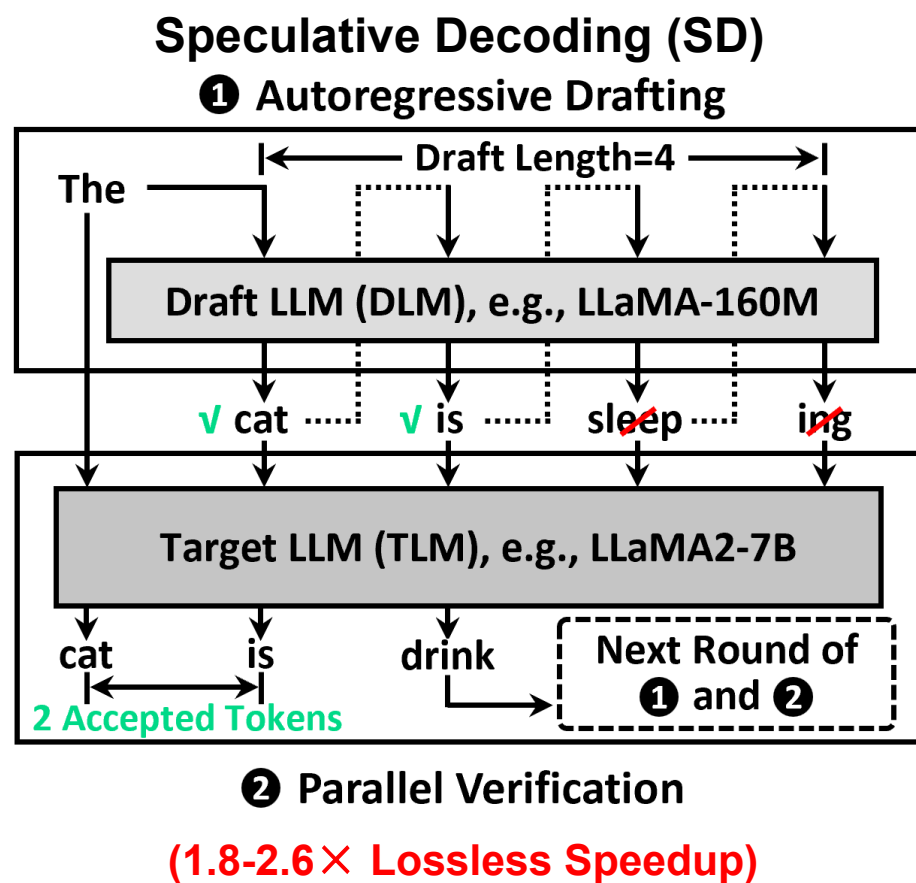
Autoregressive Decoding in LLM

- LLM reads the same weight from external memory at each AD iteration.
- External memory access (EMA) for weight occupies over 80% latency.



Speculative Decoding and Its Bottlenecks

- SD shifts the overheads of AD from a large TLM to a small DLM by parallel verification.
- TLM: weight EMA occupies >60% latency, DLM: high rejection ratio at long draft length.



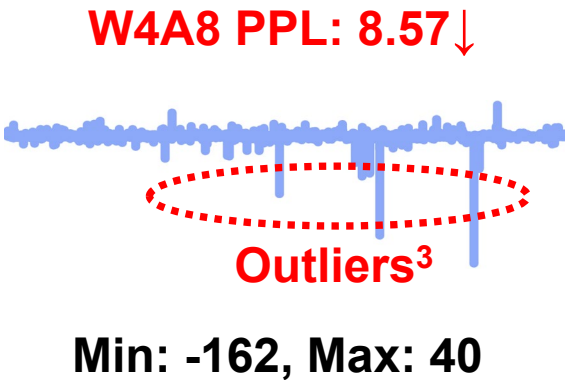
¹: Evaluated on 4 NPUs (2 TOPS and 3MB SRAM each) with LPDDR3 interface.

TLM Bottleneck-Activation Outliers

- Post-Training Quantization (PTQ) is widely adopted to reduce TLM Weight EMA.
- Activation outliers degrade accuracy at low-bit and are hard to be removed by PTQ.

Table I: Wikitext-2 PPL¹ on Various LLM and Quantization Setup

LLM	Precision (W-A-KV)		
	BF16 (Baseline)	W8A8KV8 ²	W4A8KV8 ²
LLaMA2-7B	5.47	5.92	8.57 (+3.1)
LLaMA3-8B	6.14	6.46	7.54 (+1.4)
Qwen3-8B	9.72	9.85	12.18 (+2.46)
Qwen3-1.7B	16.7	16.88	44.98 (+28.28)
Vicuna-1B	9.18	9.27	10.34 (+1.16)

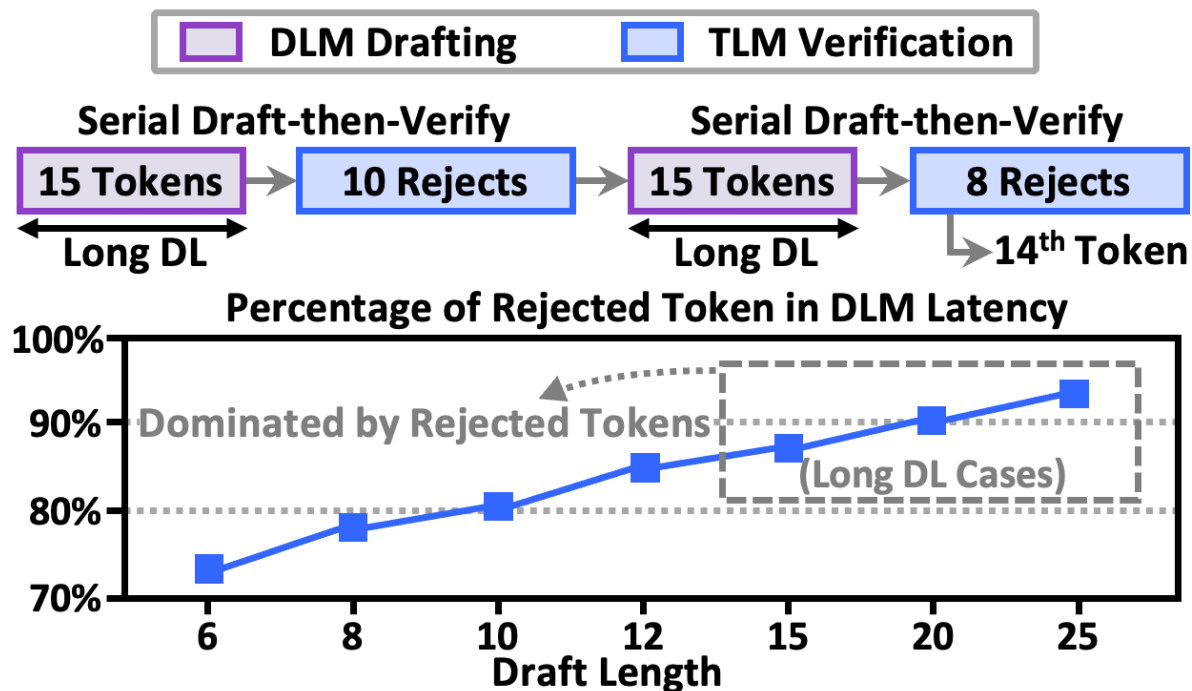
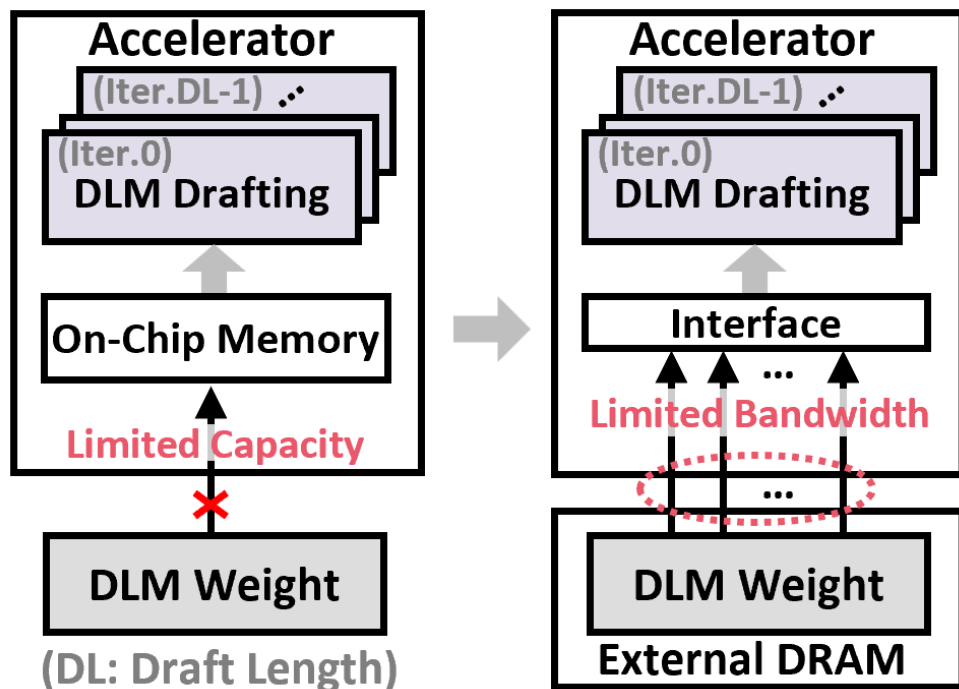


³: The 1st down_proj layer in LLaMA2-7B

¹: The lower the better, ²: GPTQ for weight, RTN for activation and KV-Cache, the others follow hardware implementation, e.g, customized BF16/INT32 Softmax/RMSNorm/ActFunc engine.

DLM Bottlenecks-EMA and High Rejection Ratio

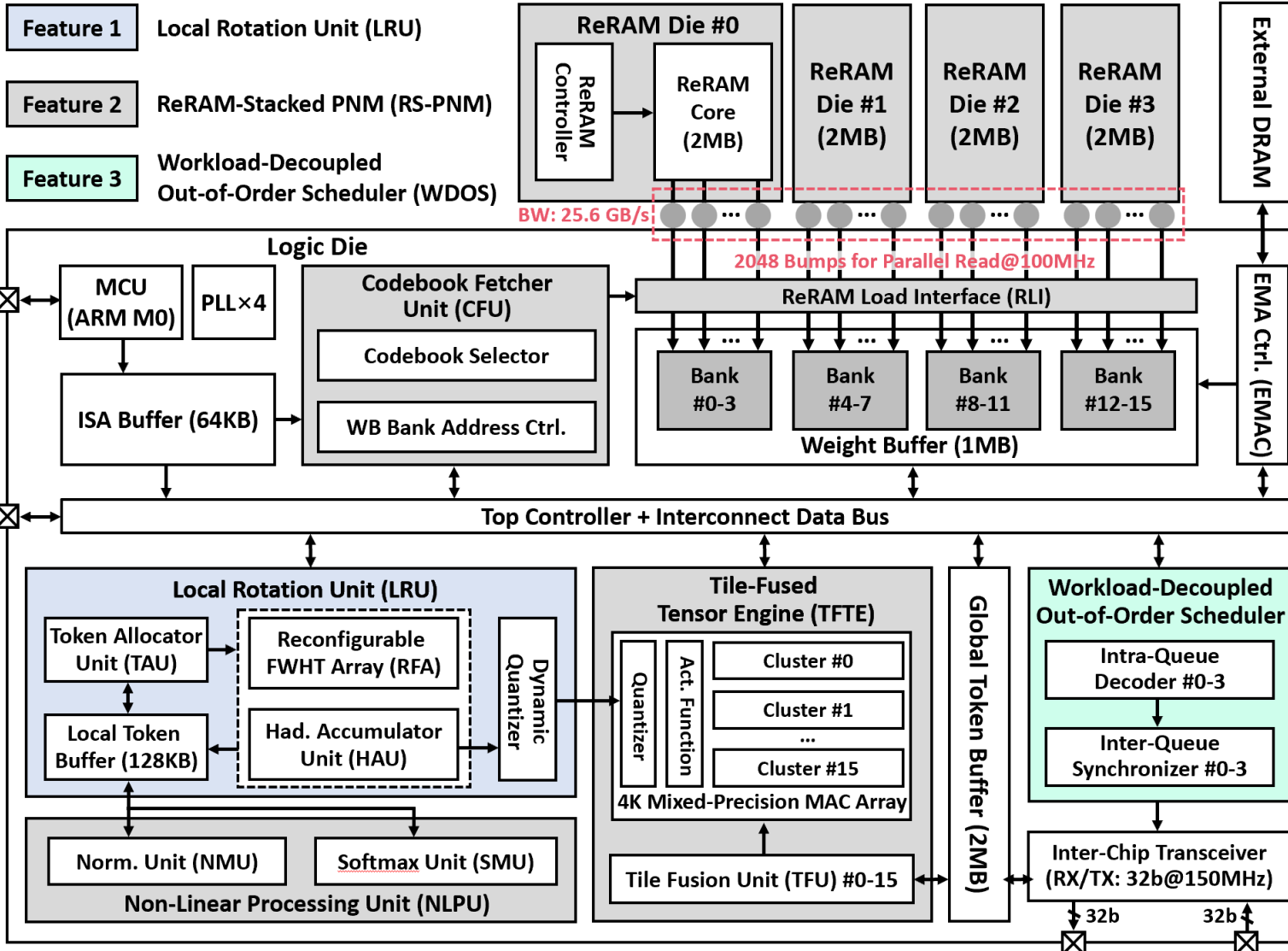
- Limited on-chip memory fails to buffer DLM even it is much smaller → EMA Issue.
- Most draft tokens are rejected at long draft length → Outweigh TLM latency savings.



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Overall Architecture



Key Specifications:

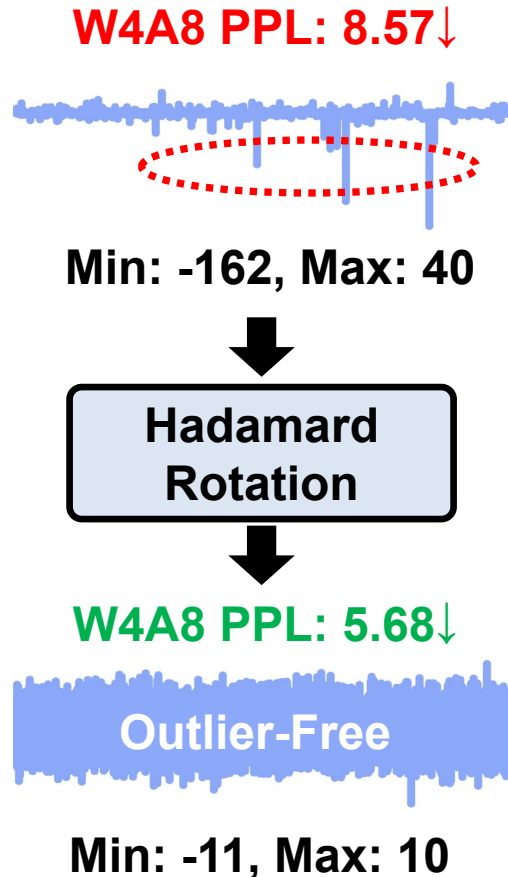
- 4 ReRAM dies are stacked on logic die via 2912 face-to-face bumps (864 ctrl. + 2048 data) with 80μm pitch.
- 8MB ReRAM with 25.6GB/s bandwidth at 100MHz and 3MB on-chip SRAM.
- A 4K mixed-precision MAC array supports W4A8/W8A8.
- 4-chip system based on unidirectional ring topology with 150 MHz, 32-bit source-synchronous parallel GPIOs

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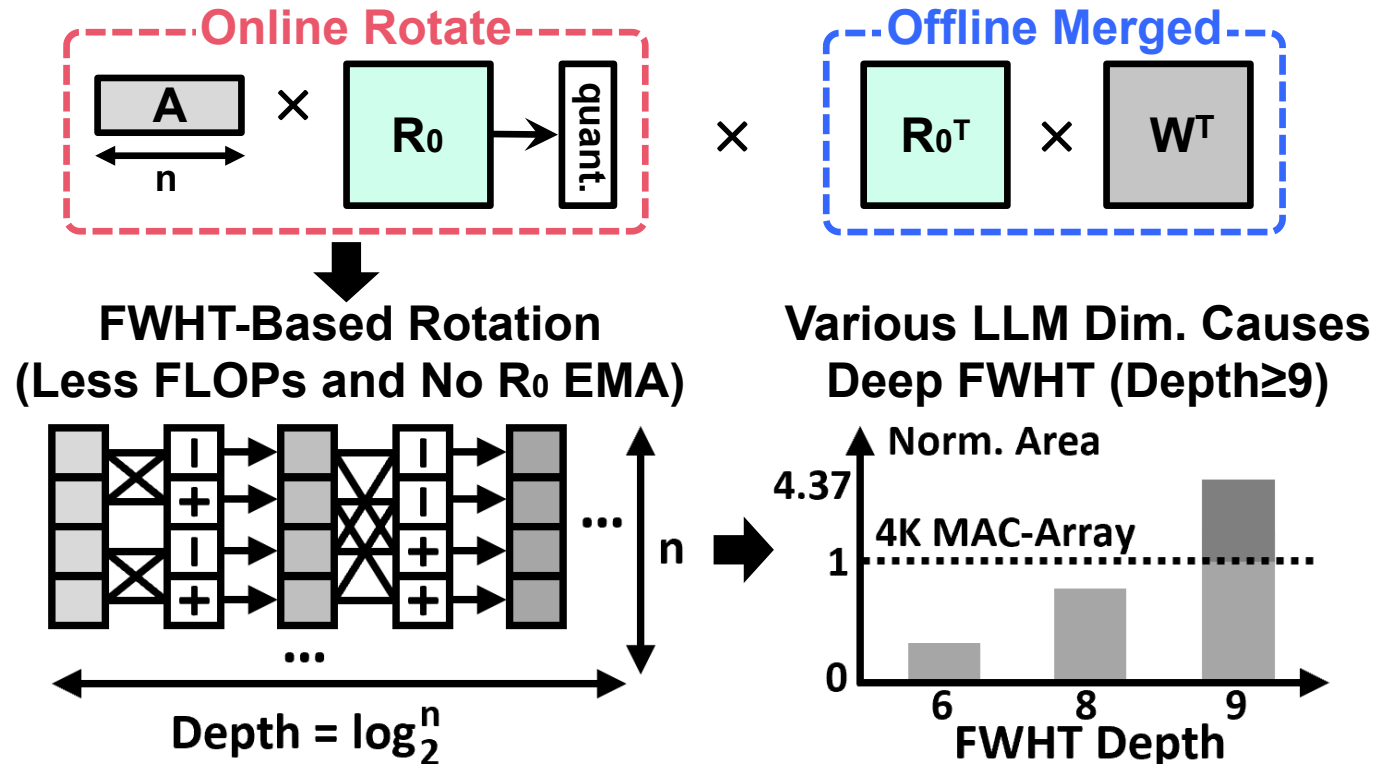
Hadamard Rotation

- Recent Hadamard rotation can remove outliers while preserving comp. invariance.
- Various LLM dimensions require deep FWHT arrays → Heavy Area overheads.



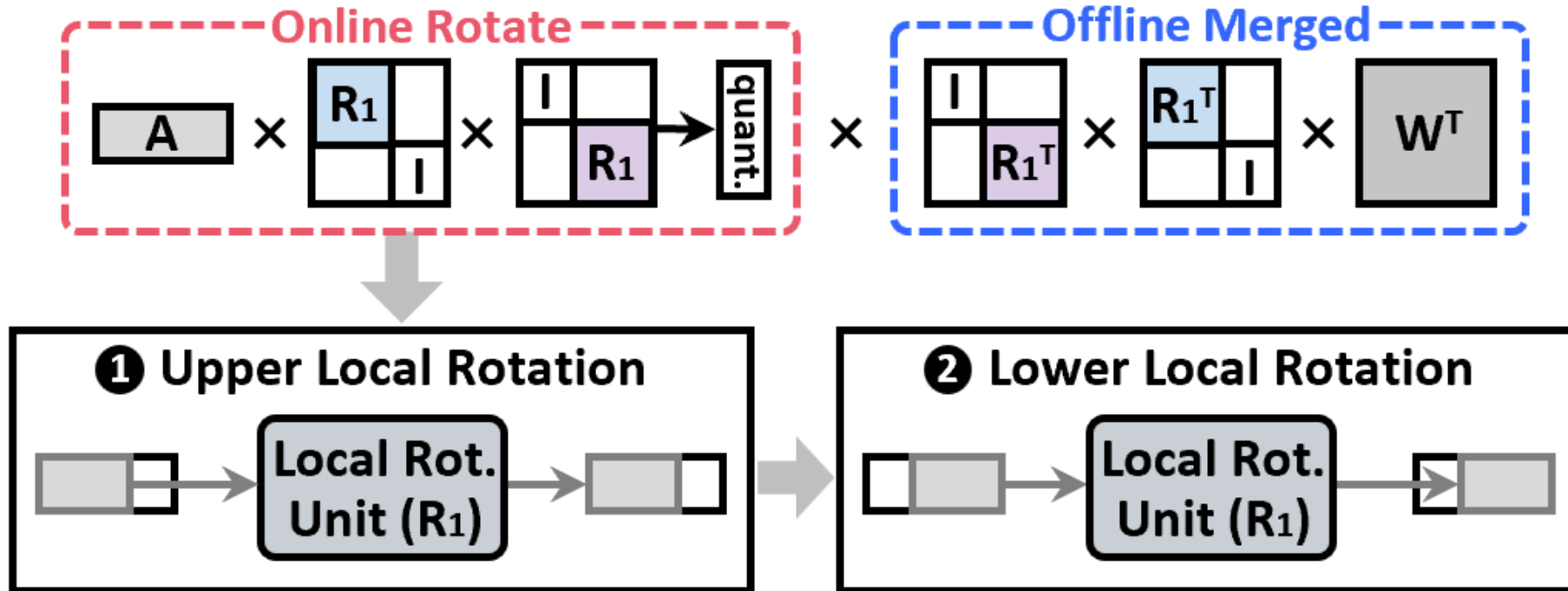
Computational Invariance in Hadamard Rotation

$$R_0 \times R_0^T = I \longrightarrow A \times W^T = (A \times R_0) \times (R_0^T \times W^T)$$



Proposed Local Rotation

- Decompose global Rot. into two overlapped local Rot. with shallow FWHT.
- Config on-chip local rotation unit (LRU) to execute two-stage online rotation.

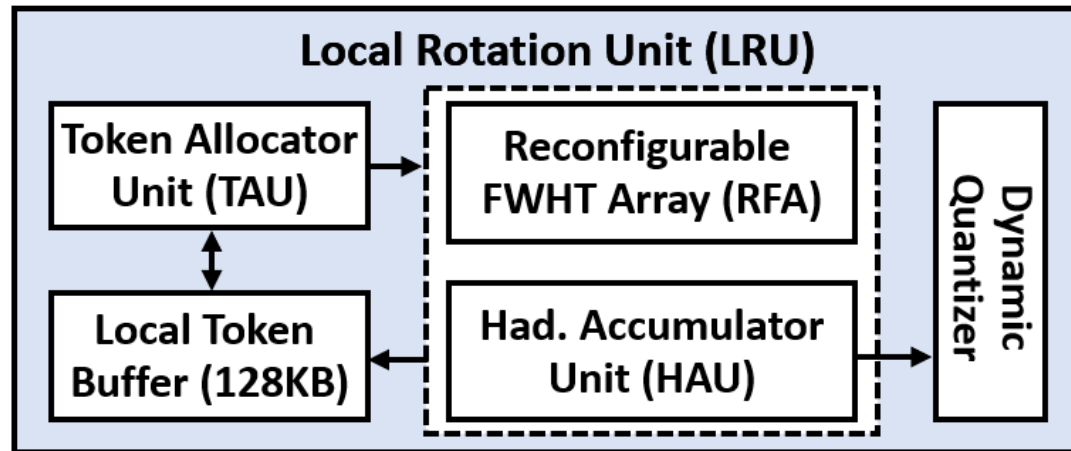


FWHT Decomposition

- Restrict the max range of FWHT array from area-heavy 2^9 to area-efficient 2^6 .
- Search (m, k) pair within 2^6 -range such that $2^k \times m \times 2$ covers original dimension n .

@Non-Power-of-Two (non-pot) Case:

Reshape $1 \times n$ into $2^k \times m$, then $H_n = H_{2^k} \otimes H_m$



Restrict FWHT max range from 2^9 to 2^6 , find (m, k) to $\min(2^k \times m \times 2 - n, m)$ s.t. $2^k \times m \times 2 \geq n$

TAU Allocates Upper $2^k \times m$ Token Features

Stage 1 (Upper Rot.)

2^k FWHT in RFA

H_m GEMM in HAU

2^k FWHT in RFA

H_m GEMM in HAU

TAU Allocates Lower $2^k \times m$ Token Features

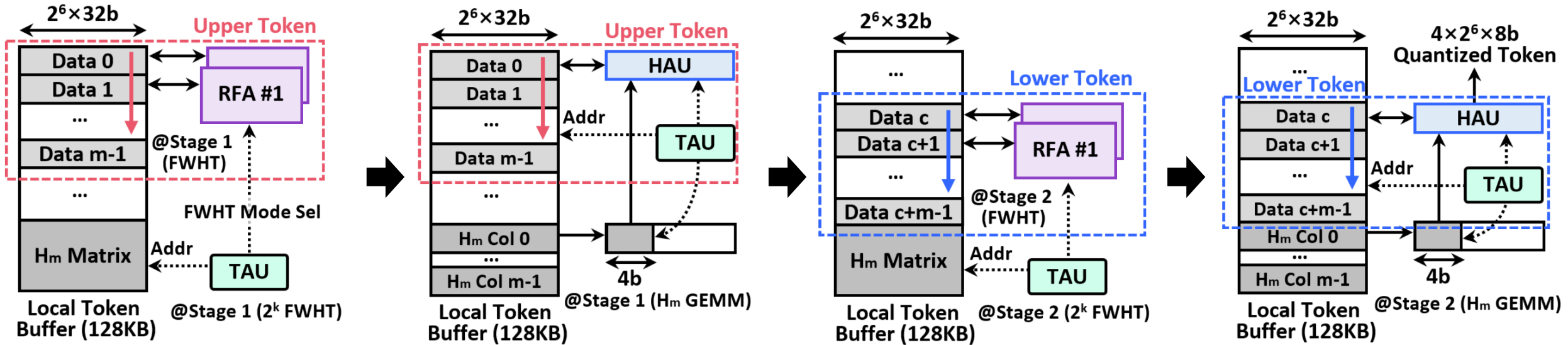
Stage 2 (Lower Rot.)

(m, k)
in ISA



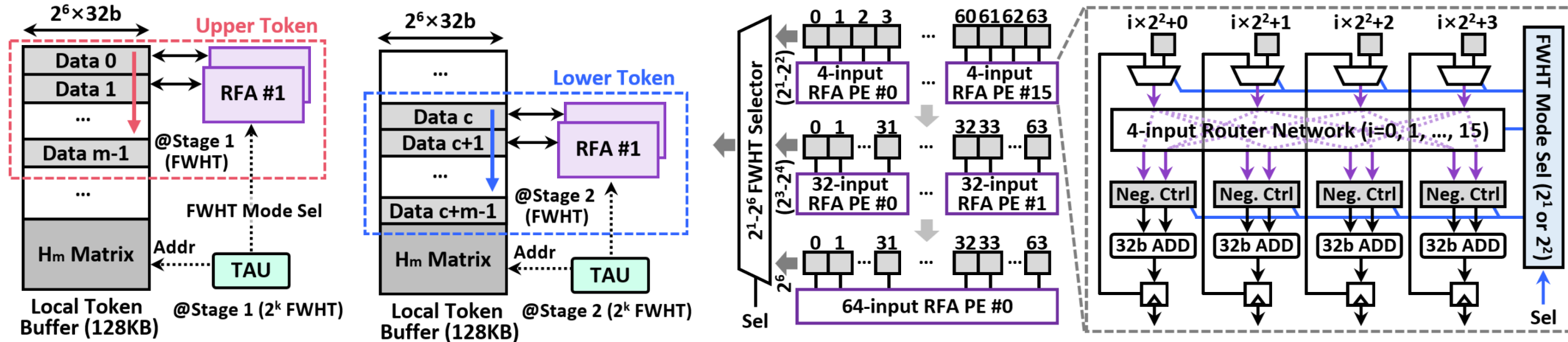
Local Rotation Unit (LRU) Workflow

- Reconfigurable FWHT Array (RFA) supports any FWHT in the range of 2^1 - 2^6 .
- Hadamard Accumulator Unit (HAU) performs MAC-free Hadamard GEMM.



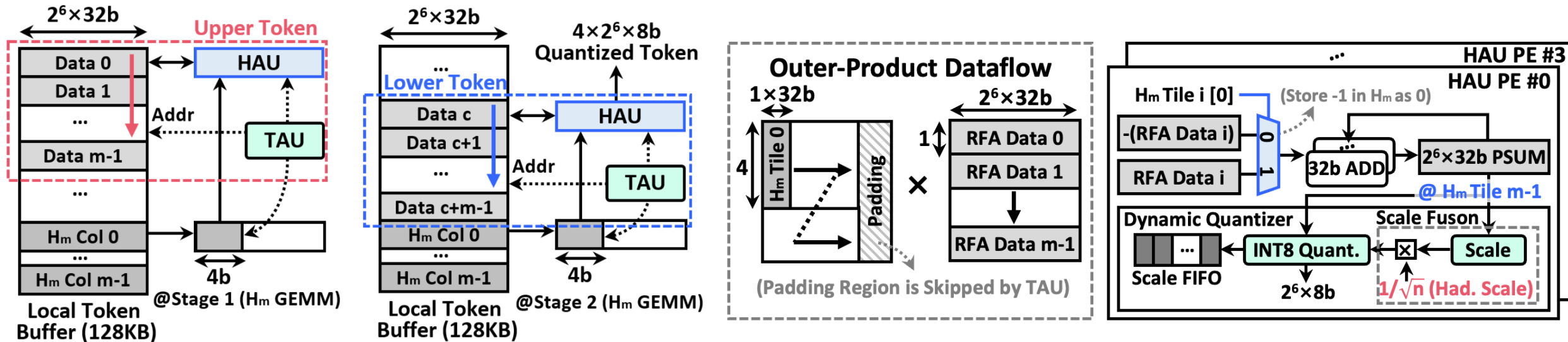
Reconfigurable FWHT Array (RFA)

- Merge adjacent FWHTs at the early stages → Reduce INT32 adders
- Input sharing → Support multi-range FWHT by switching the routers.



Hadamard Accumulator Unit (HAU)

- Fuse FP scales in Had. and dynamic quantizer → Replace FP16 MAC with INT Adder.
- Row-wise token mapping → Col-wise Had. matrix mapping w/ outer-product dataflow.



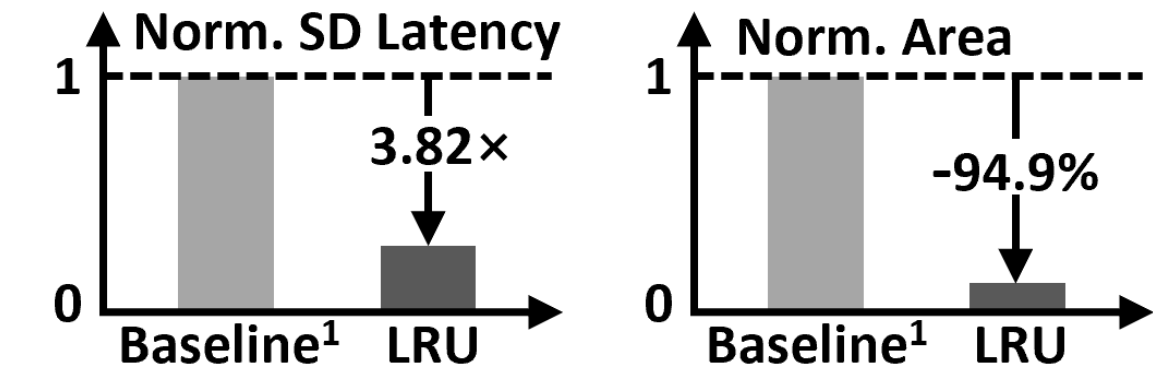
Performance Improvement

- LRU achieves the same accuracy as global rotation-based quantization
- Save 94.9% area compared with global rotation unit and achieve 3.82× speedup.

Table II: Wikitext-2 PPL on Quantization Methods

TLM	BF16	W4A8 ¹	W4A8 w/ GR ²	W4A8 w/ LRU
Vicuna-1B	9.18	10.34	9.43	9.41
LLaMA2-7B	5.47	8.57	5.68	5.68
LLaMA3-8B	6.14	7.54	6.70	6.71

¹: Baseline uses GPTQ, ²: Global rotation with FWHT.



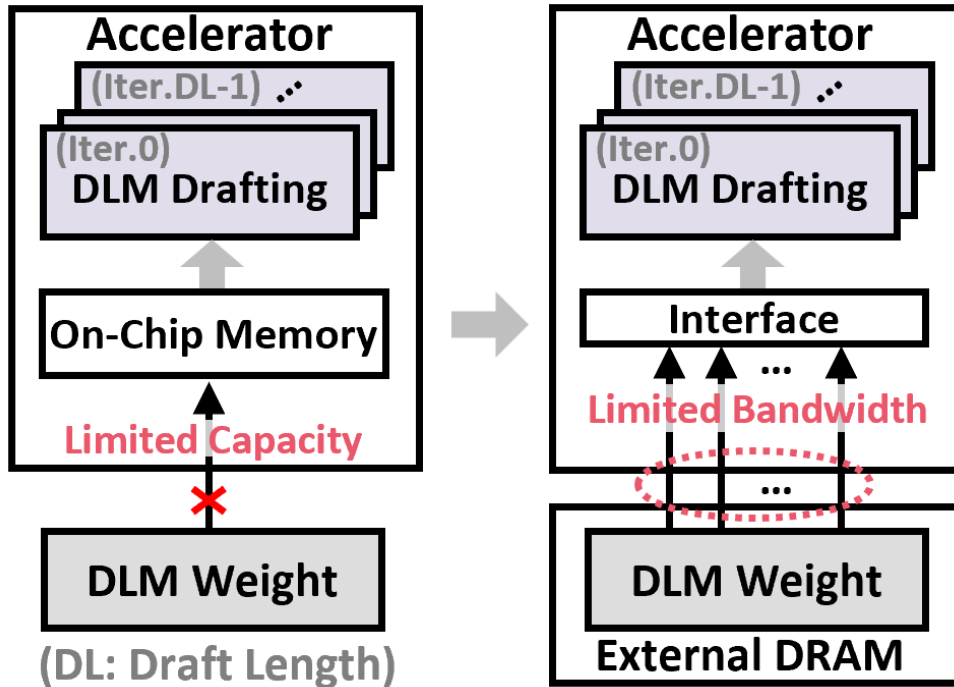
¹: TLM is LLaMA2-7B while DLM is LLaMA-160M, both are BF16. ¹: Original 2⁹ FWHT-based Hadamard rotation engine.

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DLM EMA Issue

- Limited on-chip memory fails to buffer DLM even though it is much smaller.
- ReRAM-on-Logic stacked PNM with blockwise vector quantization (VQ).



Leverage High Density and BW in ReRAM-on-Logic Stacking to Avoid DLM Weight EMA

Block-level Clustering

B ₀₀	B ₀₀	B ₀₂	B ₀₂
B ₀₁	B ₀₃	B ₀₃	B ₀₁



Index

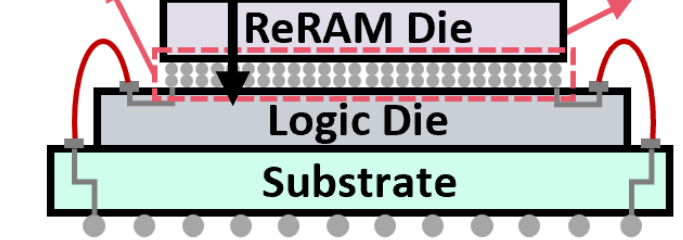
Codebook

0	0	2	2
1	3	3	1

B ₀₀	B ₀₁
B ₀₂	B ₀₃

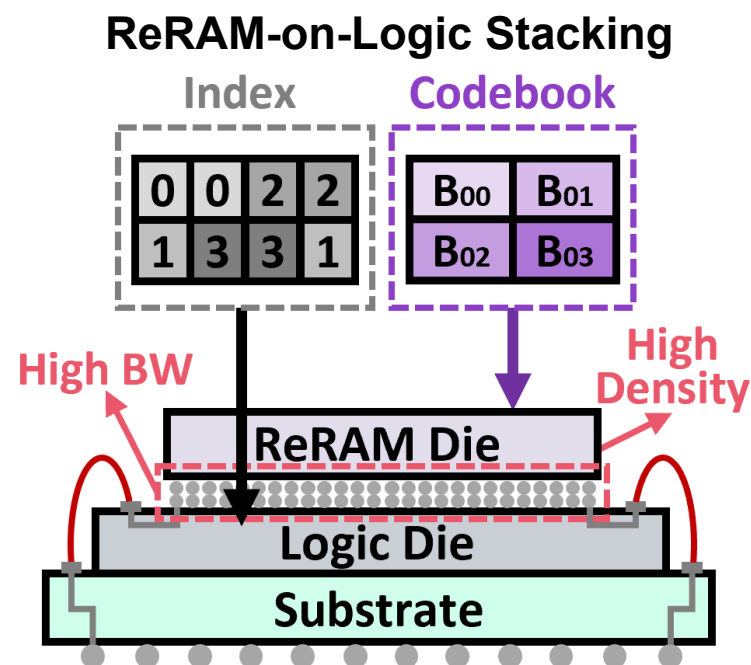
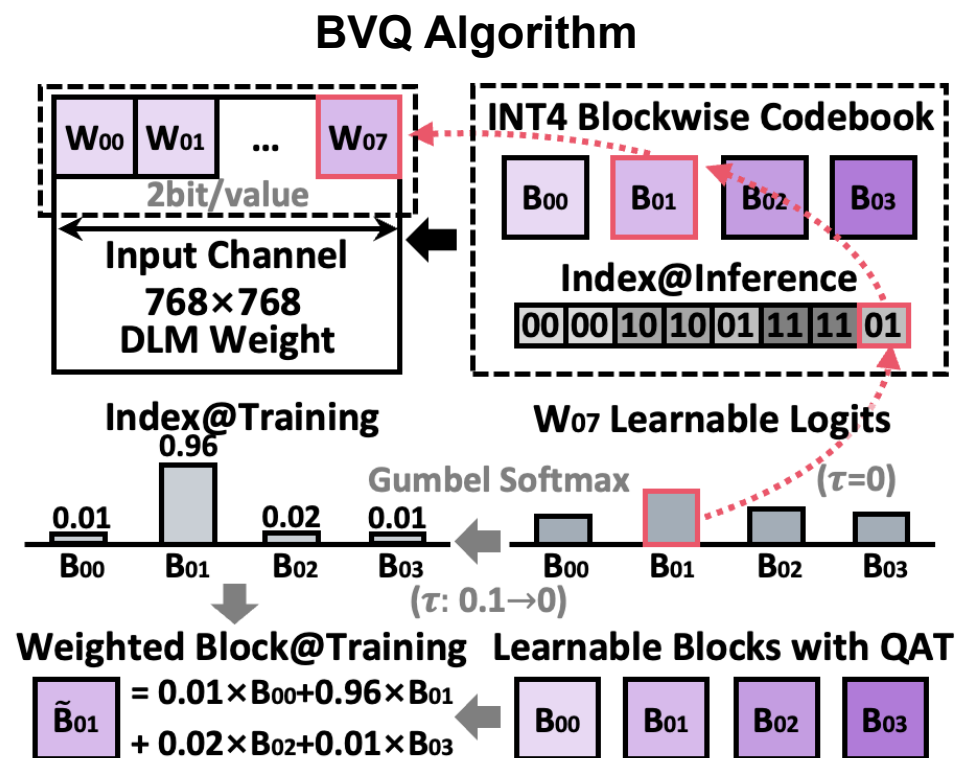
High BW

High Density

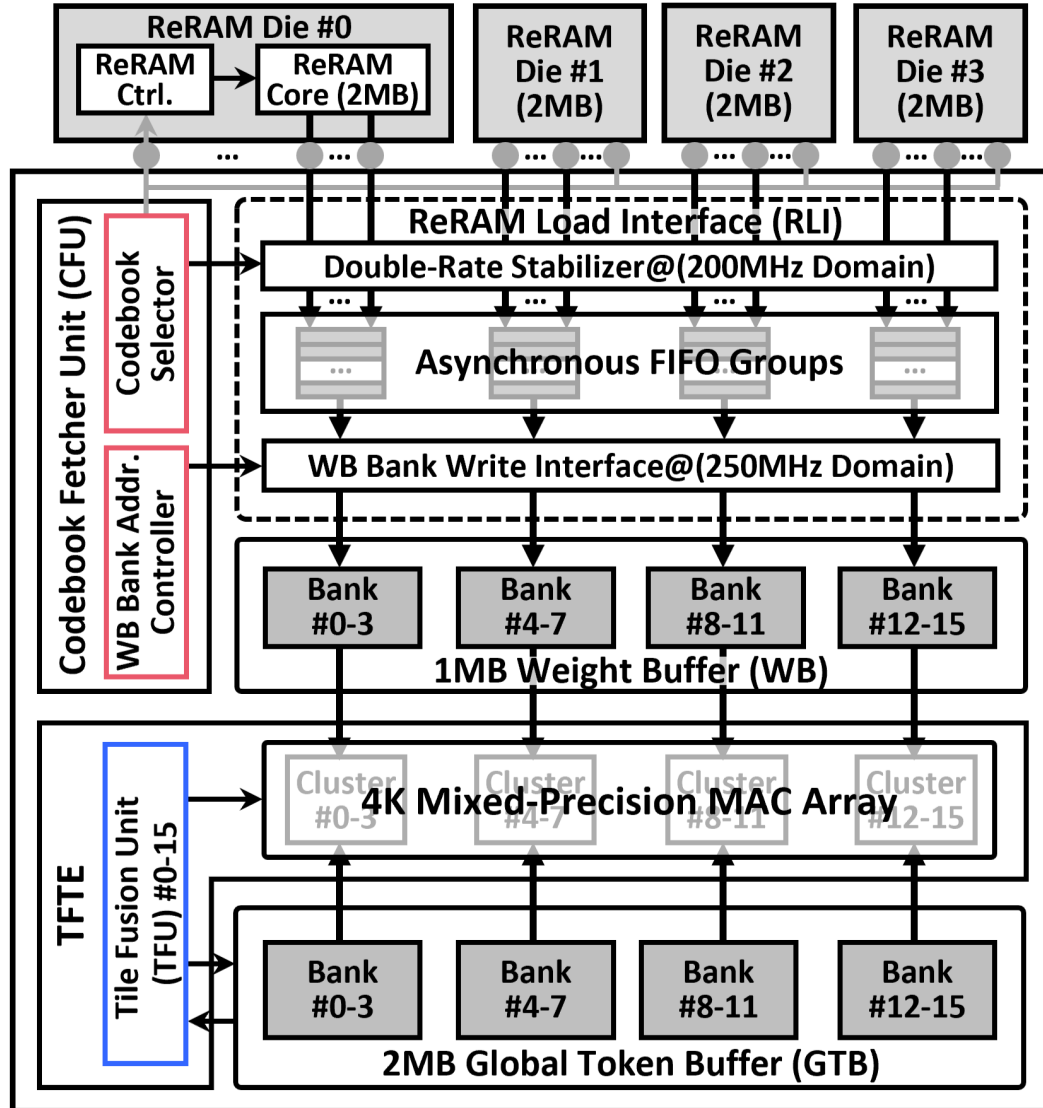


Blockwise Vector Quantization (BVQ)

- BVQ clusters weight into blockwise codebooks (CBs) and compact index.
- ReRAM-on-Logic Stacking → High memory density and bandwidth.



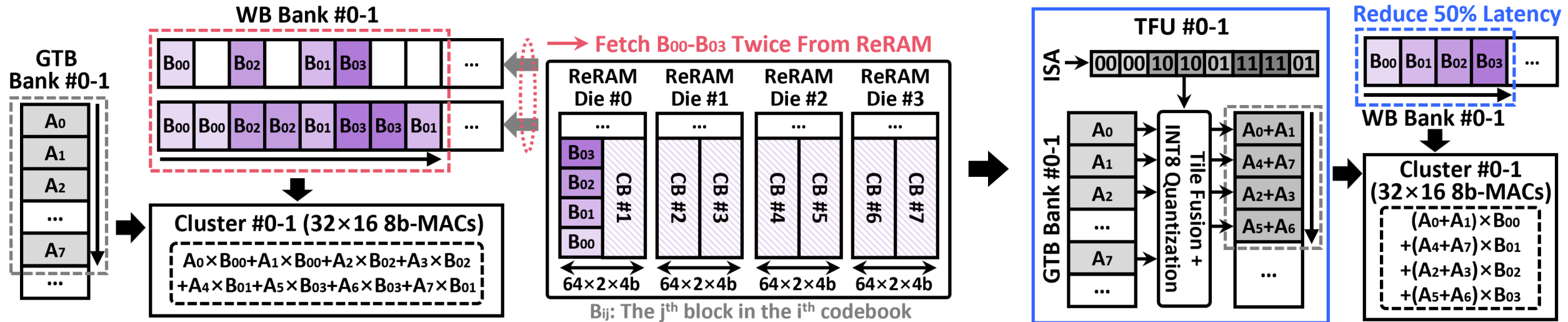
ReRAM-Stacked PNM (RS-PNM)



- **ReRAM Load Interface (RLI)** samples data at double clock rate and employs asynchronous FIFO groups for reliable clock domain crossing.
- **Codebook Fetcher Unit (CFU)** triggers ReRAM Controller and RLI to load CBs.
- **Tile-Fused Tensor Engine (TFTE)** fuses the tiles in the same CBs to reduce redundant ReRAM read.

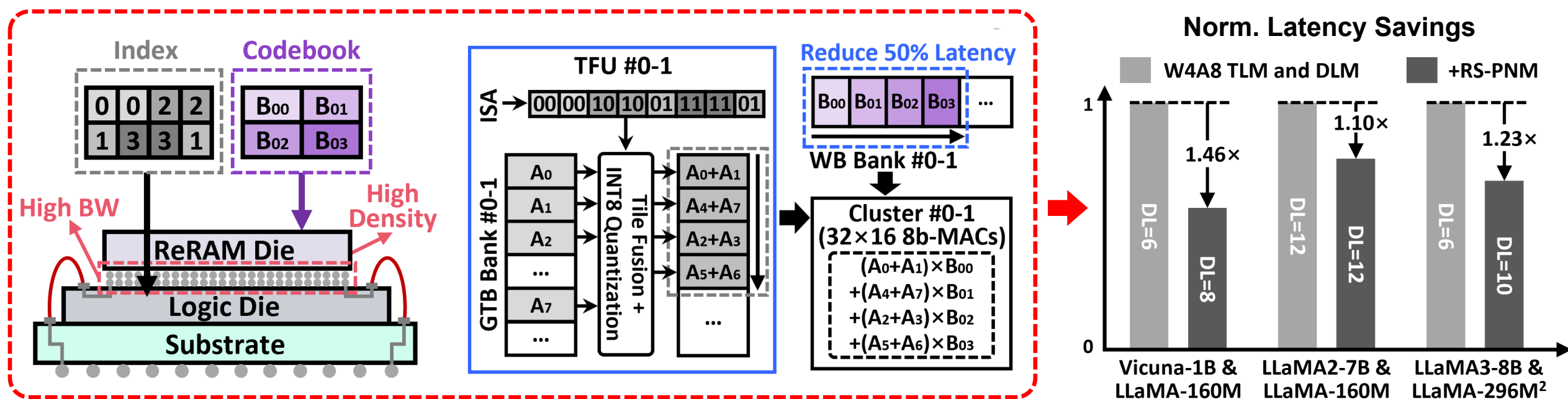
ReRAM-Stacked PNM (RS-PNM)

- Vertical CB mapping → Redundant CB read → Extra latency overhead.
- Fuse token tiles in the same CB entry → Each CB is read only once.



Performance Improvement

- Compress DLM from BF16 to 2bit/value and achieve 1.1-1.46 $\times$ speedup over LRU.

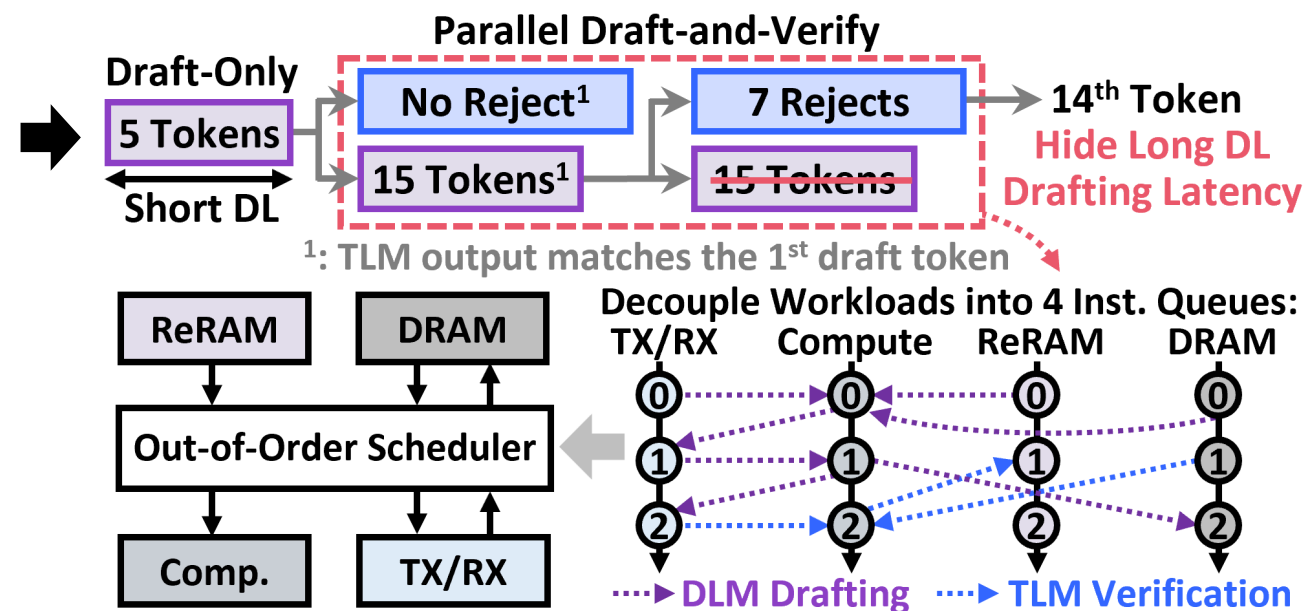
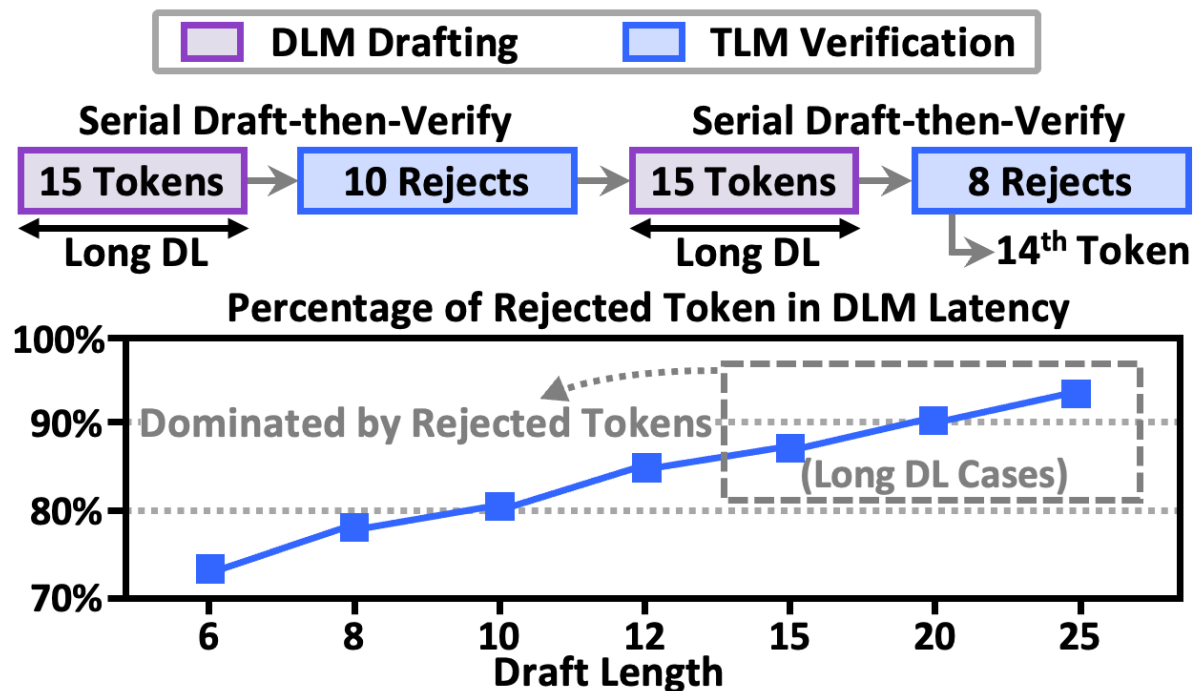


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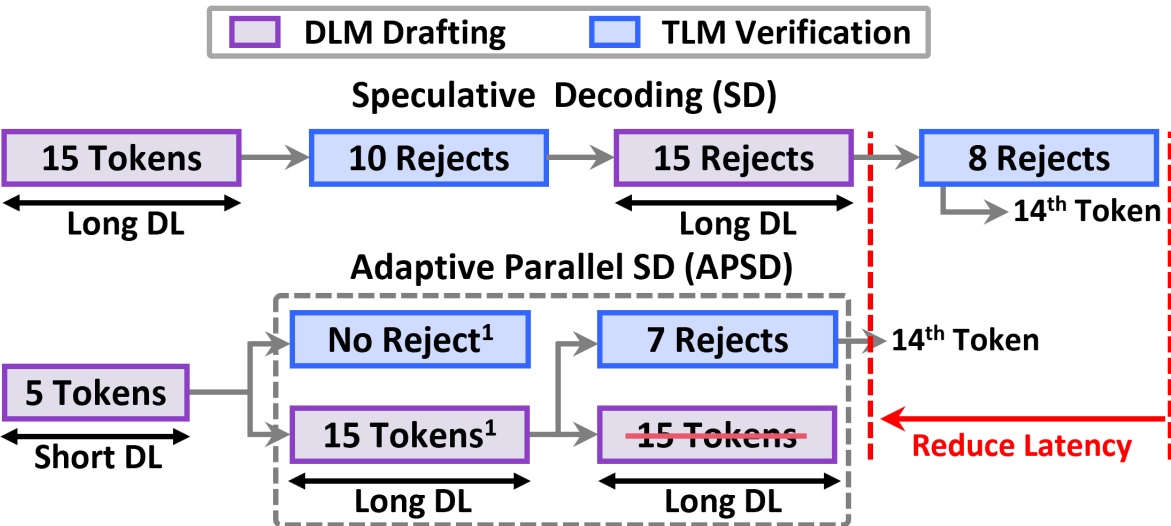
High Rejection Ratio at Long DL

- Long Draft length → High Rejection Ratio → Waste accepted tokens.
- Adaptive parallel SD with a workload-decoupled out-of-order scheduler.

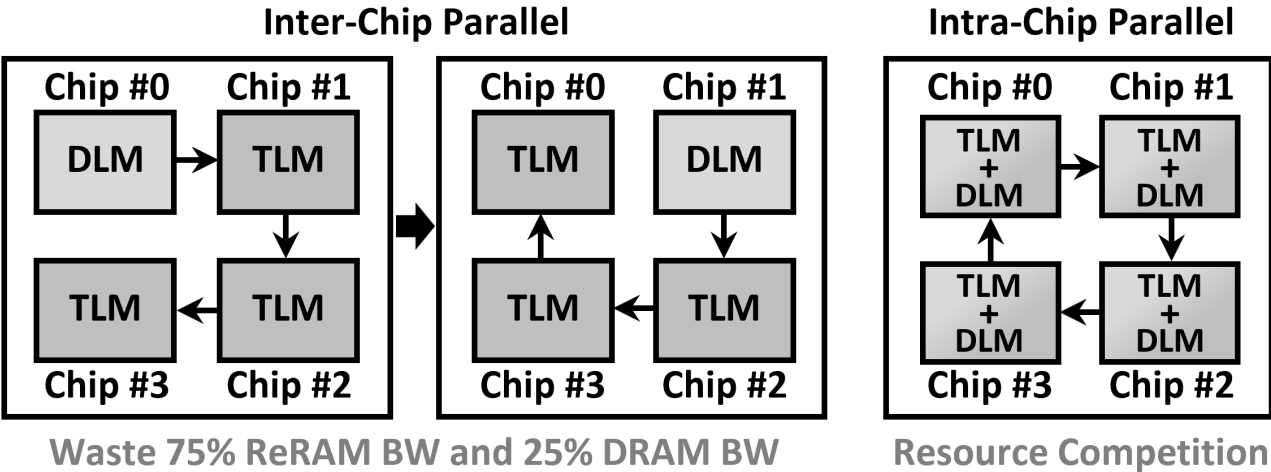


Adaptive Parallel Speculative Decoding (APSD)

- APSD adaptively switches between short DL drafting and long DL parallel draft&verify.
 - Short DL: high all-acceptance ratio, Long DL: high rejection ratio but more acc. tokens.
- Inter-chip parallel: bandwidth waste, Intra-chip parallel: resource competition.
 - At least 4-chip is required to store all DLM codebooks for APSD end-to-end inference.

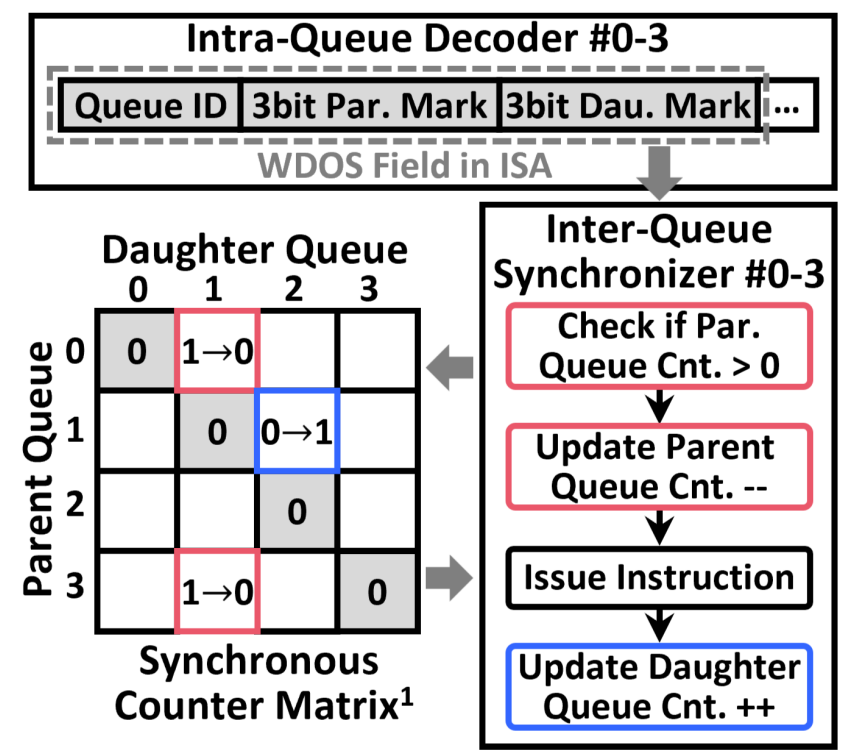
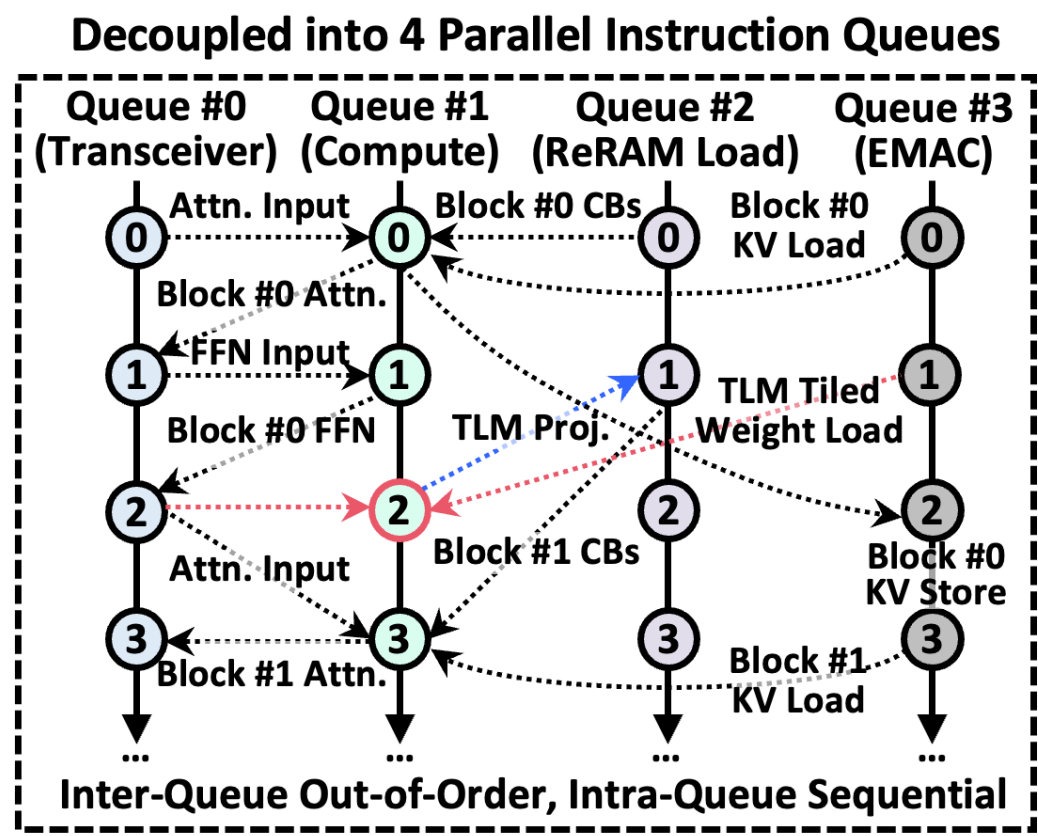


¹: TLM output token matches the 1st token from concurrent DLM drafting.



Workload-Decoupled Out-of-Order Scheduler

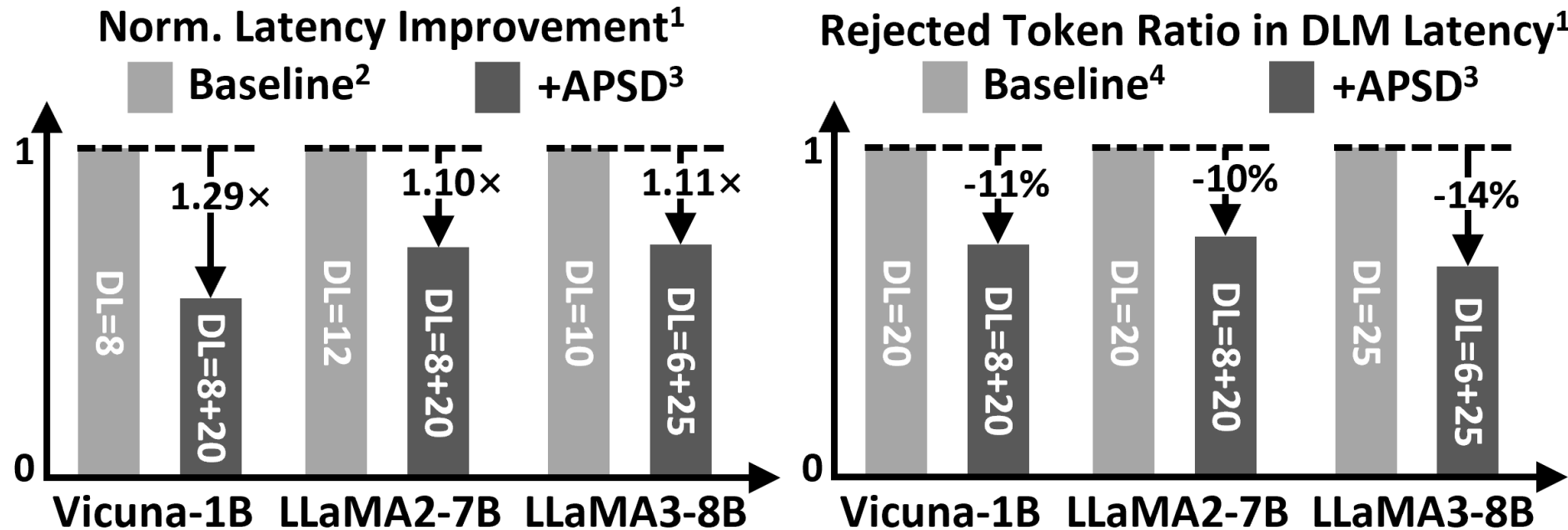
- **Workloads are decoupled into 4 parallel instruction queues**
 Queue #0: inter-chip transceiver, Queue #1: compute, Queue #2: ReRAM load, Queue #3: EMA Ctrl. (EMAC)
- **An instruction is issued when its Par. Queues are ready, then notifies its Dau. Queues.**



¹: Only changes@(Instr. 2 of Queue #1) are shown, where Par. Queues are 0&3 and Dau. Queue is 2.

Performance Improvement

- Achieve 1.1-1.29 $\times$ speedup over RS-PNM and LRU with 10-14% rejected draft token latency reduction.

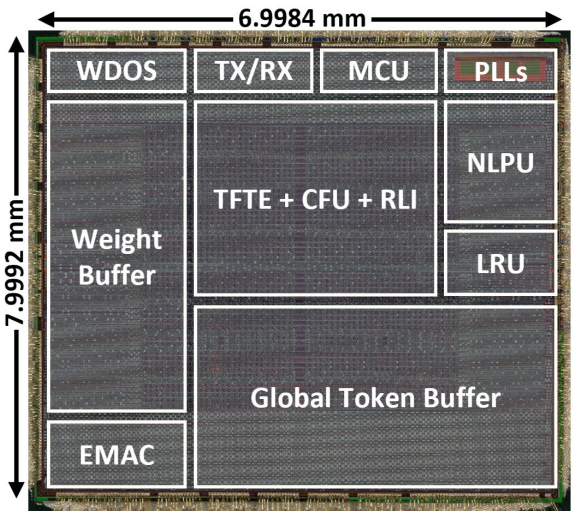


¹: Ignore DLM illustration and test on 4 chips with LPDDR3 interface, ²: SD with LRU and RS-PNM, ³: Include CILM and WDOS, ⁴: RS-PNM with the same long DL in APSD.

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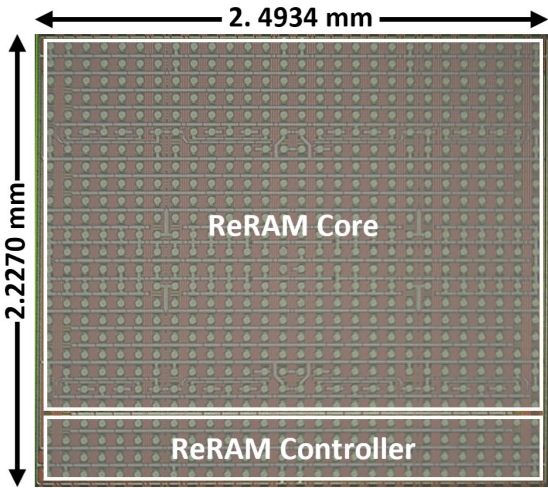
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Die Photos, Specifications and Test System

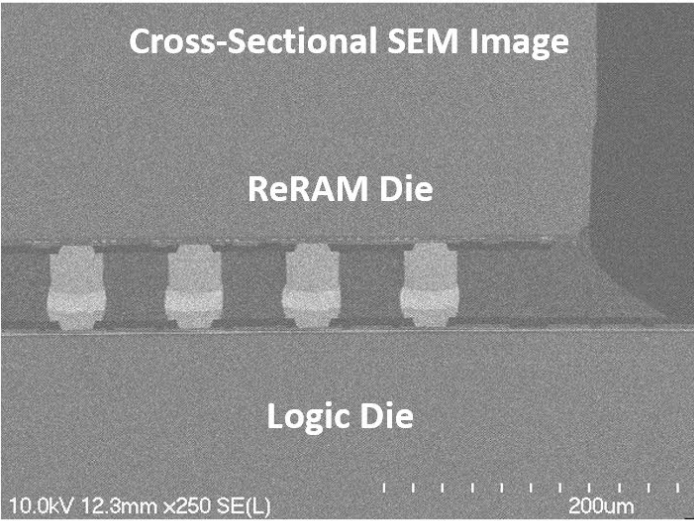
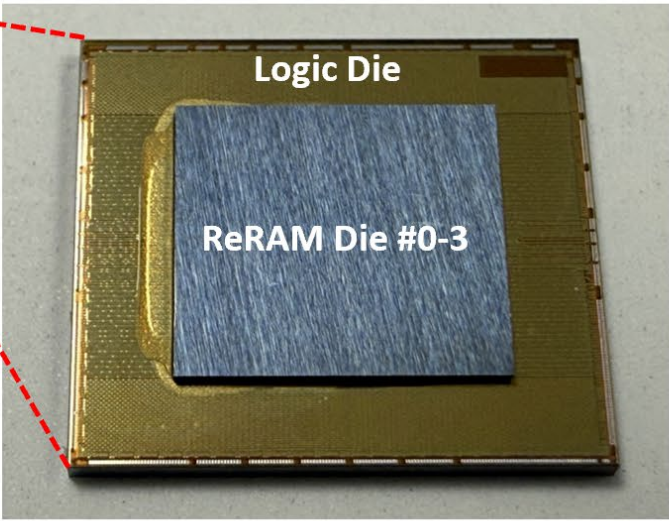


Specification (Logic Die)	
Technology	55nm
Voltage	0.89-1.40 V
Frequency	62.5-285 MHz
Die Area	55.98 mm ²
Precision	W: INT4/8, A: INT8/16/32
SRAM	3.43 MB
Performance ^{*1}	0.51 ^{*2} -2.33 ^{*3} TOPS
Power	0.265 ^{*2} -3.060 ^{*3} W

^{*1}: 1 OP = 1 Multiplication or Addition in the PE Array.
^{*2}: At 0.89V and 62.5MHz. ^{*3}: At 1.40V and 285MHz.



Specification (4 ReRAM Dies)	
Technology	55nm
Voltage	1.1 V
Frequency	100 MHz
Die Area	5.553×4 mm ²
Capacity	2×4 MB
Bump #	2912 (Data: 2048, Ctrl.: 864)
Bump Pitch	80μm
Bump Diameter	40μm
Read Power	49.54×4 mW

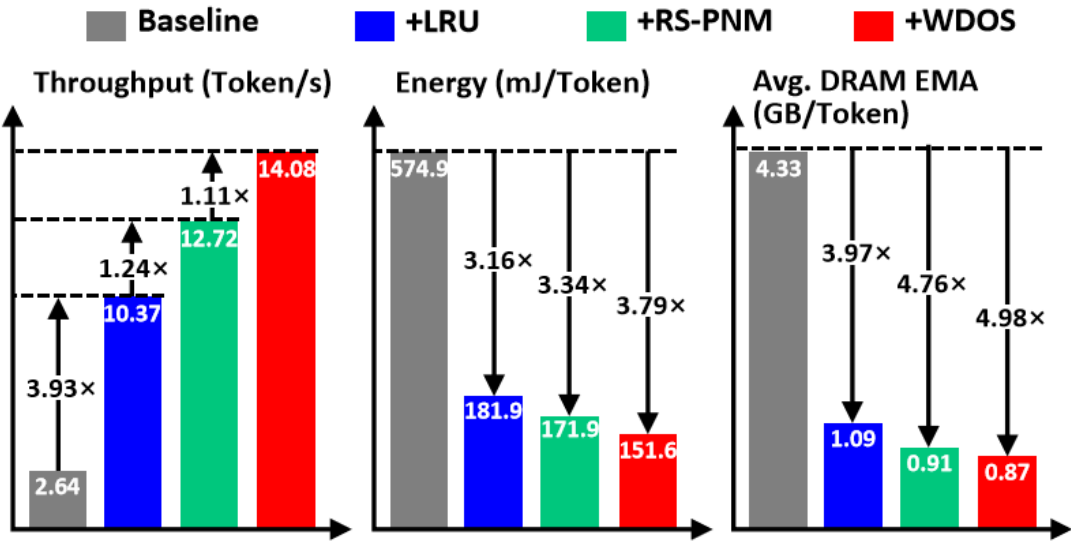


Benchmark Analysis

■ The chip achieves 4.46-7.71× speedup and 3.74-4.85× energy savings at system-level.

TLM & DLM	Vicuna-1B & LLaMA-160M	LLaMA2-7B & LLaMA-160M	LLaMA3-8B & LLaMA-296M
Data Precision	TLM: W4A8, DLM: W4A8 (2bit/value with BVQ)*1		
TLM Perplexity (↓)*2	9.41 (+0.23)	5.68 (+0.21)	6.71 (+0.57)
Avg. ReRAM Access*3*4	36.75 MB/Token	0.0432 GB/Token	0.0399 GB/Token
Avg. DRAM Access*3*4	97.68 MB/Token	0.6637 GB/Token	0.8674 GB/Token
Avg. EMA Saving*7	6.67×	4.79×	4.99×
Decoding Throughput*3*4	135.69 Token/s	17.82 Token/s	14.08 Token/s
Speedup*5	7.17×	4.46×	5.33×
Energy Consumption*3*4	18.26 mJ/Token	123.41 mJ/Token	151.59 mJ/Token
Energy Saving*5	4.85×	3.74×	3.95×

Improvement Breakdown Analysis on LLaMA3-8B&LLaMA-296M*3-5



*1: Each DLM is trained on ShareGPT dataset. *2: Evaluated on Wikitext-2 dataset with 2048 prefill length and the baseline is BF16 TLM. *3: EMA is simulated assuming 4-chip parallelism and an LPDDR3 interface and the on-chip performance is measured at 1.40V, 285MHz. *4: Evaluated on MT-Bench dataset. *5: The baseline is BF16 SD.

Performance Comparison

	VLSI'25 [6]	VLSI'25 [7]	ISSCC'25 [4]	ISSCC'25 [5]	ISSCC'24 [3]	This Work
Technology	28nm	28nm	28nm	28nm	28nm	55nm with ReRAM Die on Logic Wafer Stacking
Memory Capacity	24KB SRAM	272KB SRAM	384KB SRAM	500KB SRAM	500KB SRAM	3.43MB SRAM + 8MB ReRAM
Memory BW	NA	NA	NA	1.6 GB/s (DRAM)	1.6 GB/s (DRAM)	25.6 GB/s (ReRAM)
Supply Voltage (V)	0.6-1.0	0.6-1.0	0.63-1.0	0.58-1.0	0.7-1.1	0.89-1.40 (logic)
Frequency (MHz)	85-410	110-500	50-460	25-200	50-200	63.5-285 (Logic), 100 (ReRAM)
Die Area (mm ²)	1.56	12.65	3.52	20.25	20.25	55.98 (Logic), 22.21 (ReRAM)
Precision	W: INT4 A: INT4/INT8	W: INT8/NUQ1~4 A: FP16	W: INT8 A: BF16	W: INT1-16/1.58b A: INT4/8/16	INT8	W: INT4/INT8 A: INT8/16/32
Peak Performance* ¹ (TOPS or TFLOPS)	0.63-2.52 (Hybrid A4/A8)	0.12-0.96 (NUQ1~4@Decoding)	3.58 (Dense)	13.1 (87.5% Sparsity)	3.41 (Dense)	2.33* ² (Dense W4A8/W8A8)
Benchmark Model and Perplexity (↓)	LLaMA2-7B (6.16)	OPT-1.3B (15.1) LLaMA2-7B (6.26)	GPT2-0.1B (30.85) OPT-1.3B (15.50) LLaMA2-7B (5.66)	LLaMA-3B (1.58bit: 10.3)	GPT2-0.7B (14.96)	Vicuna-1B (9.41) LLaMA2-7B (5.68) LLaMA3-8B (6.71)
System-Level Decoding* ^{3,4} Throughput (Token/s)	LLaMA2-7B (5.46)	OPT-1.3B (32.75) LLaMA2-7B (6.76)	GPT2-0.1B (286.69) OPT-1.3B (28.13) LLaMA2-7B (5.63)	LLaMA-3B (1.58bit: 74.92)	GPT2-0.7B (69.9)	Vicuna-1B (135.69) LLaMA2-7B (17.82) LLaMA3-8B (14.08)
System-Level Energy* ^{3,4} Consumption (mJ/Token)	LLaMA2-7B (223.69)	OPT-1.3B (47.09) LLaMA2-7B (181.48)	GPT2-0.1B (5.36) OPT-1.3B (55.09) LLaMA2-7B (218.25)	LLaMA-3B (1.58bit: 16.03)	GPT2-0.7B (21.90)	Vicuna-1B (18.26) LLaMA2-7B (123.41) LLaMA3-8B (151.59)

*¹: EMA is excluded. *²: @1.40V, 285MHz. *³ *³: EMA is simulated assuming 4-chip parallelism and an LPDDR3 interface, whereas on-chip performance is measured w/ ReRAM access included in this work. *⁴: Based on MT-Bench dataset with average prefill/decode lengths of 454/512 tokens.

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Summary

- Speculative decoding provides an opportunity to optimize the EMA of large-scale LLM indirectly through a small and highly tuneable draft model.
- ReRAM-on-Logic stacking extends memory capacity in a low-power and cost-effective manner.
- This work leverages bumping-based ReRAM-on-Logic stacking technology to design an LLM accelerator with three HW/SW co-design features:
 - **Local Rotation Unit with Decompose FWHT**
(Accurate W4A8 TLM with little area overheads and $3.82\text{-}3.93\times$ speedup)
 - **ReRAM-Stacked PNM with Blockwise Vector Quantization**
(Compress DLM to 2bit/value with $1.1\text{-}1.46\times$ speedup)
 - **Adaptive Parallel SD with Workload-Decoupled Out-of-Order Scheduler**
(Attain $1.1\text{-}1.29\times$ speedup and 10-14% rejected token ratio reduction)

Revolver: Low-Bit GenAI Accelerator for Distilled-Model and CoT with Phase-Aware-Quantization and Rotation-Based Integer-Scaled Group Quantization

***Sangjin Kim*, Jungjun Oh, Byeongcheol Kim,
Yuseon Choi, Gwangtae Park, and Hoi-Jun Yoo**

**Semiconductor System Lab.
School of EE, KAIST**

Outline

I. Introduction

II. Motivation

III. Revolver Architecture

- 1) Phase-aware Precision Selection (PAPS)
with Multi-Precision Loader (MP-Loader)
- 2) Local-Rotation with Harmonic Aligned Permutation (LR-HAP)
and Paired Unit-FHT Block (PUB)
- 3) Sliced Integer-based Dequantization Unit (SIDU)

IV. Chip Implementation Results

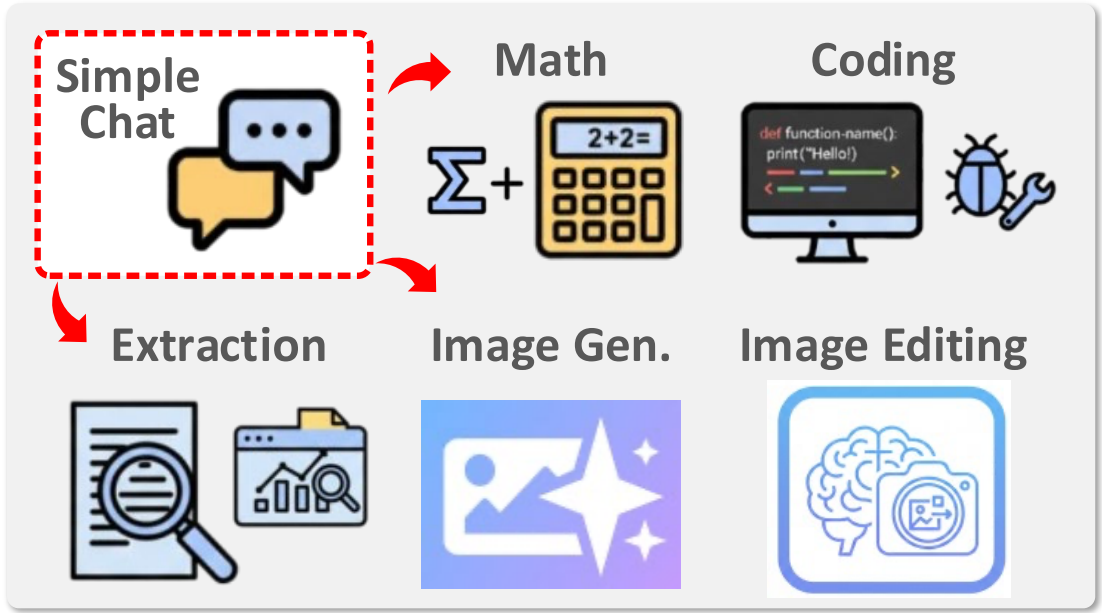
V. Conclusion

Characteristics on Recent GenAI Model

❑ Versatile Capabilities by Strong Large Foundation Model

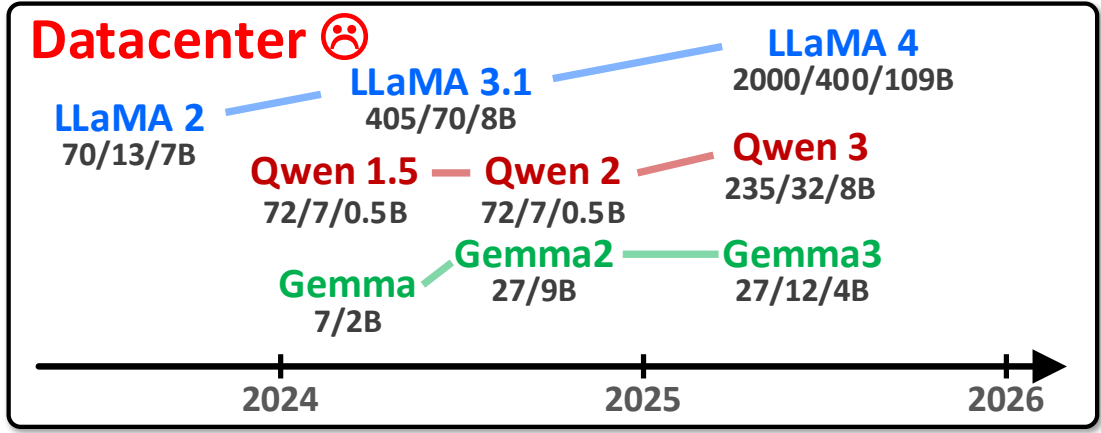
- Seamlessly extend workload from simple chat to specialized real-world task
- Base on the large model’s performance ➔ Not deployable on edge

Versatile Capabilities of Modern AI



Seamlessly Extending to Specialized Workflows

Scaling Model Size for Better Performance



Edge Environment

Limited DRAM Capacity (~12GB)

Weight EMA Bottleneck

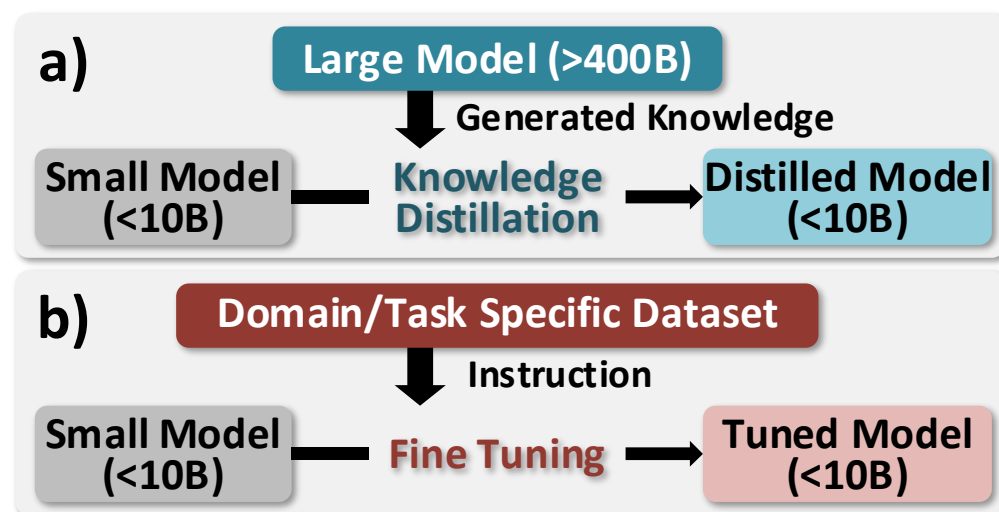
Large Models ➔ Not Deployable on Edge

Trend) Overcoming Capacity for Edge

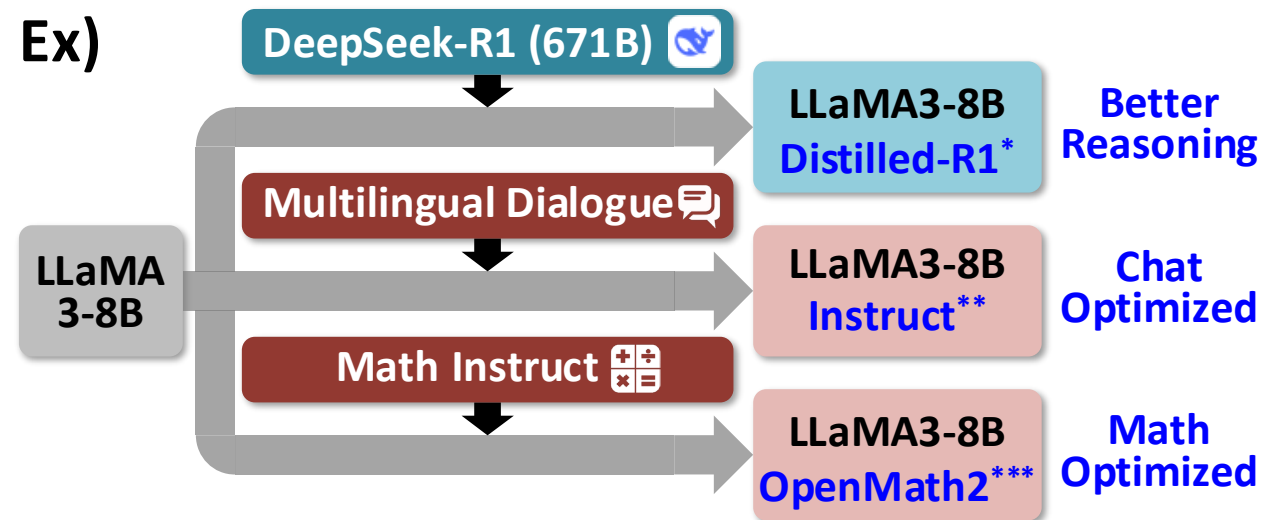
□ Trend 1) Distillation & Tuning: Improving Quality of Small Model

- a) **Knowledge Distillation:** Improving by using large model as a teacher
- b) **Fine-Tuning:** Specialization with domain- or task-specific dataset

Trend 1) Distilled & Tuned Model



Guide from Large Model/Dataset



* *deepseek-ai/DeepSeek-R1-Distill-Llama-8B*

** *meta/Meta-Llama-3-8B-Instruct*

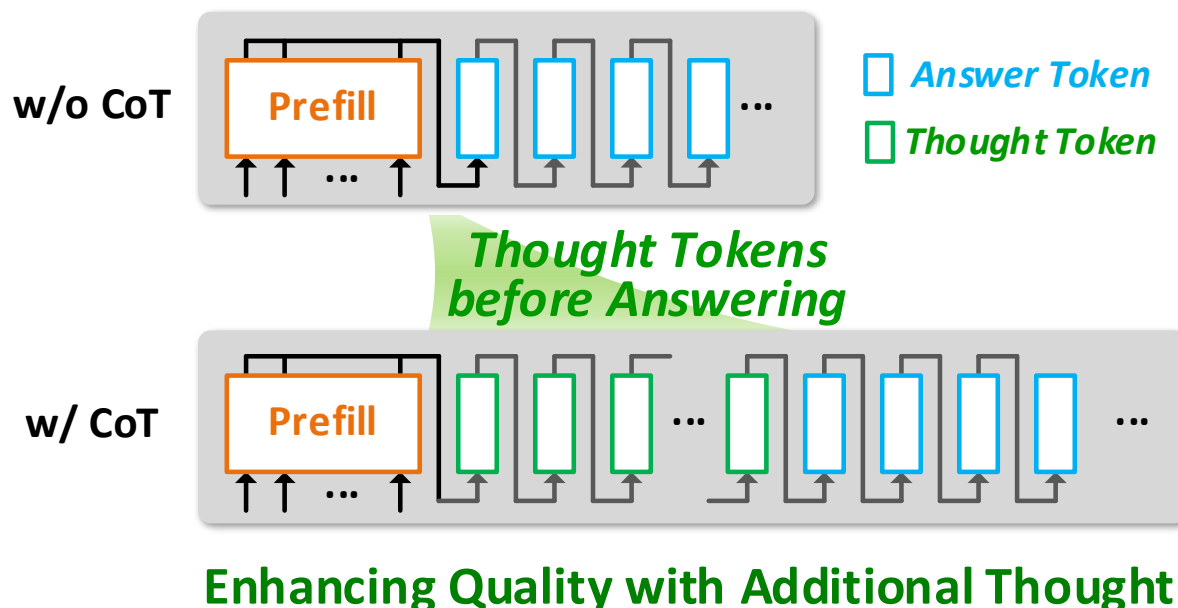
*** *nvidia/OpenMath2-Llama3.1-8B*

Trend) Overcoming Capacity for Edge

□ Trend 2) Chain-of-Thought: Thought Tokens for Better Accuracy

- Additional thought tokens before the final answer
- Enable model to reach the correct answer through step-by-step reasoning

Trend 2) Chain-of-Thought (CoT)



Ex)

David has three sisters. Each of them has one brother. How many brothers does David have?

w/o CoT

David has 3 brothers. **X**

w/ CoT

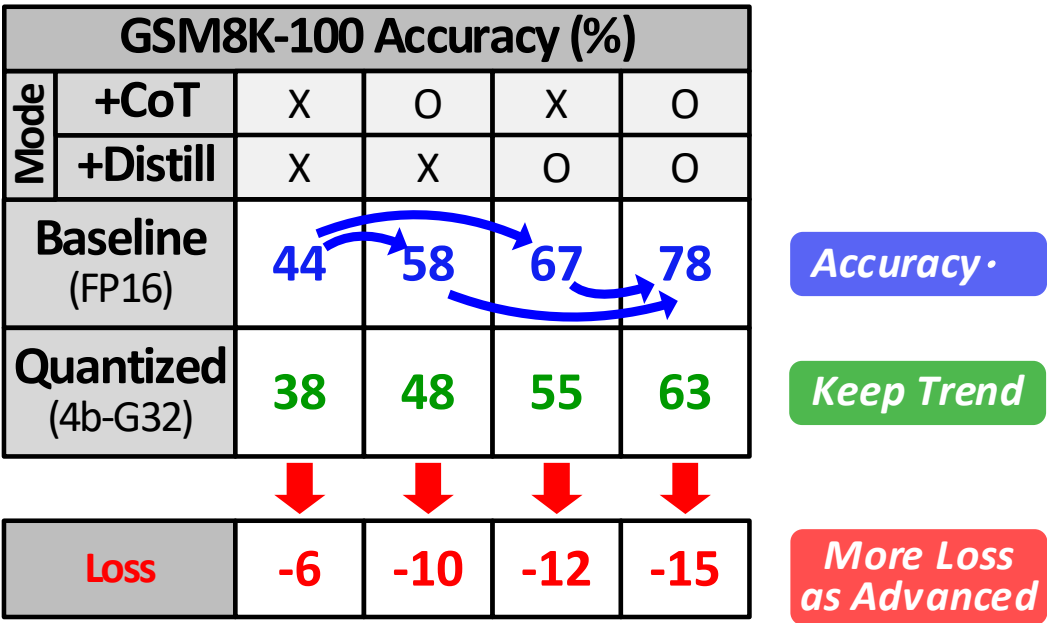
1. David is a male.
2. Three sisters all share the same brother, David.
3. There are no other brothers mentioned.
The answer is 0. **✓**

Challenge) Complicated Precision Demand

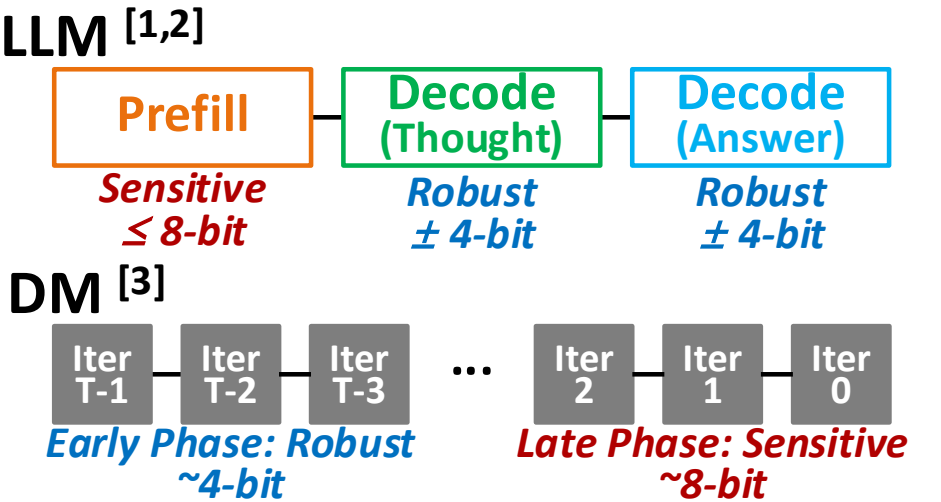
❑ Complex Precision Requirements

- **Distillation & CoT**: Improving performance but more quantization loss
- **Phase-aware Demand**: Varying precision demand depending on phase ^[1-3]
Especially when supporting both LLM and DM

C1) More Accurate but **More Drop**



C2) **Phase-aware** Varying Demand



[1] H. M. Chen, "Progressive mixed-precision decoding ~~~"
[2] J. Zhao, "LLM ~~~ Phase-aware Partition and Adaptive Quant."
[3] Y. Kim, "Leveraging Early-stage Robustness in DM ~~~"

GenAI Accelerator for Distilled Model and CoT

□ Unified LLM and DM Optimizations for Edge Environment

- 1) **Phase-aware Quantization**: different precision for different phase
- 2) **Rotation-based Quantization**: flattening data distribution for quantizability
- 3) **Asymmetric Group Quantization**: group-wise scale factor & zero-point

Revolver: Unified LLM/DM Accelerator for *Distilled Model* and *CoT*



1) **Phase-aware
Optimization**



2) **Rotation-based
Quantization**



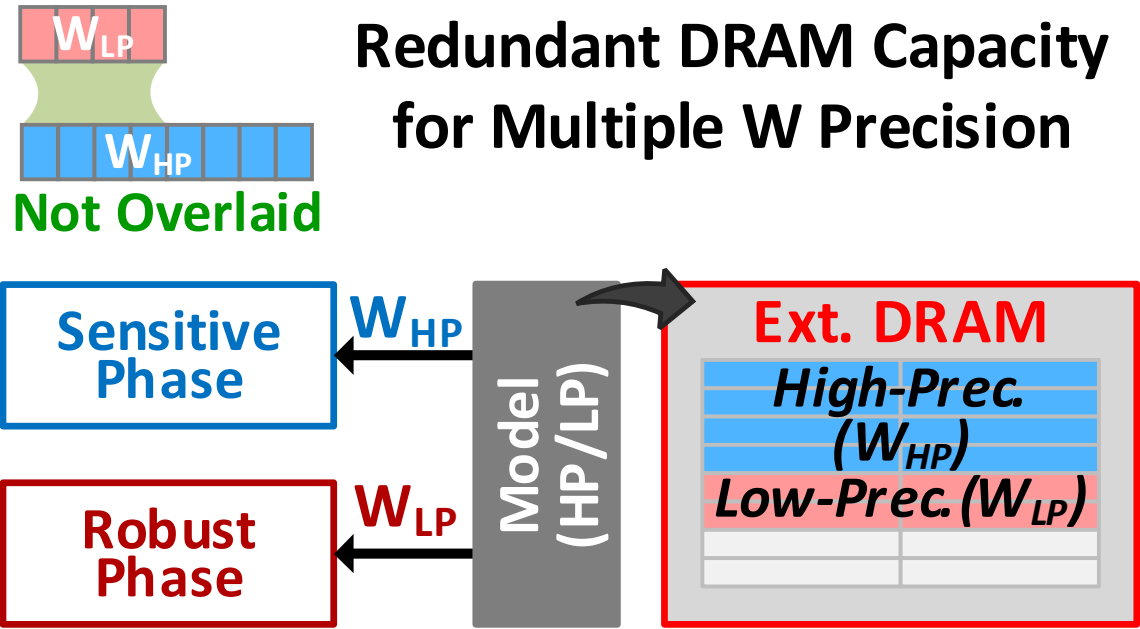
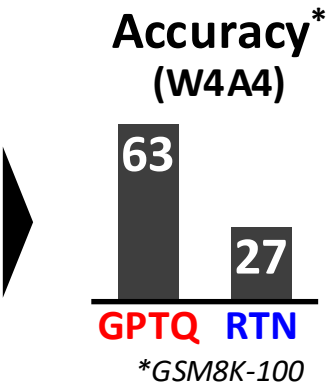
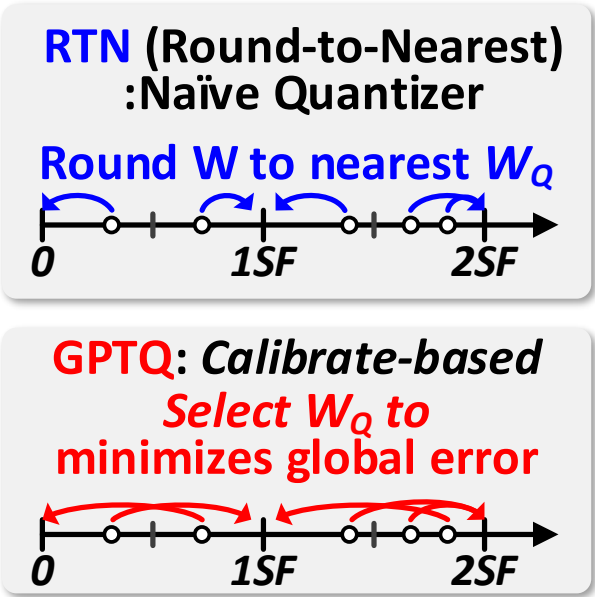
3) **Asymmetric
Group-Quantization**

Design Challenge 1: Multiple Weight Copy

❑ DRAM Capacity Overhead by Multi-Precision Weight

- Calibration-based quantizers* are essential for performance
- Independently optimize low-/high-bit → weight copies are not directly overlaid

Calibration-based Quant. (GPTQ)



Calibration → Essential for Low-bit Quant.

* E Frantar, "GPTQ", 2022

Design Challenge 2: Rotation Unit

- ❑ **Non-PoT*-Sized Channel of Large Model due to Capacity Budget**
 - Rotation unit design: **Mat.-Mult.(MM)** vs **Fast-Hadamard Transform (FHT)**
 - Practical implementation: **FHT+MM** with Kronecker Decomposition

Rotation Implementation

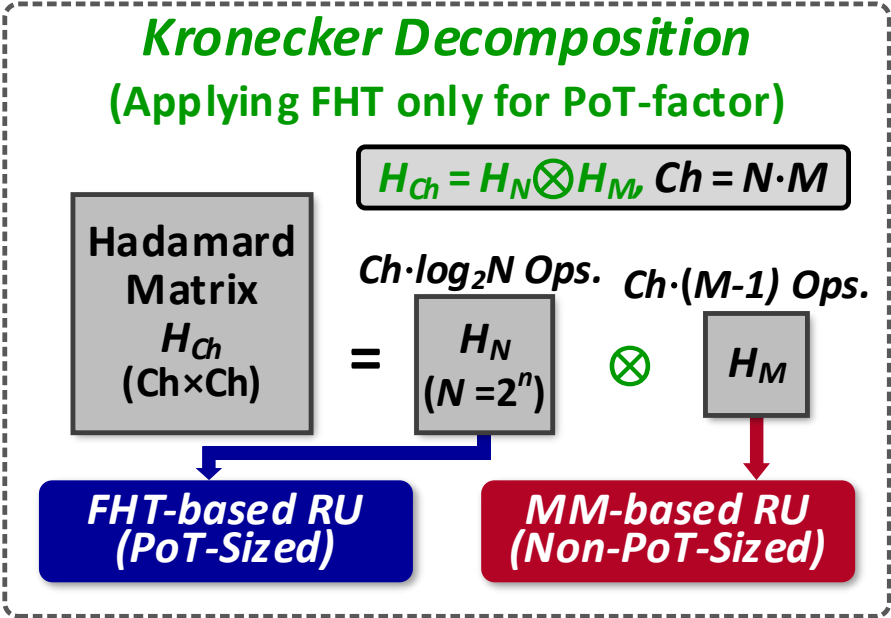
	MM-based	FHT-based
# of FP-ADD	$O(N^2)$ ☹️	$O(N \cdot \log_2 N)$ 😊
Condition of N	Any Sized N 😊	$N = 2^n$ (Int n) ☹️

Impractical

Practical Ch Size in GenAI Model

LLaMa-2			LLaMa-3		PixArt
7B	7B	70B	8B	70B	Σ
11008	13824	29672	14336	29672	4608

Non-PoT Ch Size

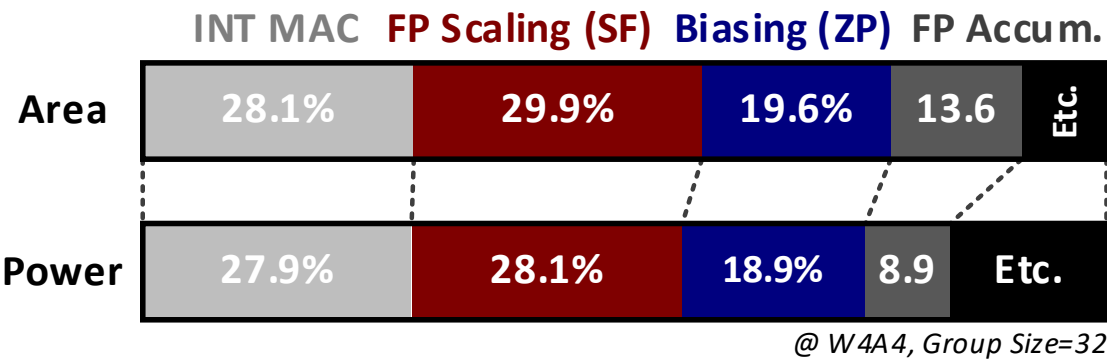


➔ Both FHT + MM Unit Required

* PoT: Power-of-Two

Design Challenge 3: Dequantization Logic

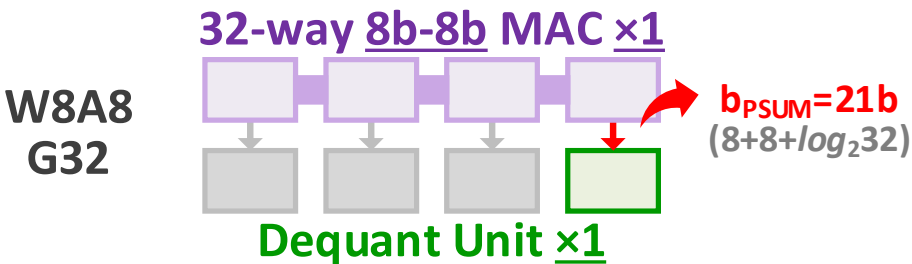
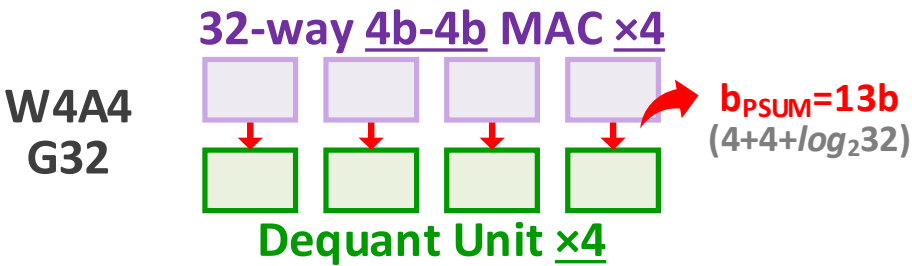
- ❑ Large Dequantization Overhead w/ Asymmetric Group Quant.
- Per-group FP logic to handle scale factor (SF) and zero point (ZP)
 - Bit-scalable arch.: operation ratio & partial-sum (PSUM) bit-width mis-match



Dequantization Cost

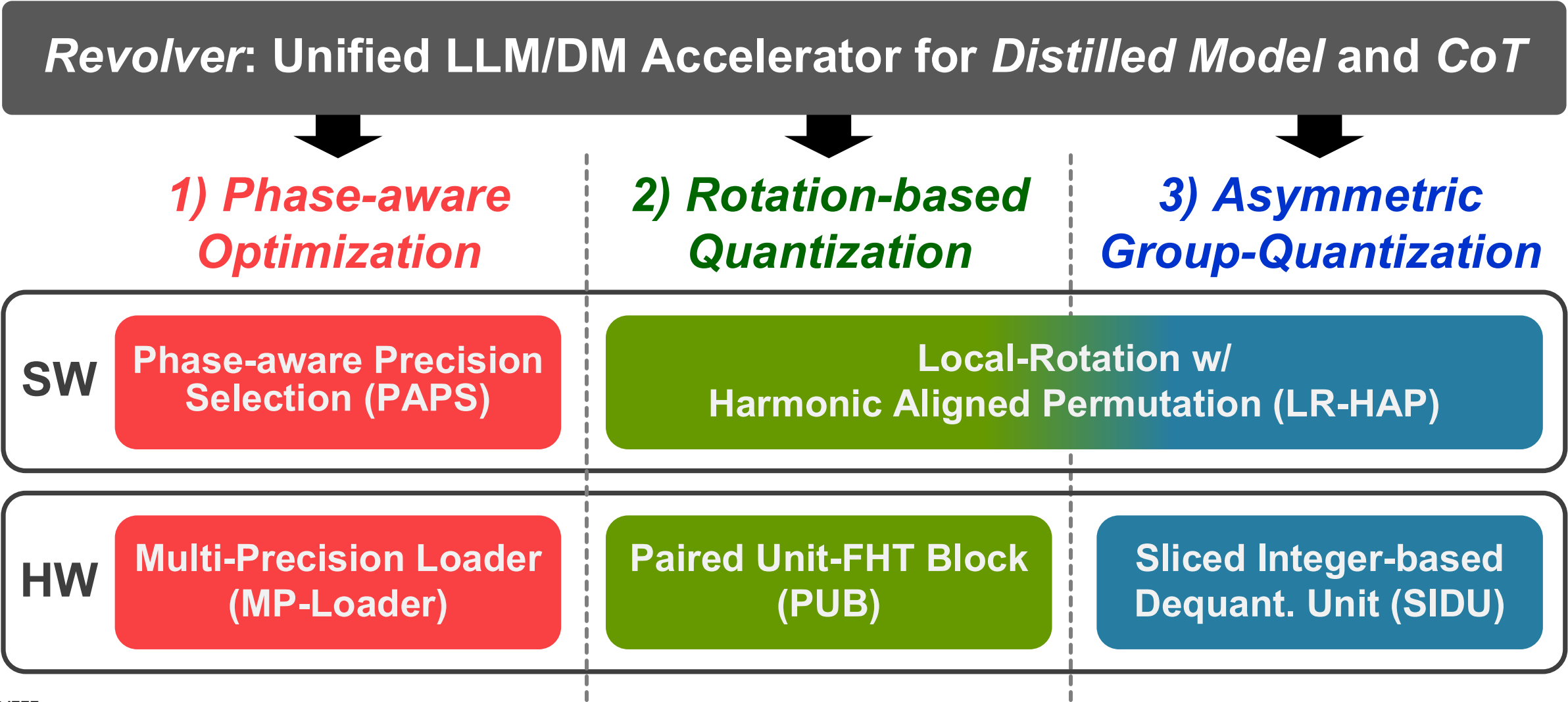
- FP Multiplying Logic for Scaling Factor (SF)
- Biasing Logic for Zero Point (ZP)
- Inter-group FP Accumulation

Dequant Unit for Bit-scalable MAC



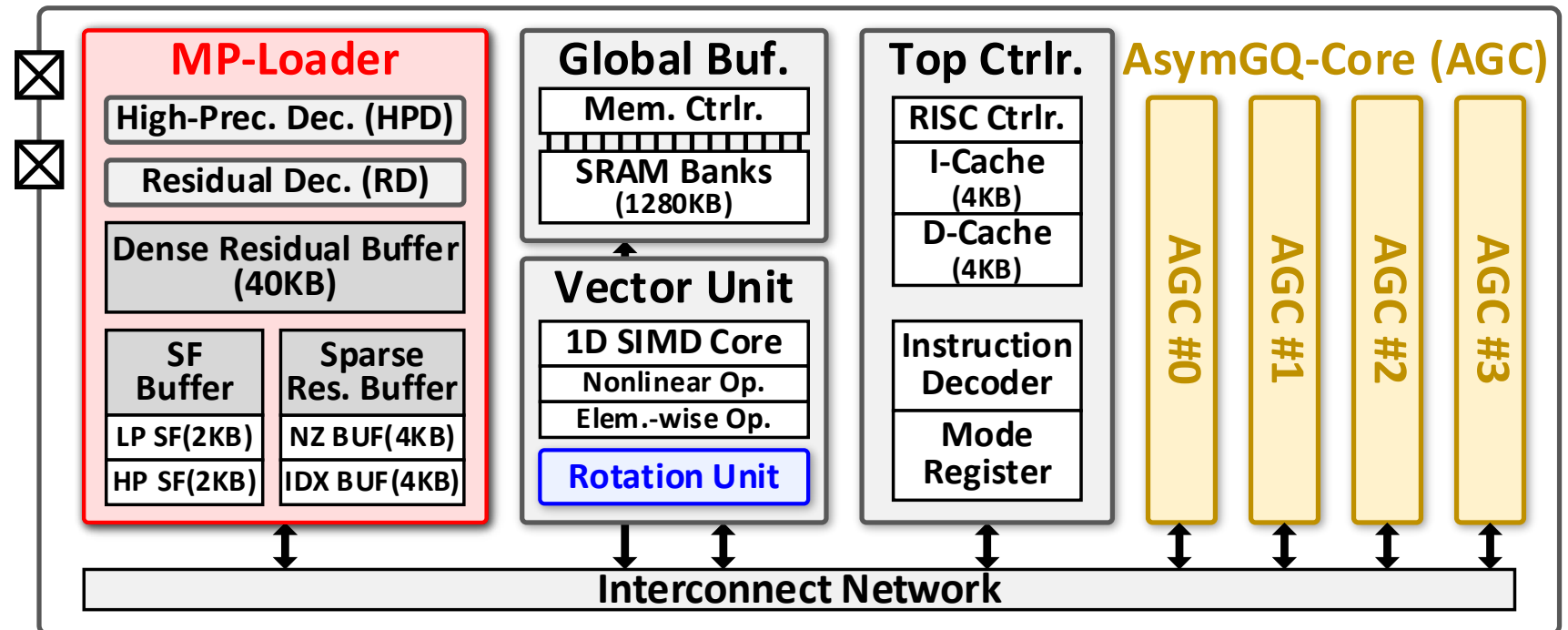
➔ Utilization Drop & Varying b_{PSUM}

Proposed Processor: *Revolver*



Overall Architecture of *Revolver*

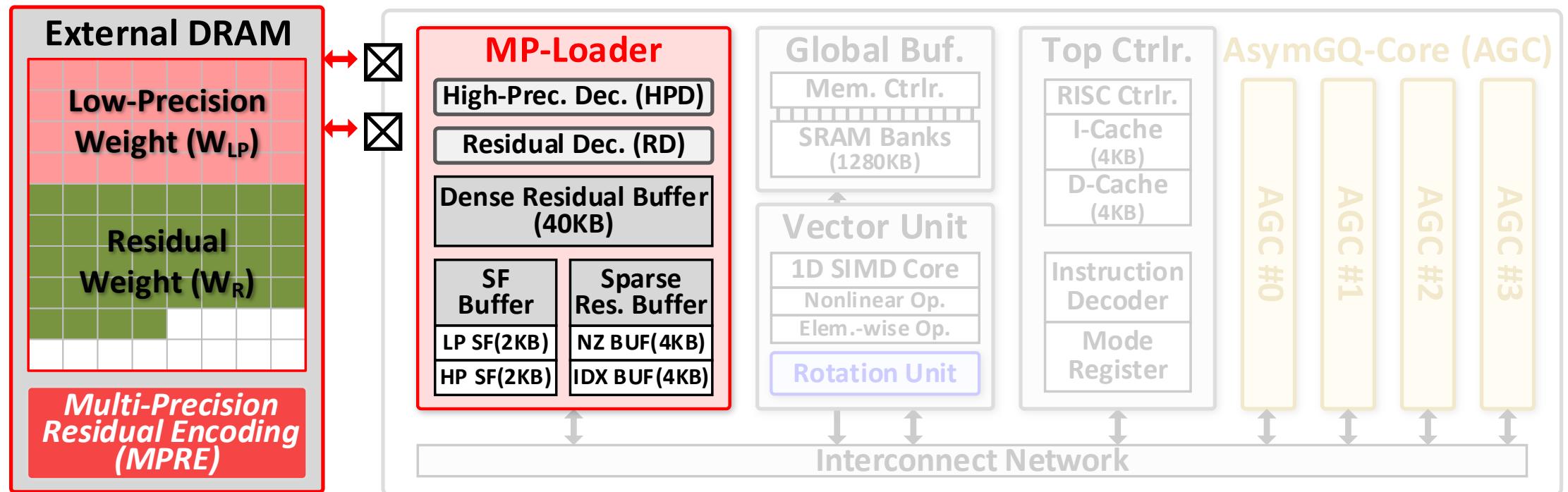
1. Multi-Precision Loader (MP-Loader)
2. Rotation Unit (RU)
3. 4 × AsymGQ-Core (AGC)
4. 1280KB Global Buffer
5. Top RISC Controller
6. Vector Unit



Overall Architecture of *Revolver*

1. Multi-Precision Loader (MP-Loader)

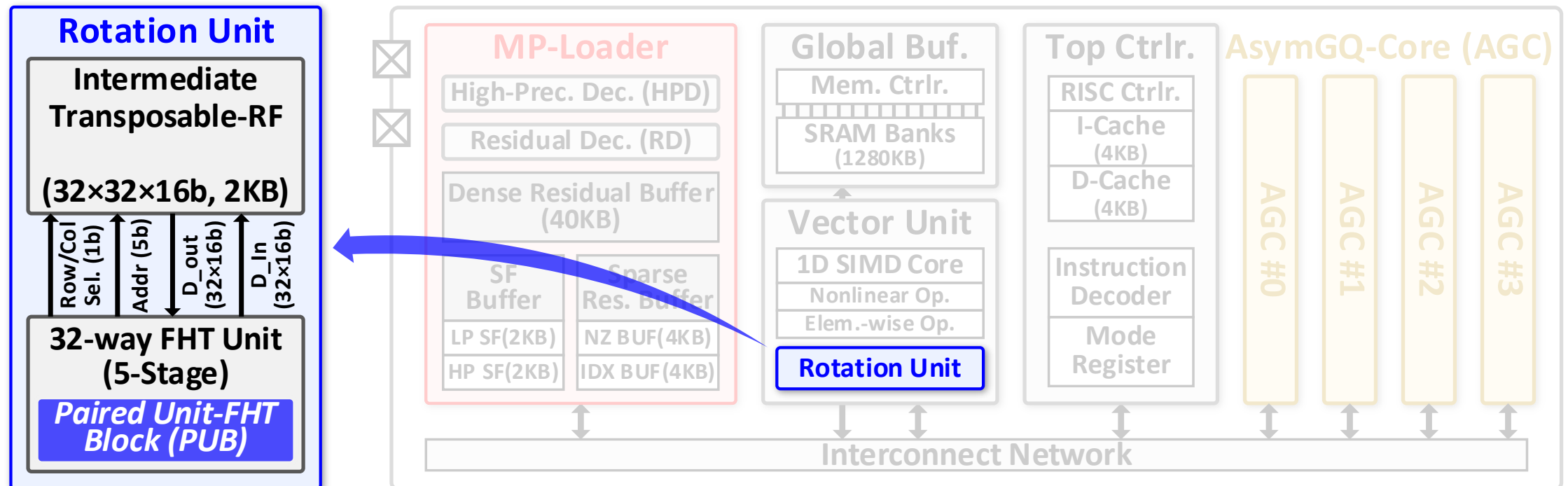
- Supports selective fetching of multi-precision weight (W_{LP} & W_{HP})
- Dual copy stored using **Multi-Precision Residual Encoding (MPRE)**



Overall Architecture of *Revolver*

2. Rotation Unit (RU)

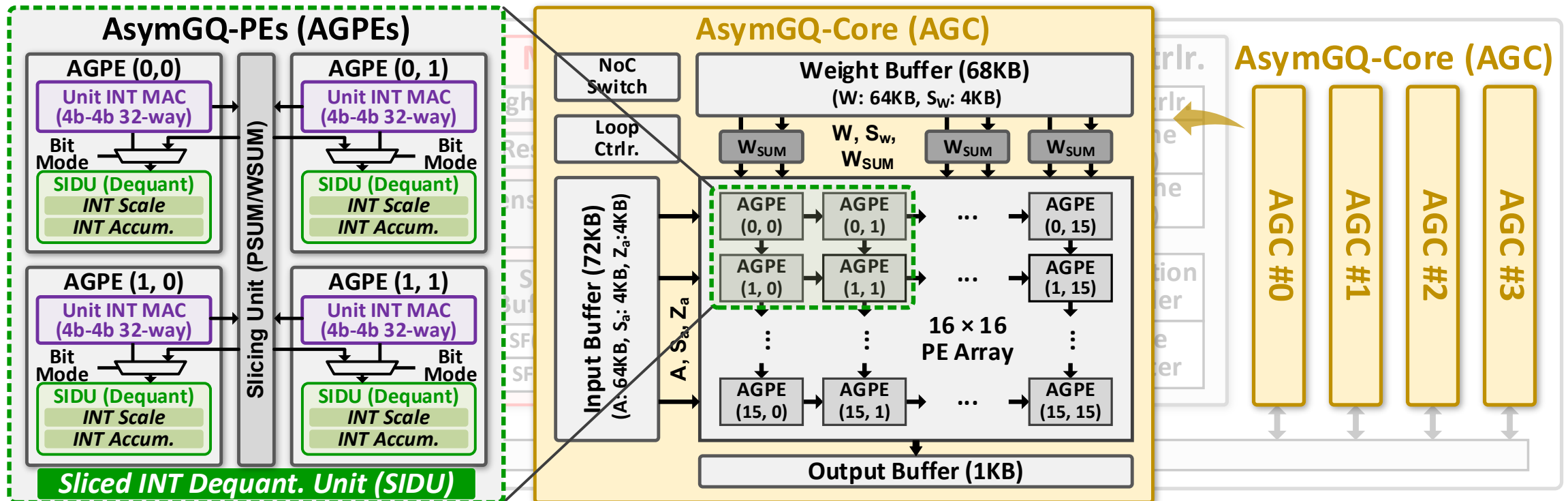
- 32-way FHT unit implemented with **Paired Unit-FHT Block (PUB)**
- 2KB **Transposable RF** for 2-stage operation: supports ~1024-way FHT



Overall Architecture of *Revolver*

3. AsymGQ-Core (AGC)

- 16×16 Asymmetric GQ-PE (AGPE) / Single row of weight sum unit (W_{SUM})
- **PE**: 32-way 4b-4b MAC + **Sliced INT Dequantization Unit (SIDU)**



Proposed Processor: *Revolver*

Revolver: Unified LLM/DM Accelerator for *Distilled Model* and *CoT*

**1) Phase-aware
Optimization**

2) Rotation-based
Quantization

3) Asymmetric
Group-Quantization

SW

**Phase-aware Precision
Selection (PAPS)**

Local-Rotation w/
Harmonic Aligned Permutation (LR-HAP)

HW

**Multi-Precision Loader
(MP-Loader)**

Paired Unit-FHT Block
(PUB)

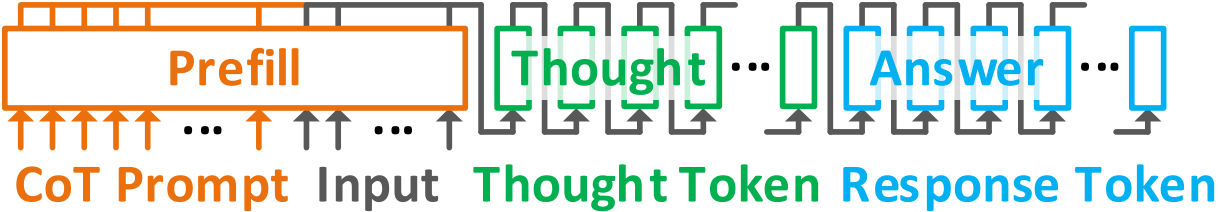
Sliced Integer-based
Dequant. Unit (SIDU)

Phase-aware Precision Selection (PAPS)

□ Different Precision according to Characteristics of LLM & DM

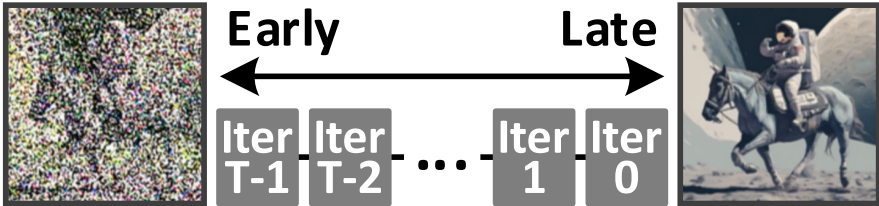
- LLM: High-precision W for *prefilling* & Low-precision W for *decoding*
 - DM: Low-precision W & IA at *early* & High-precision W & IA at *late iteration*
- ➔ Reducing average precision for computation & EMA ☺

LLM



Phase	Prefill	Decode-Thought	Decode-Answer
Prec.	W: HP IA: HP	W: LP IA: HP	
Task	Understanding	Reasoning	Response Gen.
Acc.	Sensitive	Robust	
Focus	Computation	Weight EMA	

DM



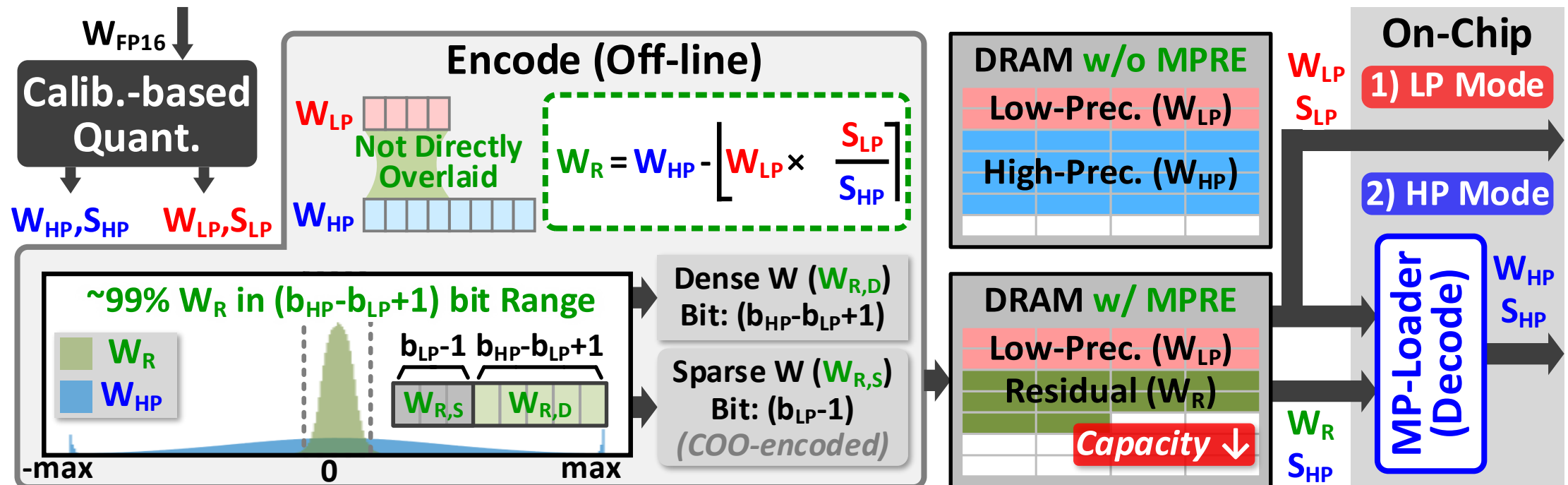
Iter.	Early	Late
Prec.	W: LP IA: LP	W: HP IA: HP
Task	Naïve Outlining	Detailing
Acc.	Robust	Sensitive
Focus	Computation	Computation

Multi-Precision Residual Enc. (MPRE)

❑ Hardware-level Support of Precision Adaptivity 😊

⇔ (Baseline 1) Separate W_{HP} & W_{LP} : External DRAM capacity overhead ☹️

⇔ (Baseline 2) Overlaying W_{HP} & W_{LP} : Error-prone or requiring re-training* ☹️

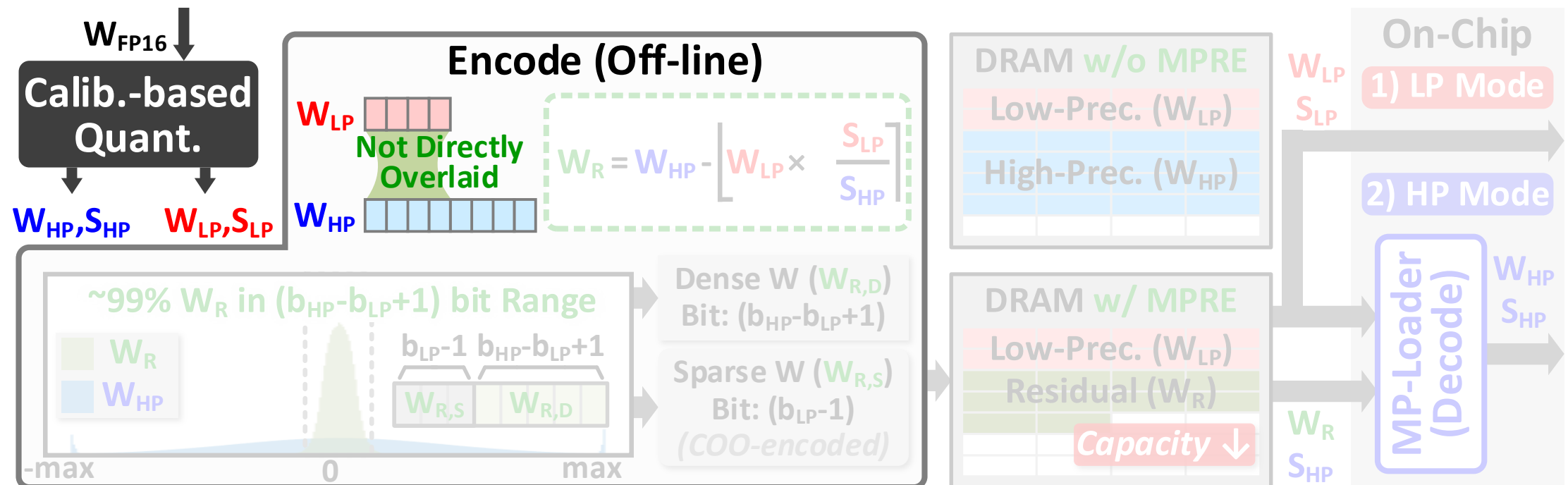


* P. Nair, "Matryoshka Quantization", 2025

Multi-Precision Residual Enc. (MPRE)

□ Hardware-level Support of Precision Adaptivity

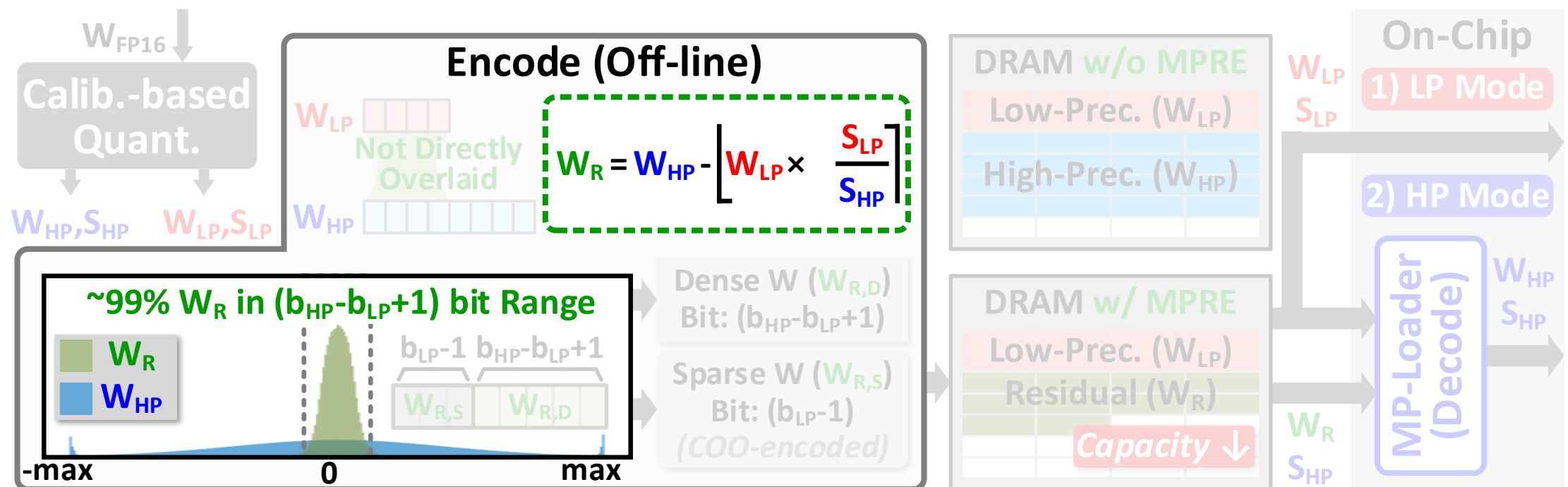
- Directly using calibrated weight (W_{HP} & W_{LP}) and scale factor (S_{HP} & S_{LP})
- Basically, W_{HP} & W_{LP} are not directly overlaid



Multi-Precision Residual Enc. (MPRE)

□ Hardware-level Support of Precision Adaptivity

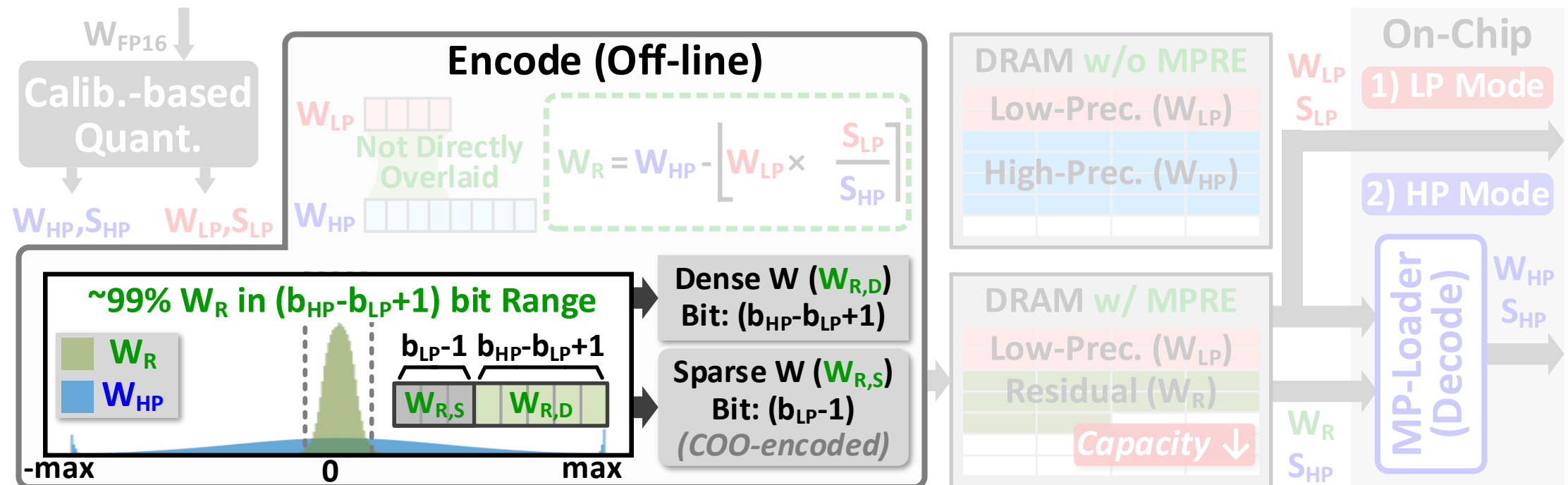
- BUT, overall trends of W_{HP} & W_{LP} remain consistent
- W_R (scaled difference of W_{HP} & W_{LP}) $\rightarrow$ Narrow W_R distribution than W_{HP}



Multi-Precision Residual Enc. (MPRE)

□ Hardware-level Support of Precision Adaptivity

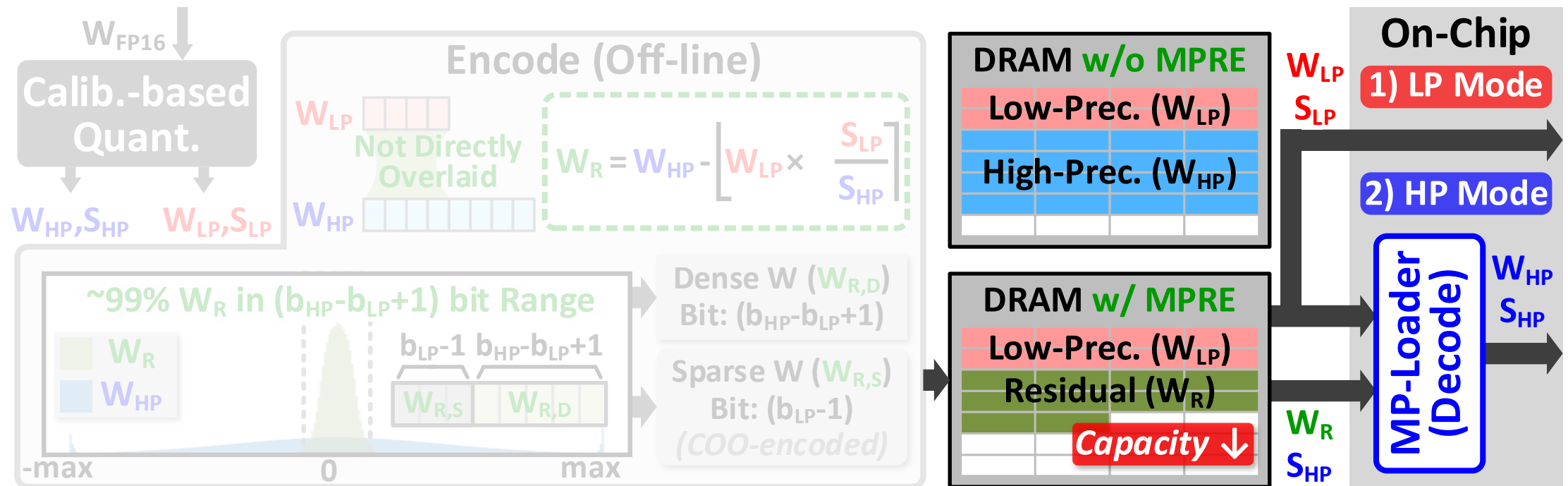
- Upper $(b_{LP} - 1)$ -bits of $\mathbf{W}_R \rightarrow \sim 99\%$ Sparse & COO-encoded weight ($\mathbf{W}_{R,S}$)
- Remaining Lower $(b_{LP} - b_{LP} + 1)$ -bits of $\mathbf{W}_R \rightarrow$ Dense weight ($\mathbf{W}_{R,D}$)



Multi-Precision Residual Enc. (MPRE)

□ Hardware-level Support of Precision Adaptivity

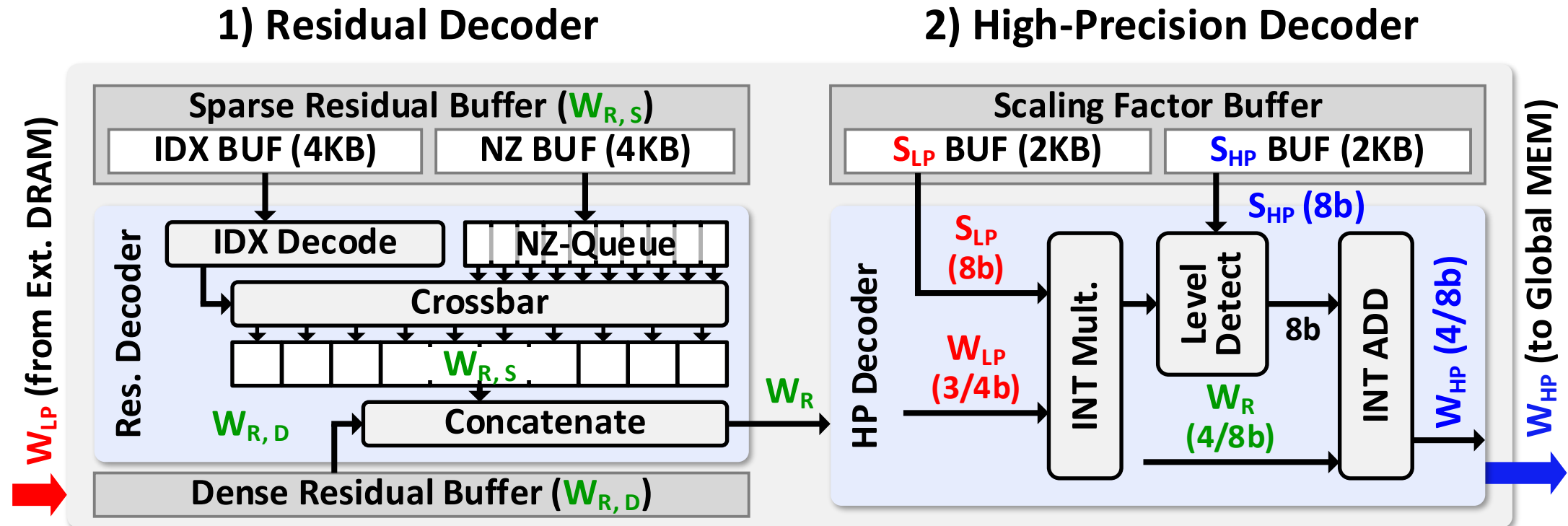
- MP-Loader supports LP mode (fetching W_{LP}) / HP mode (fetching $W_{LP} + W_R$)
- DRAM capacity reduction 😊 compared to the separate W_{HP} & W_{LP}



Implementation of MP-Loader

□ Streaming Reconstruction with 2-stage Decoder

- 1) **Residual Dec.:** Sparse COO decoding logic for ($\mathbf{W}_{R,S} + \mathbf{W}_{R,D} \rightarrow \mathbf{W}_R$)
- 2) **High-Prec. Dec.:** Scaling & addition logic for ($\mathbf{W}_R + \mathbf{W}_{LP} \rightarrow \mathbf{W}_{HP}$)



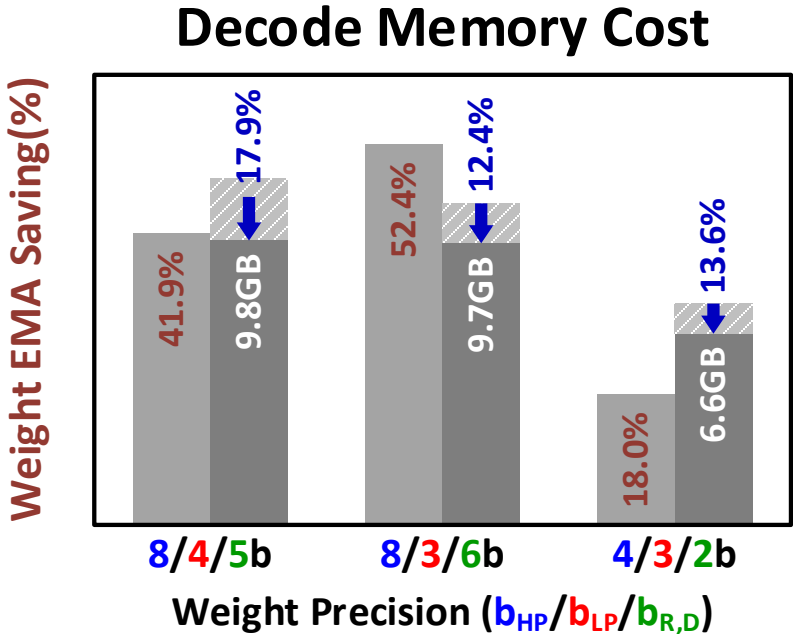
Performance Improvement w/ PAPS & MP-Loader

- ❑ 18.0% ~ 41.9% Weight EMA ↓
- ❑ 12.4% ~ 17.9% DRAM Capacity for Weight ↓
- ❑ 50.4% System Energy (Chip + DRAM Dynamic + Static) ↓

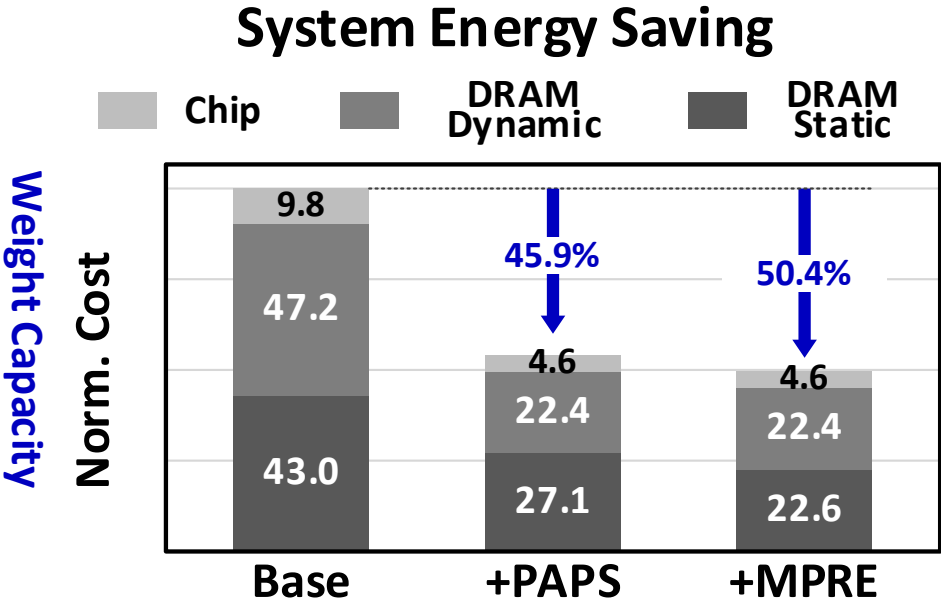
Performance of PAPS*				
	bit _W	bit _{IA}	Speed Up	Acc.**
w/o PAPS	8	8	×1.00	77.3
	4	4	×1.72	73.0
w/ PAPS	8/4	8/4	×1.72	77.0
	8/3	8/4	×2.10	75.7

* LLaMA3-8B-R1(Distilled)[8]+CoT[12]

**Accuracy(%) on GSM8K-300 [1]



* LLaMA3-8B, Output Length= 256



* LLaMA3-8B, In/Out Length=256/256

** W: HP/LP = 8/3b, IA: HP/LP = 8/4b

Proposed Processor: *Revolver*

Revolver: Unified LLM/DM Accelerator for Distilled Model and CoT

*1) Phase-aware
Optimization*

*2) Rotation-based
Quantization*

*3) Asymmetric
Group-Quantization*

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Phase-aware Precision
Selection (PAPS)

Local-Rotation w/
Harmonic Aligned Permutation (LR-HAP)

HW

Multi-Precision Loader
(MP-Loader)

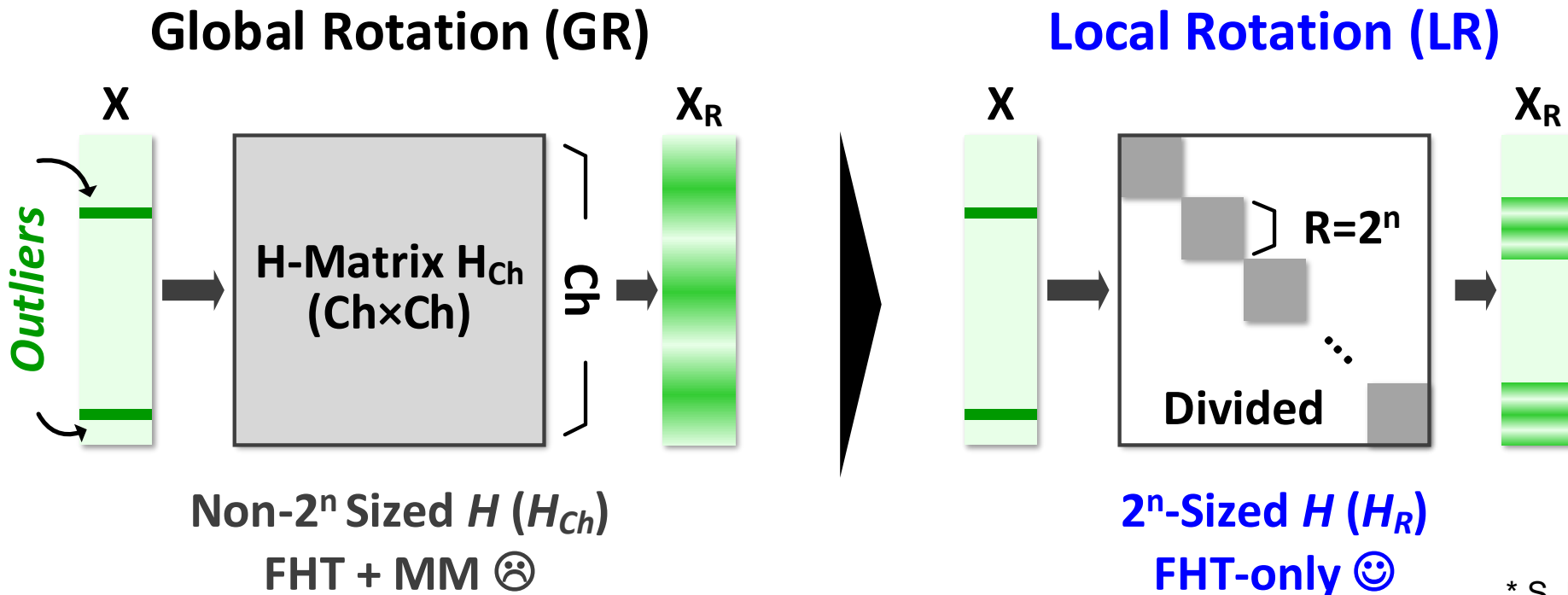
Paired Unit-FHT Block
(PUB)

Sliced Integer-based
Dequant. Unit (SIDU)

Rotation Optimization 1) Local Rotation (LR)

□ Local Rotation (LR) to Avoid Arbitrary-sized Rotation

- Dividing channels with multiple PoT (2^n)-sized groups
- No accuracy degradation by LR under the group quantization*
- **FHT + MM ☹️ → FHT-Only rotation unit without MM overhead 😊**

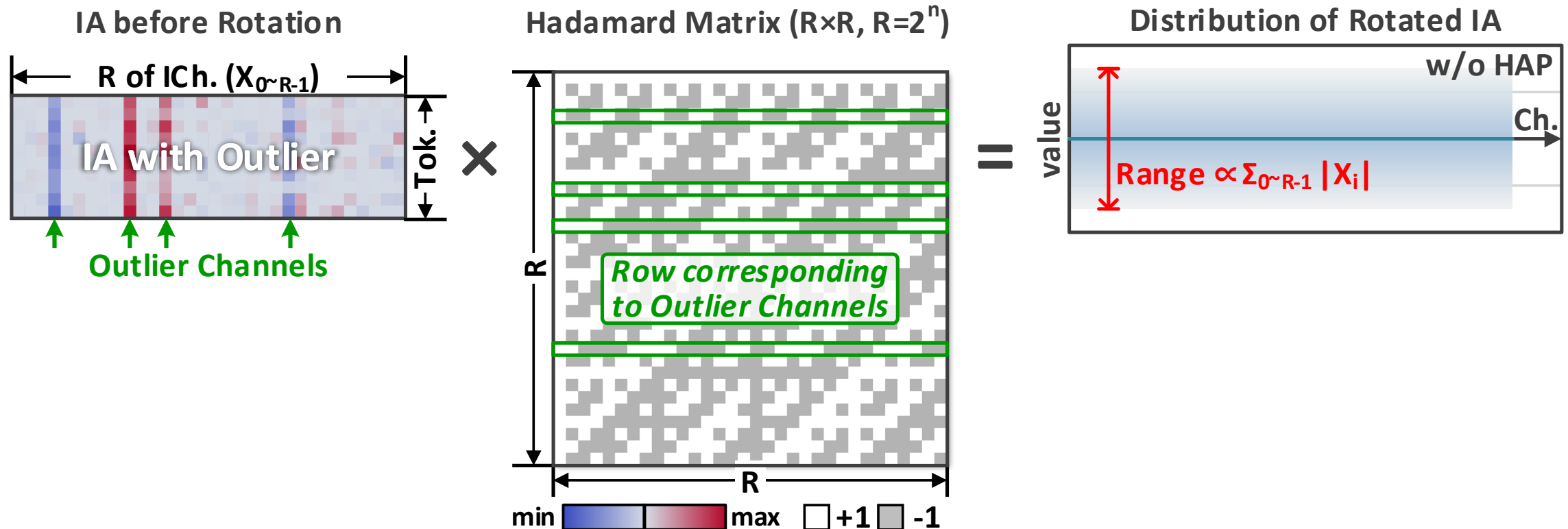


* S. Kim, "LightRot", 2025

Optimization 2) Harmonic Aligned Permutation (HAP)

□ (Baseline) Uniformly Distributing Outlier across All Channels

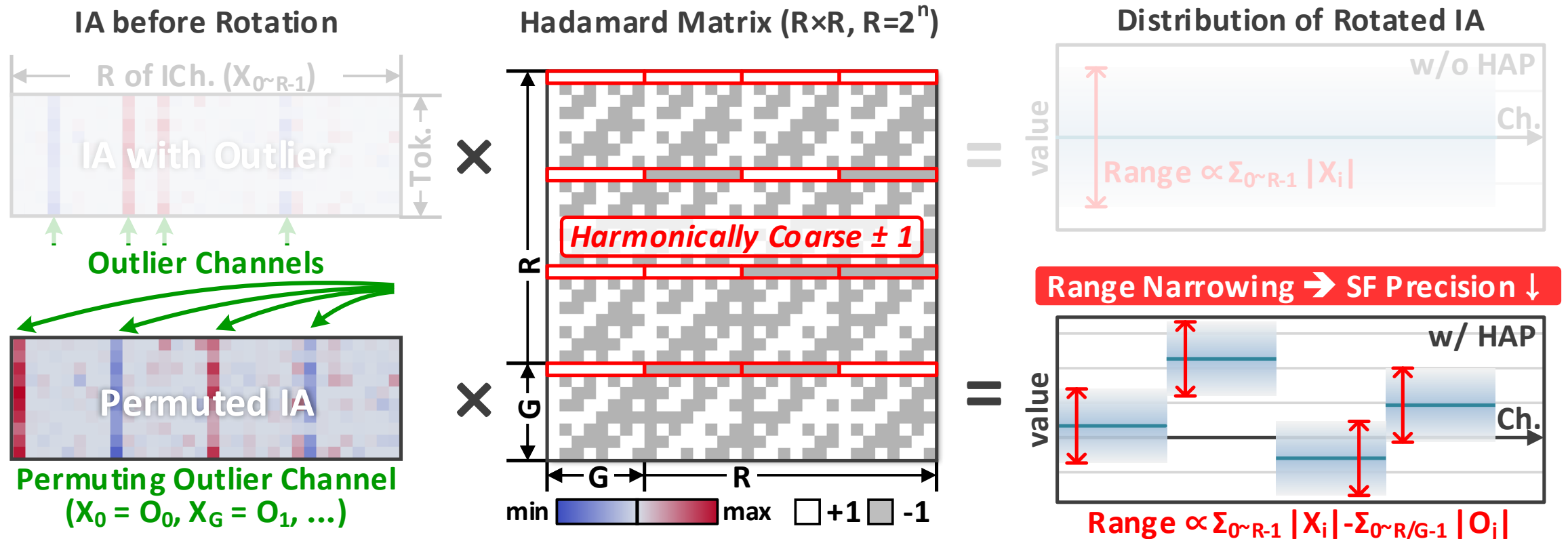
- Outlier channels multiplied with *fine-grained mixed sign (+1/-1) pattern*
- Rotated IA range affected by all channel (*Defined by abs-sum of all channel*)



Optimization 2) Harmonic Aligned Permutation (HAP)

□(HAP) Reshaping Rotated IA Distribution by Permutation

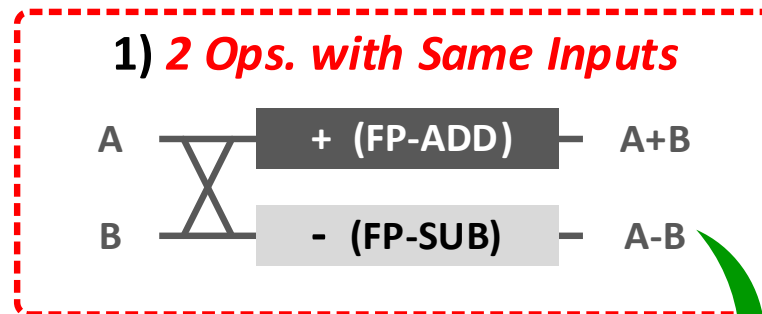
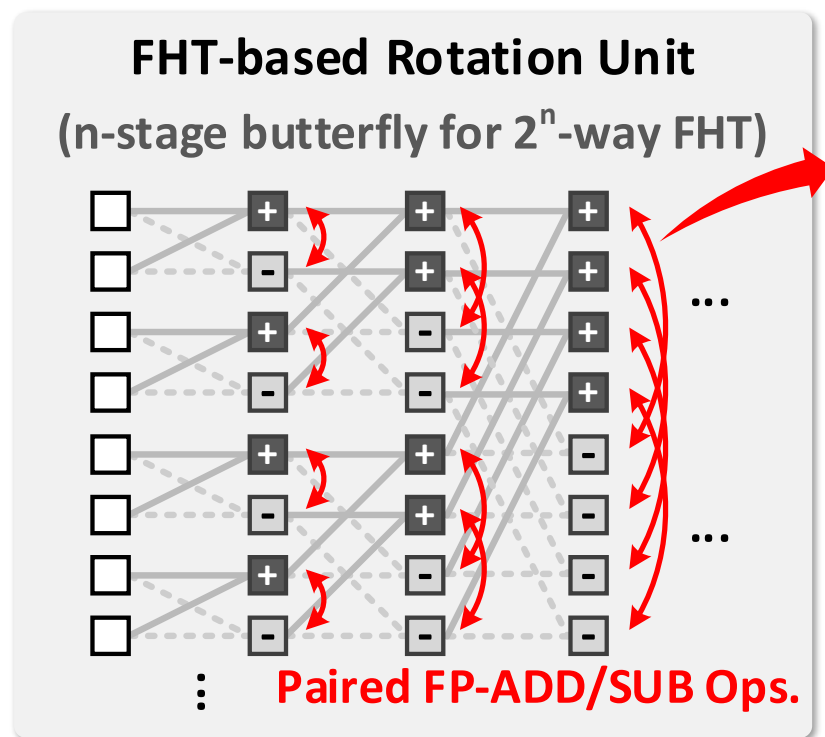
- Aligning outliers to harmonic row $\rightarrow$ coarse-grained sign pattern with size G
- Canceling out outliers to bias & narrowing range $\rightarrow$ **Lower SF precision 😊**



Optimization 3) Paired Unit-FHT Block (PUB)

□ Motivation: Paired FP-ADD/SUB Operation in FHT Logic

- 1) Always Compute $A+B$ and $A-B$ together for the same inputs A and B
- 2) Always 1 same signed and 1 different signed operation



2) *1 Same & 1 Diff. Signed Ops.*

A	B	A+B	A-B
+	+	<i>Same Sign</i>	<i>Diff. Sign</i>
-	-	<i>Same Sign</i>	<i>Diff. Sign</i>
+	-	<i>Diff. Sign</i>	<i>Same Sign</i>
-	+	<i>Diff. Sign</i>	<i>Same Sign</i>

Optimization 3) Paired Unit-FHT Block (PUB)

- ❑ Motivation: Paired FP-ADD and SUB Operation in FHT Logic
 - Different-signed Operation: full norm. (*leading-1 detector, variable shifter*)
 - Same-signed Operation: simplified norm. (*overflow checker, 1-b shifter*)

FHT-based Rotation Unit
(n-stage butterfly for 2^n -way FHT)

Paired FP-ADD/SUB Ops.

1) 2 Ops. with Same Inputs

A + (FP-ADD) = A+B

B - (FP-SUB) = A-B

2) 1 Same & 1 Diff. Signed Ops.

A	B	A+B	A-B
+	+	Same Sign	Diff. Sign
-	-	Same Sign	Diff. Sign
+	-	Diff. Sign	Same Sign
-	+	Diff. Sign	Same Sign

Operation after Mantissa Alignment

Diff. Signed → Unsigned-SUB

Full Norm. Required

Same Signed → Unsigned-ADD

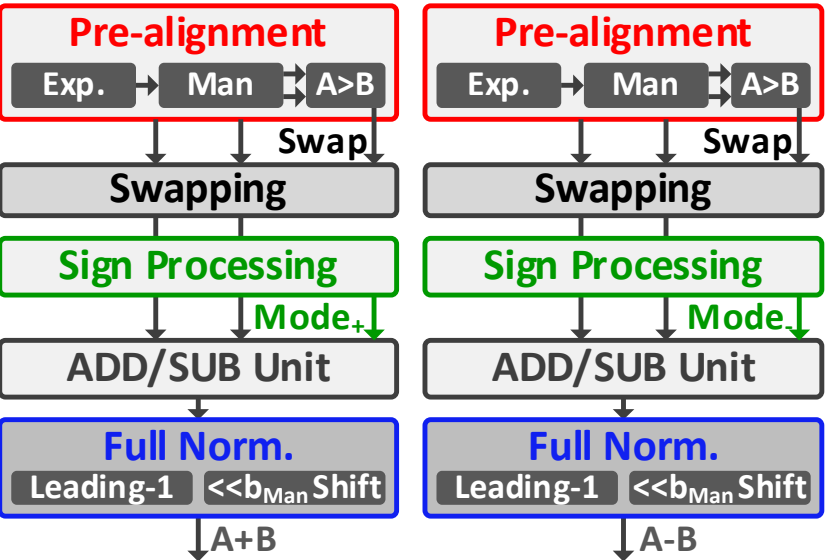
At Most 1-b Norm.

PUB Micro-architecture

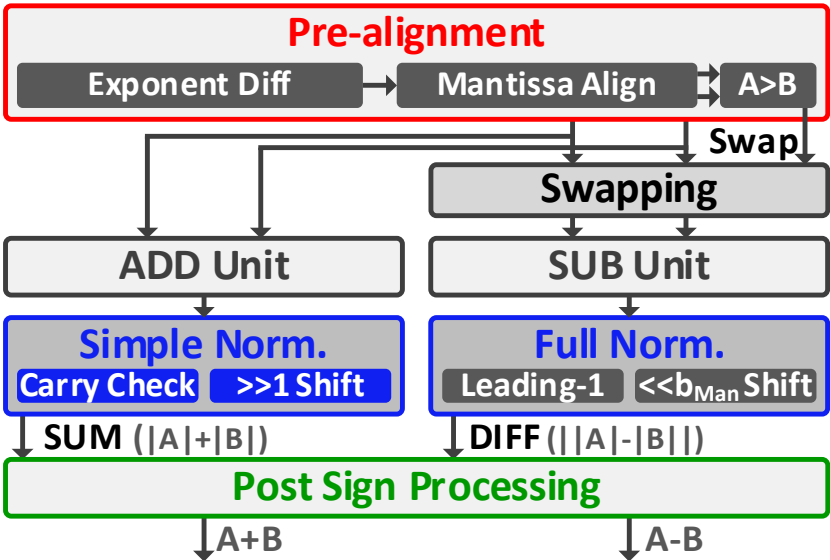
❑ Fusing ADD/SUB-Unit with Logic Sharing & Simplification

- Sharing **Pre-alignment Logic** & **Simplified Norm.** for same-signed path
- **Post Sign Processing** ensures correct signed output
- ➔ Logic cost reduction for FP-ADD/SUB of FHT-unit 😊

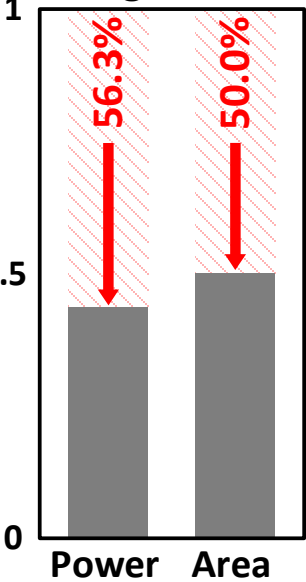
Baseline (Separate FP-ADD/FP-SUB Unit)



Proposed PUB



FP-ADD/SUB Logic Cost

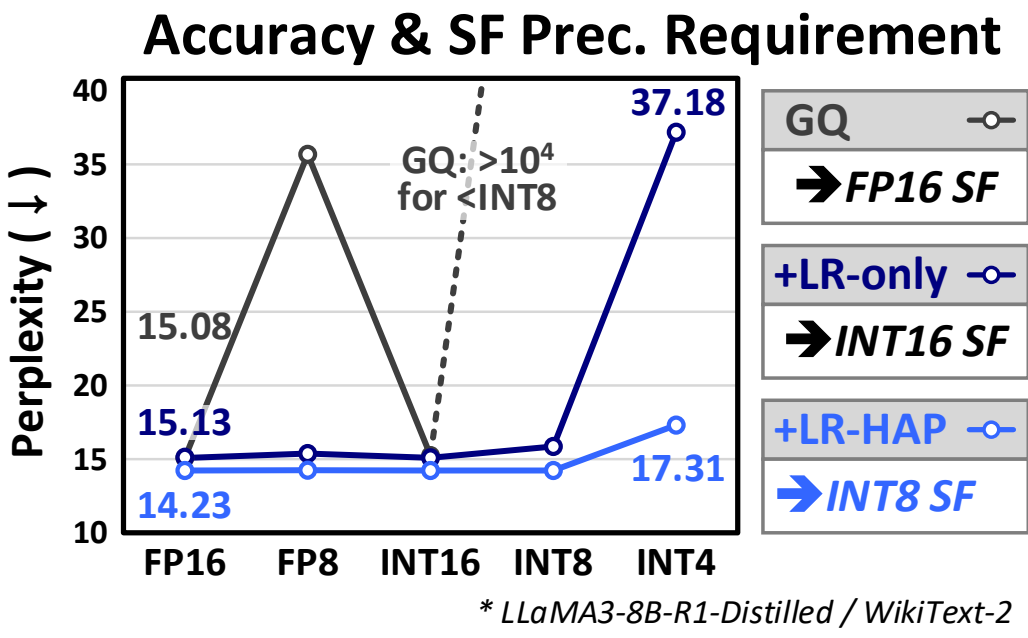
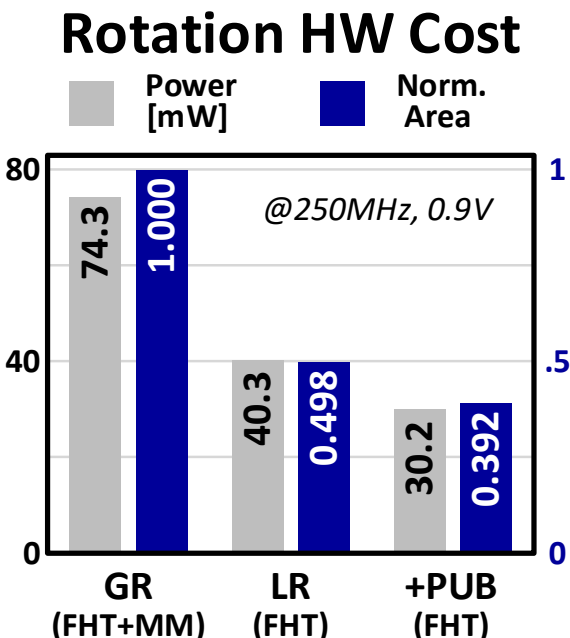
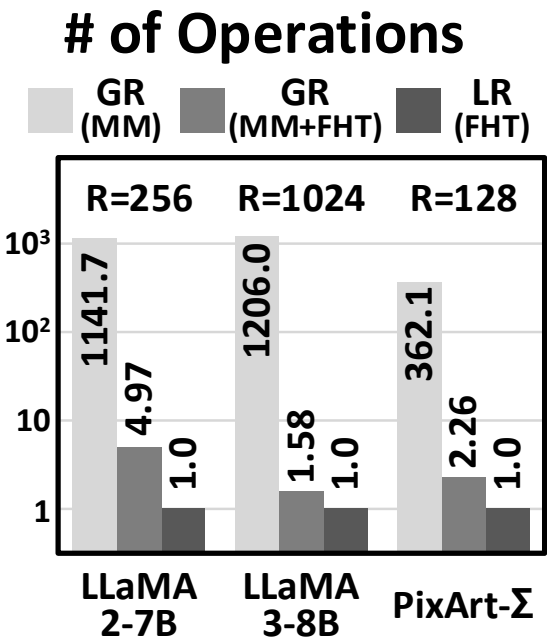


Performance of LR-HAP with PUB

Rotation Cost Saving

- LR: **2.26 ~ 4.97× operation** ↓ than *global rotation (GR)* with *FHT+MM*
- LR + PUB: **59.4% power** and **60.8% area** ↓

SF Precision: FP16 (GQ) → INT16 (LR) → INT8 (LR-HAP)



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*2) Rotation-based
Quantization*

**3) Asymmetric
Group-Quantization**

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Selection (PAPS)

Local-Rotation w/
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Multi-Precision Loader
(MP-Loader)

Paired Unit-FHT Block
(PUB)

Sliced Integer-based
Dequant. Unit (SIDU)

Quantization and Dequantization Operation

□ Group Quantized IA & W

- **IA**: Asymmetric GQ for better quantizability + synergy with rotation (LR-HAP)
- **W**: Symmetric GQ for lower dequantization overhead

Quantization

IA: Asymmetric Group Quantization

$$A_Q = \lfloor (A + Z_a) / S_a \rfloor$$

S_a : Scaling Factor
 Z_a : Zero Point

W: Symmetric Group Quantization

$$W_Q = \lfloor W / S_w \rfloor$$

S_w : Scaling Factor

Quantization and Dequantization Operation

□ Dequantization Operations for Group Quantized IA & W

- **PSUM**: Inner-product of W & A within a group → scaled with **SF** of W & IA
- **WSUM**: Sum of W within group → scaled with **ZP** of IA & bias for PSUM

Quantization

IA: Asymmetric Group Quantization

$$A_Q = \lfloor (A + Z_a) / S_a \rfloor$$

S_a : *Scaling Factor*
 Z_a : *Zero Point*

W: Symmetric Group Quantization

$$W_Q = \lfloor W / S_w \rfloor$$

S_w : *Scaling Factor*

Dequantization (with Computation)

Intra-group: Operations with Quantized W & IA

PSUM: Group-wise W-A Inner-product

$$PSUM_g = \sum_{i \in g} A_{Q,i} \cdot W_{Q,i}$$

WSUM: Group-wise W Sum

$$WSUM_g = \sum_{i \in g} W_{Q,i}$$

Inter-group: Accumulation with Scaling/Biasing

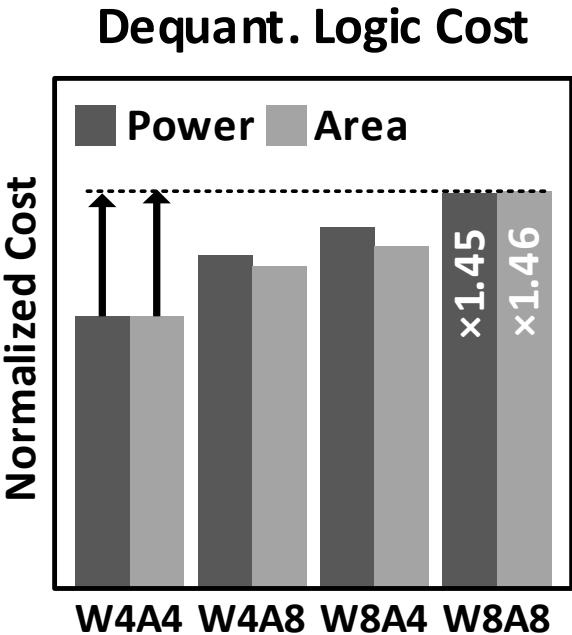
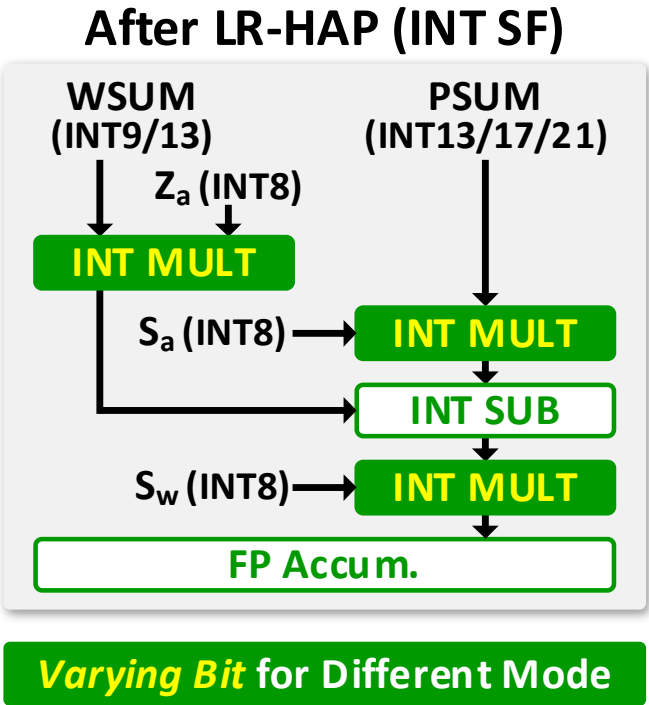
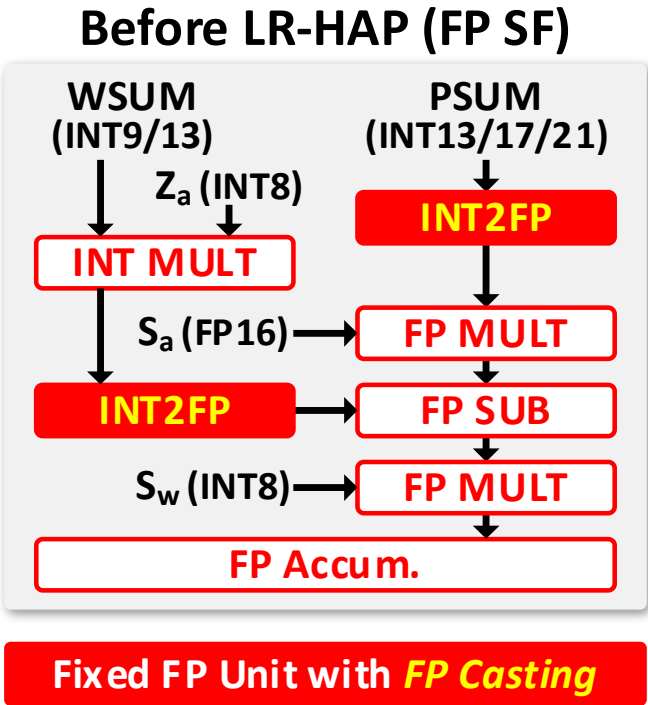
$$\sum_{g \in G} S_{w,g} \cdot (S_{a,g} \cdot PSUM_g - Z_{a,g} \cdot WSUM_g)$$

Sliced Integer-based Dequantization Unit (SIDU)

□ Varying b_{PSUM}/b_{WSUM} w/ Bit-width of W & A

- **FP SF** (w/o LR-HAP): FP dequant with INT-to-FP casting
 - **INT SF** (w/ LR-HAP): Varying bit-width handling required
- ➔ *Simply using high-bit logic (W8A8) resulting in large cost*

Dequantization Operation			
Mode	PSUM	WSUM	#Op
W4A4	13b	9b	×4
W4A8	17b	9b	×2
W8A4	17b	13b	×2
W8A8	21b	13b	×1

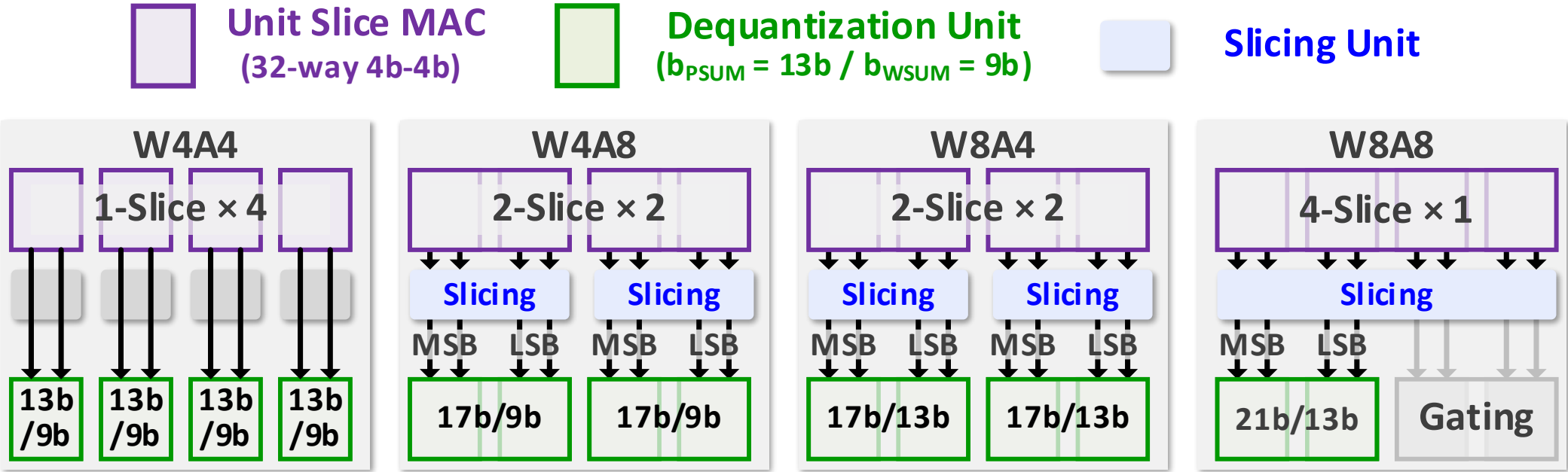


Sliced Integer-based Dequantization Unit (SIDU)

Multi-Precision Supporting with Bit-Slicing

- Low-precision mode: $b_{PSUM}/b_{WSUM} \downarrow$, # of operation $\uparrow$
 - High-precision mode: $b_{PSUM}/b_{WSUM} \uparrow$, # of operation $\downarrow$
- ➔ Integrate *low-bit logic* (W4A4) & combine for *high-bit op*.

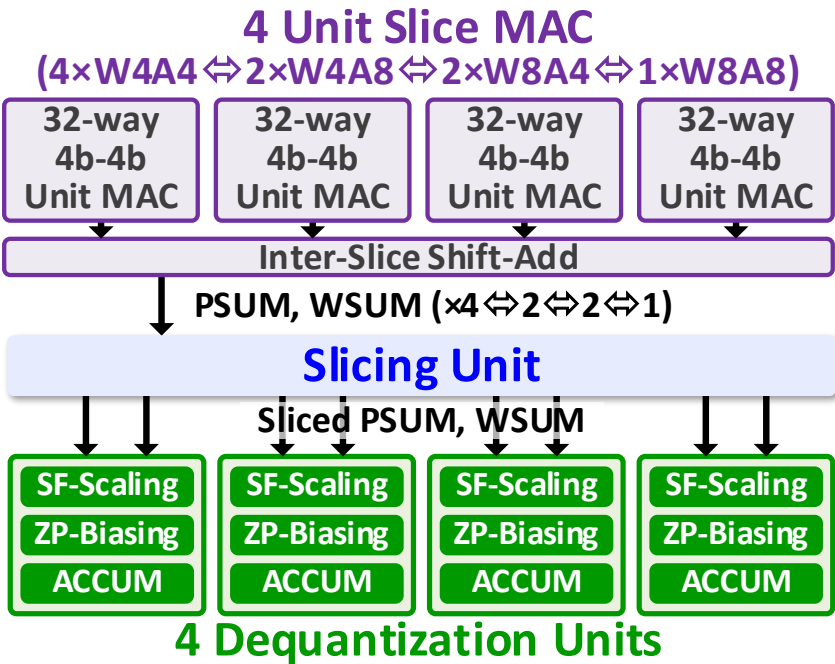
Dequantization Operation			
Mode	PSUM	WSUM	#Op
W4A4	13b	9b	×4
W4A8	17b	9b	×2
W8A4	17b	13b	×2
W8A8	21b	13b	×1



Micro-architecture of SIDU

- ❑ **2×2 PE Combined for Slice-based Operation**
 - **4 Unit Slice MAC** supports multi-precision operations
 - **4 Dequantization unit** for scaled-/biased-accumulation

Dequantization Operation			
Mode	PSUM	WSUM	#Op
W4A4	13b	9b	×4
W4A8	17b	9b	×2
W8A4	17b	13b	×2
W8A8	21b	13b	×1

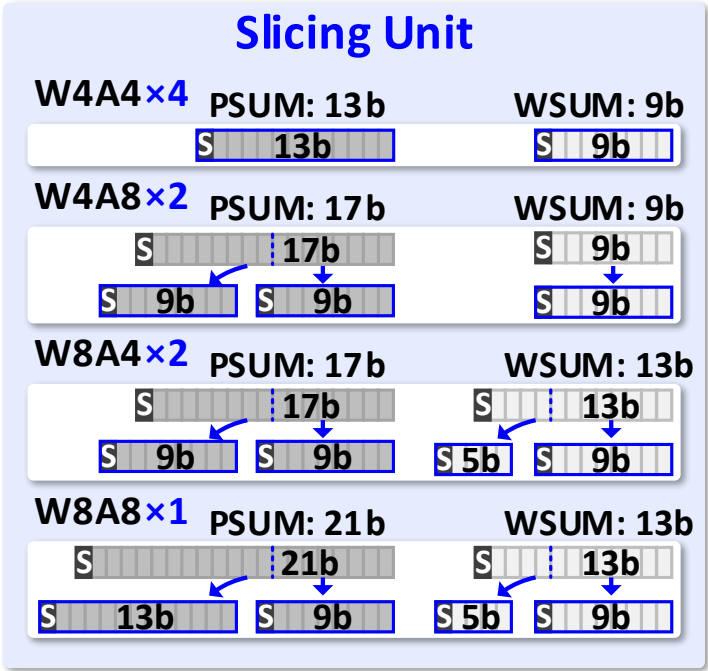
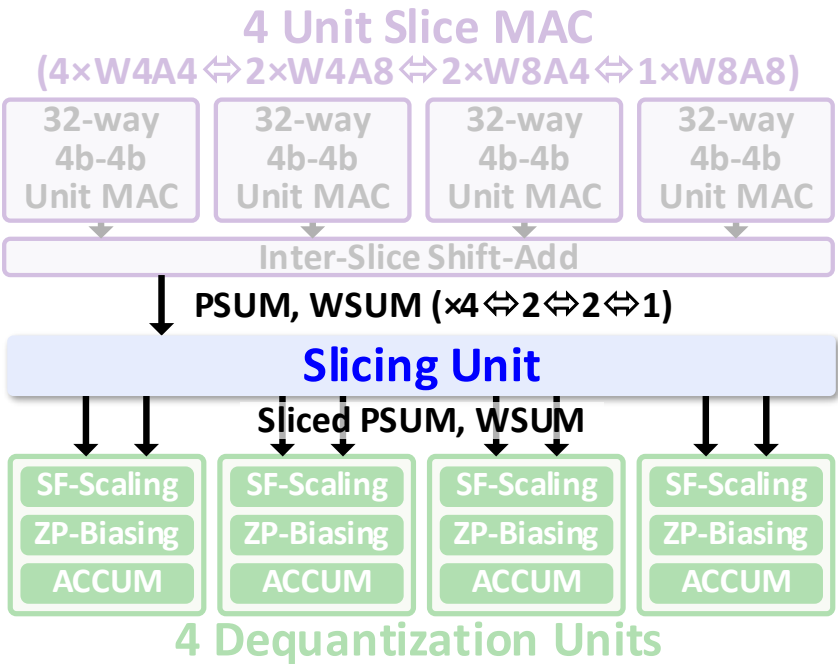


Micro-architecture of SIDU

Bit Slicing PSUM/WSUM for Multiple Mode

- **Slicing Unit** split PSUM/WSUM @ *high-bit mode*
- Each PSUM/WSUM require up to 2 slices

Dequantization Operation			
Mode	PSUM	WSUM	#Op
W4A4	13b	9b	×4
W4A8	17b	9b	×2
W8A4	17b	13b	×2
W8A8	21b	13b	×1

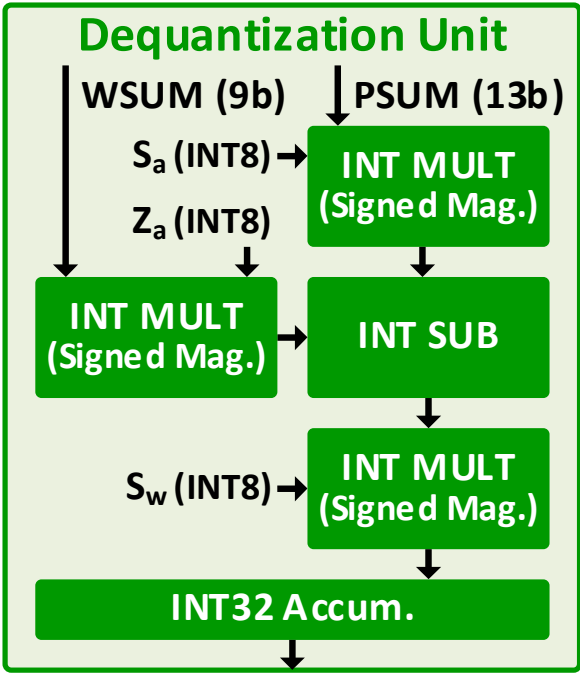
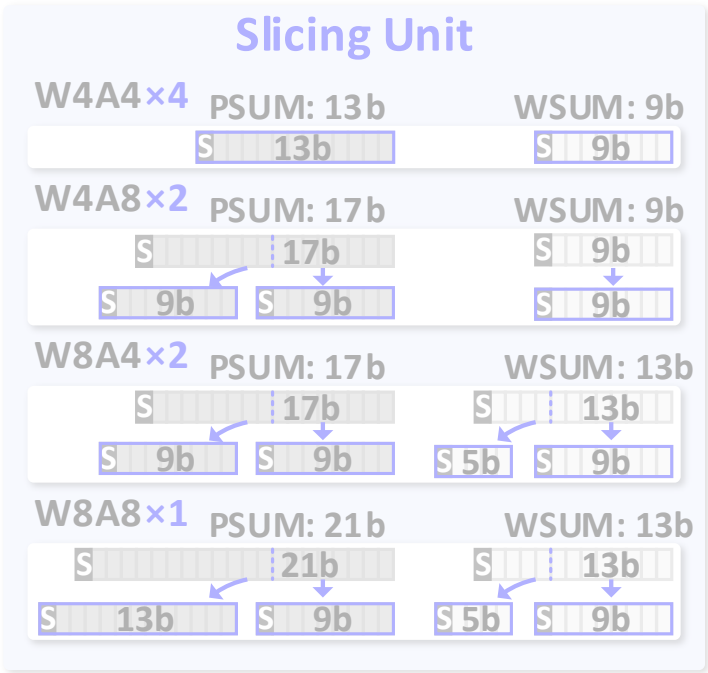
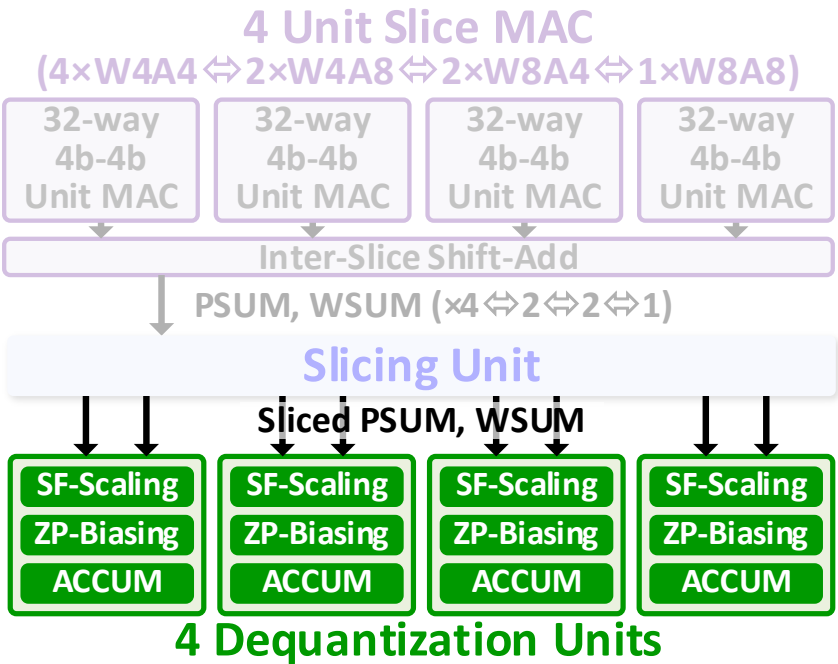


Micro-architecture of SIDU

Minimal Bit-width Dequantization Unit

- Low-bit logic for W4A4 with PSUM = 13b / WSUM = 9b
- Dequantization logic cost minimized 😊

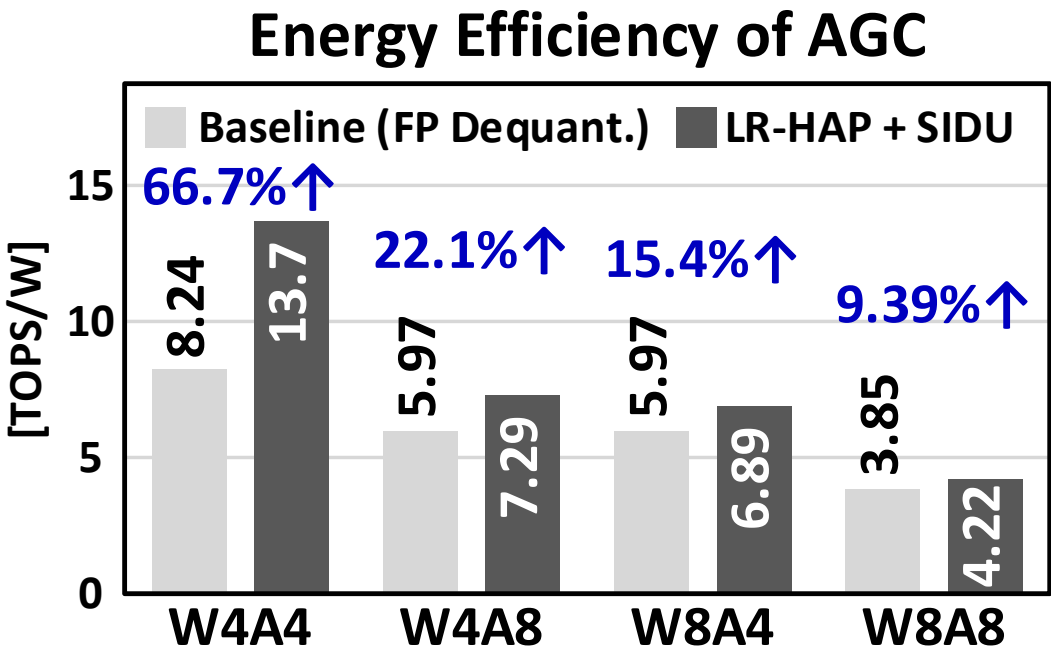
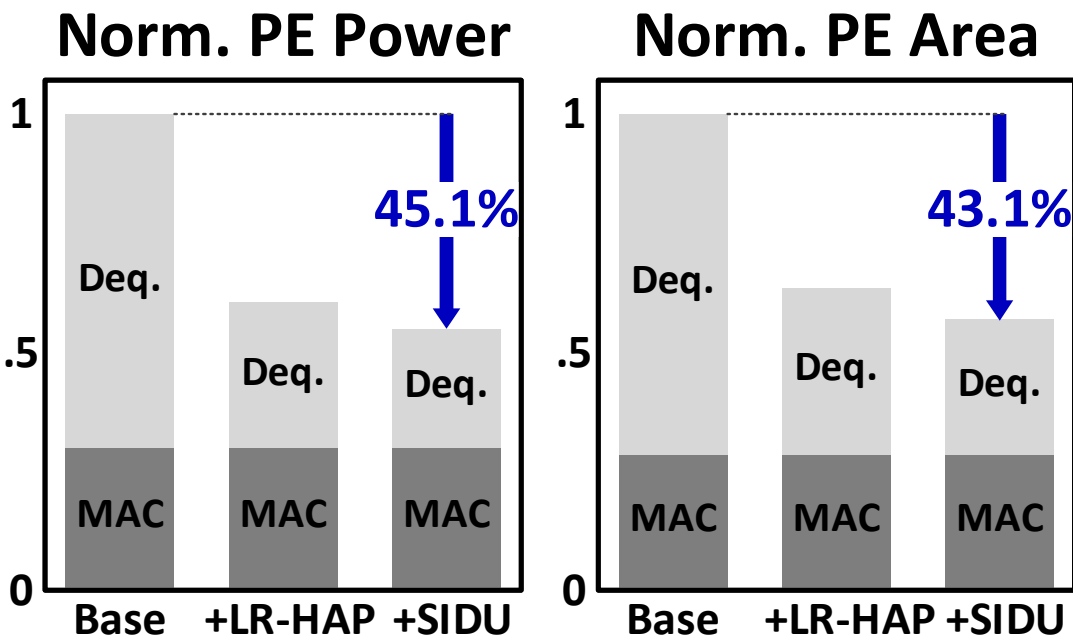
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Performance of SIDU

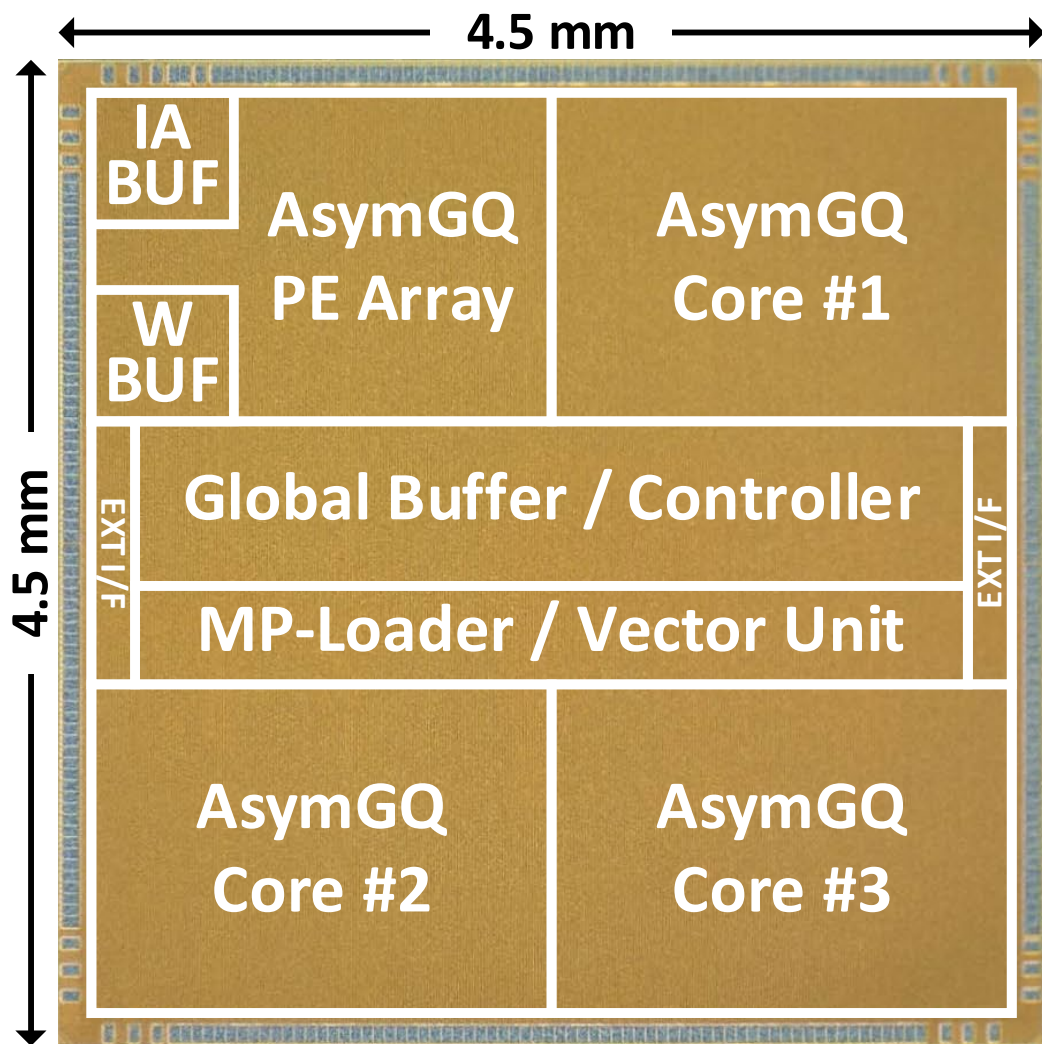
Efficient Integer-based Dequantization with LR-HAP + SIDU

- 45.1% PE Power ↓ & 43.1% PE Area ↓
- Core energy efficiency: ~66.7% ↑ for W4A4 (G=32) Operation



* Measured @ W4A4, Core Frequency=250MHz, VDD = 0.9V

Chip Photography and Summary



1 MAC = 2OPs, I/O Voltage = 1.8V

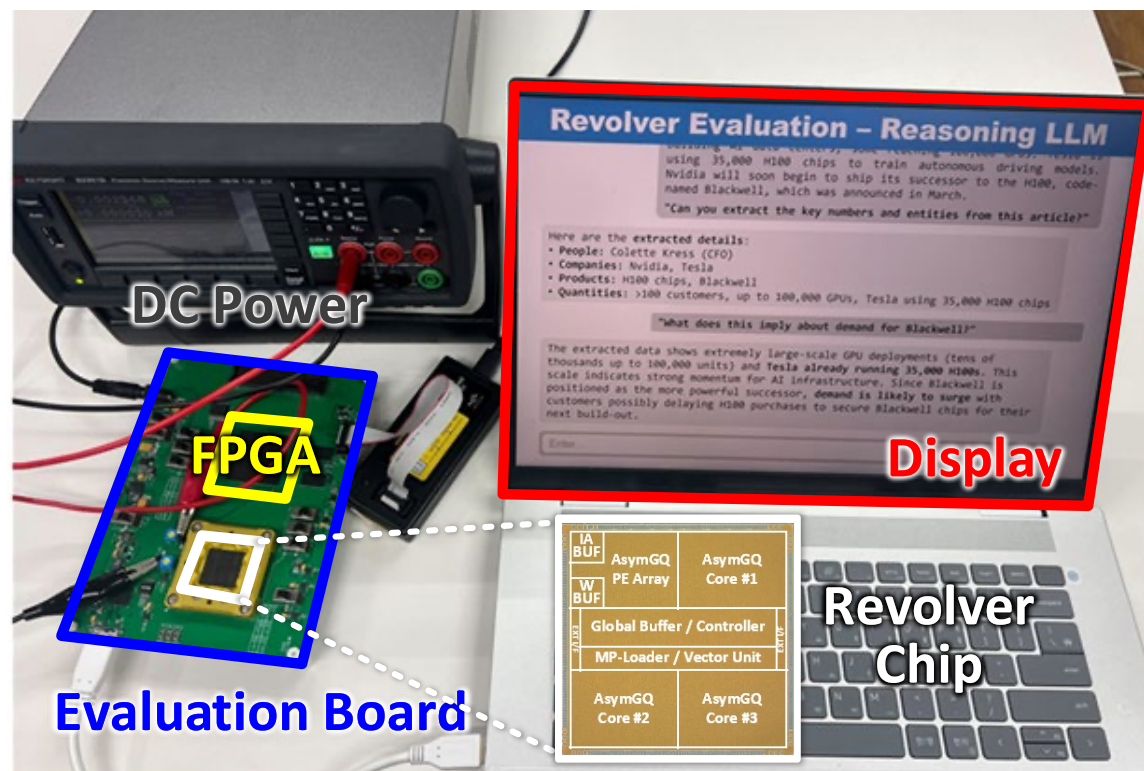
Specifications			
Technology	Samsung 28nm		
Chip Area	4.5 mm × 4.5 mm (20.25 mm ²)		
Supply Voltage	0.68-0.9 V		
Frequency	50-250MHz		
SRAM	Core	Global	Others
	560 KB	1280 KB	60 KB
Data Representation	GEMM	Per-group Factor	Accum.
	INT 4/8/12/16	SF: INT8 (IA/W) ZP: INT8 (IA)	INT32

Measured Performances				
Quantization		Per-Channel (Sym)	Per-Group (32, Sym)	Per-Group (32, Asym)
Energy Efficiency [TOPS/W]	BP ¹	21.7 (INT4) 5.43 (INT8)	15.5 (INT4) 4.94 (INT8)	13.7 (INT4) 4.22 (INT8)
	BE ²	27.3 (INT4) 6.83 (INT8)	21.2 (INT4) 6.37 (INT8)	19.3 (INT4) 5.66 (INT8)
Throughput [TOPS]	BP ¹	16.4 (INT4) / 4.1 (INT8)		
	BE ²	3.3 (INT4) / 0.8 (INT8)		

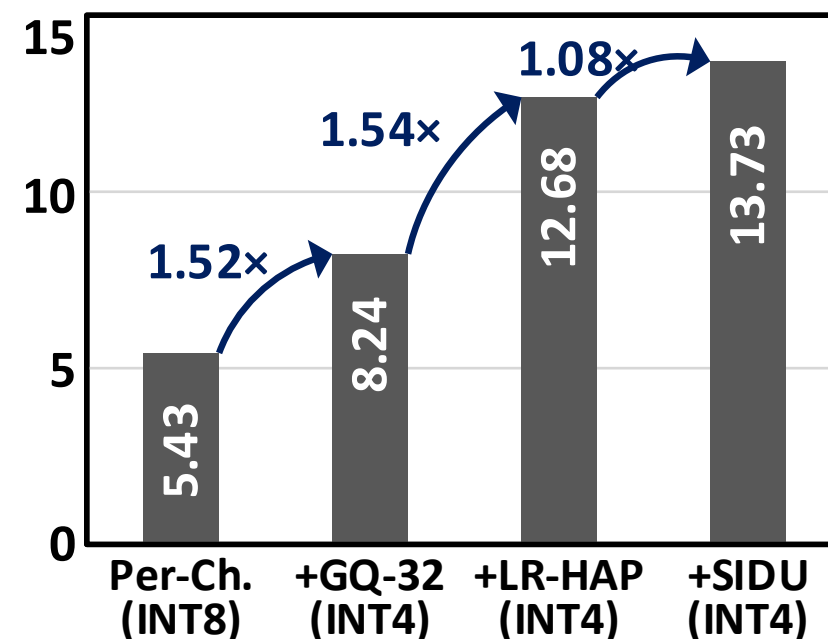
1) Best Performance, 250MHz, 0.9V
2) Best Efficiency, 50MHz, 0.68V
3) Excluding DRAM

Measurement System & Results

- ❑ Evaluation Board Integrates Revolver Chip and FPGA
- ❑ **2.53×** Higher Energy Efficiency over Per-channel Quantization



Energy Efficiency [TOPS/W]



*Operate @250MHz, 0.9V.

Evaluation on LLM Workload

- ❑ Multi-step Math Reasoning with Distilled Model + CoT
- ❑ Multi-turn Chat with Fine-tuned Model for Dialog Usage

	Multi-step Reasoning			Multi-turn Chat	
Dataset	GSM8K-300 Test + CoT			MT-Bench	
Model	LLaMa3-8B-DeepSeek-R1 (Distillation: R1-671B → 8B)			LLaMA3-8B-Instruct (Finetune for Dialogue Usage)	
Rotation	X	O	O	X	O
PAPS (HP/LP)	X	8/4	8/3	X	8/4
Precision	W:INT8 IA:INT8 SF: FP16	W: INT4/8 A: INT4/8 SF: INT8	W: INT3/8 A: INT4/8 SF: INT8	W:INT8 IA:INT8 SF: FP16	W: INT4/8 A: INT4/8 SF: INT8
Accuracy ¹⁾	77.7	75.7(-1.3)	73.0(-4.7)	46.9	41.5 (-5.4)
Energy Saving ^{3),4),6)}	×1.00	×2.46		×1.00	×2.46
Speedup ^{2),5),7)}	×1.00	×1.72	×2.10	×1.00	×1.72

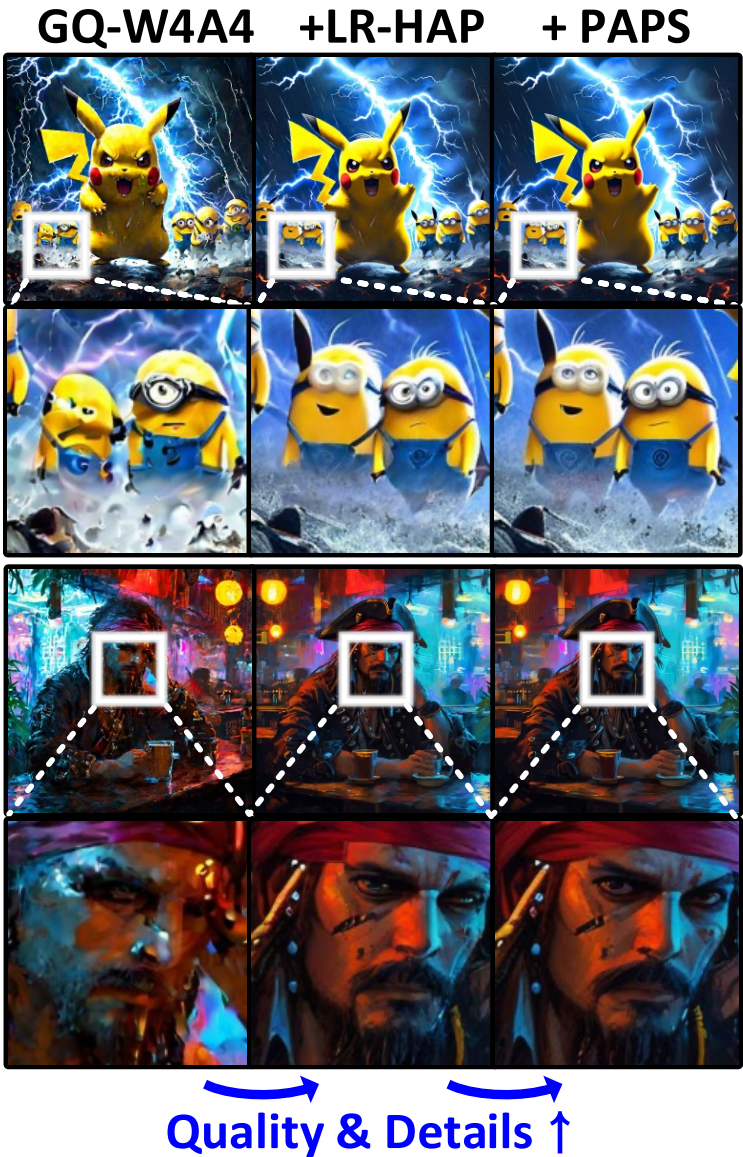
1) Reasoning: Flexible-Match Accuracy (%) / Multi-Turn Chat: Adjusted win-rate (%) over FP Baseline
2) @ 250MHz, 0.9V. 3) @ 50MHz, 0.68V. 4) Excluding DRAM. 5) Including DDR4 DRAM. 6) Prefill. 7) Decode.

Evaluation on DM Workload

❑ Text-to-Image Generation with Finetune Model using High-Quality Data

	Diffusion Model		
Dataset	HPD-v2		
Model	PixArt-Σ (Weak-to-Strong Tuning) (Finetune PixArt-α w/ High-Quality Data)		
Rotation	X	O	O
PAPS (HP/LP)	X	X	8/4
Precision	W:INT8 IA:INT8 SF: FP16	W:INT4 IA:INT4 SF: INT8	W: INT4/8 A: INT4/8 SF: INT8
Accuracy ¹⁾	30.93	30.51 (-0.42)	30.83 (-0.10)
Energy Saving ^{3),4)}	×1.00	×3.47	×2.48
Speedup ^{2), 5)}	×1.00	×2.83	×1.94

1) HPS-v2 Score. 2) @ 250MHz, 0.9V. 3) @ 50MHz, 0.68V.
4) Excluding DRAM. 5) Including DDR4 DRAM.



Performance Comparisons

		ISSCC'23 [25]	ISSCC'25 [15]	ISSCC'25 [16]	This Work
Technology [nm]		12	28	28	28
Die Area [mm ²]		4.60	20.25	12.96	20.25
Function		NLP Transformer	DM	GenAI	GenAI (+Distilled, +CoT)
Quantization	Precision	A/W-FP4/8	A/W-INT 4/8/12/16	A-INT8+FP12 W-INT4/8/12/16	A/W-INT4/8/12/16
	Symmetry	A/W-Sym	A/W-Sym	A-Sym/W-Asym	A-Asym/W-Sym
	Granularity	Per-Group16/32	Per-Group32	Per-Channel	Per-Group32
	Rotation Support	X	X	X	O (LR-HAP)
	Dequant. Logic	FP	FP-INT	-	Fully-INT
	Phase-aware Opt.	X	X	X	O (LLM, DM)
Efficiency ¹⁾ [TOPS/W]		6.61 (FP4)/3.01 (FP8)	19.0 (INT4)/4.7 (INT8)	13.4 (INT4-8)	13.7 (INT4)/4.22 (INT8)
Performance ¹⁾ [TOPS]		0.73 (FP4)/0.37 (FP8)	31.3 (INT4)/7.8(INT8)	2.05 (INT4-8)	16.4 (INT4)/4.1 (INT8)
LLM: Multi-step Reasoning & Multi-turn Chat / LLaMa3-8B					
Energy (Prefill) _{1),3),4)} [mJ/Inf.]		-	>1707.9 ⁵⁾	>1478.7 ⁵⁾	837.9
Latency (Decode) _{1),3),6)} [ms/Tok.]		-	2007.5	2007.5	979.8 (PAPS-8/3) 1197.8 (PAPS-8/4)
Diffusion Model: Text-to-Image Generation / PixArt-Σ					
Energy Efficiency _{1),3)} [mJ/Step]		-	>1946.8 ⁵⁾	>1522.9 ⁵⁾	825.8 (4b) 1093.2 (PAPS-8/4)

1) @ MAX Performance Operating Condition. 2) Excluding DRAM. 3) Including DRAM (DDR4 Dynamic Power). 4) Input Token Length = 256.
5) Energy estimated with reported peak efficiency (as a lower bound). 6) Assuming 32Gb/s External Bandwidth.

Conclusions

- ❑ **Revolver: Efficient and Accurate Low-bit GenAI Accelerator**
- ❑ **Phase-aware + Rotation-based + Asymmetric Group Quantization**
 - 1) **Phase-aware Precision Selection (PAPS) with Multi-Prec. (MP)-Loader**
→ 50.4% system energy saving & 17.9% DRAM capacity reduction
 - 2) **Local Rotation with Harmonic Aligned Permutation (LR-HAP) and Paired Unit-FHT Block (PUB)**
→ ~60% rotation unit cost reduction & allowing INT SF
 - 3) **Sliced Integer-based Dequantization Unit (SIDU)**
→ 66.7% higher core energy-efficiency

**Edge GenAI Accelerator for Distilled Models and CoT
with Unified Optimization for both LLM & DM together**

→ 2.36× Energy & 2.05× Latency Reduction vs Previous Work

A 51.6 μ J/Token Subspace-Rotation-Based Dual-Quantized Large-Language-Model Accelerator with Fused Scale-Activation INT Datapath and Rearranged Bit-Slice LUT Computation



Bo Liu, Zihan Zou, Xinming Yan, Xilong Kang,
Xinyang Chen, Bo Hu, Jiaming Lin, Haoran Du,
Jun Yang, Xin Si, Hao Cai



Southeast University, Nanjing, China



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Introduction – LLMs & Quantization

Large Language Models (LLMs)



Coding



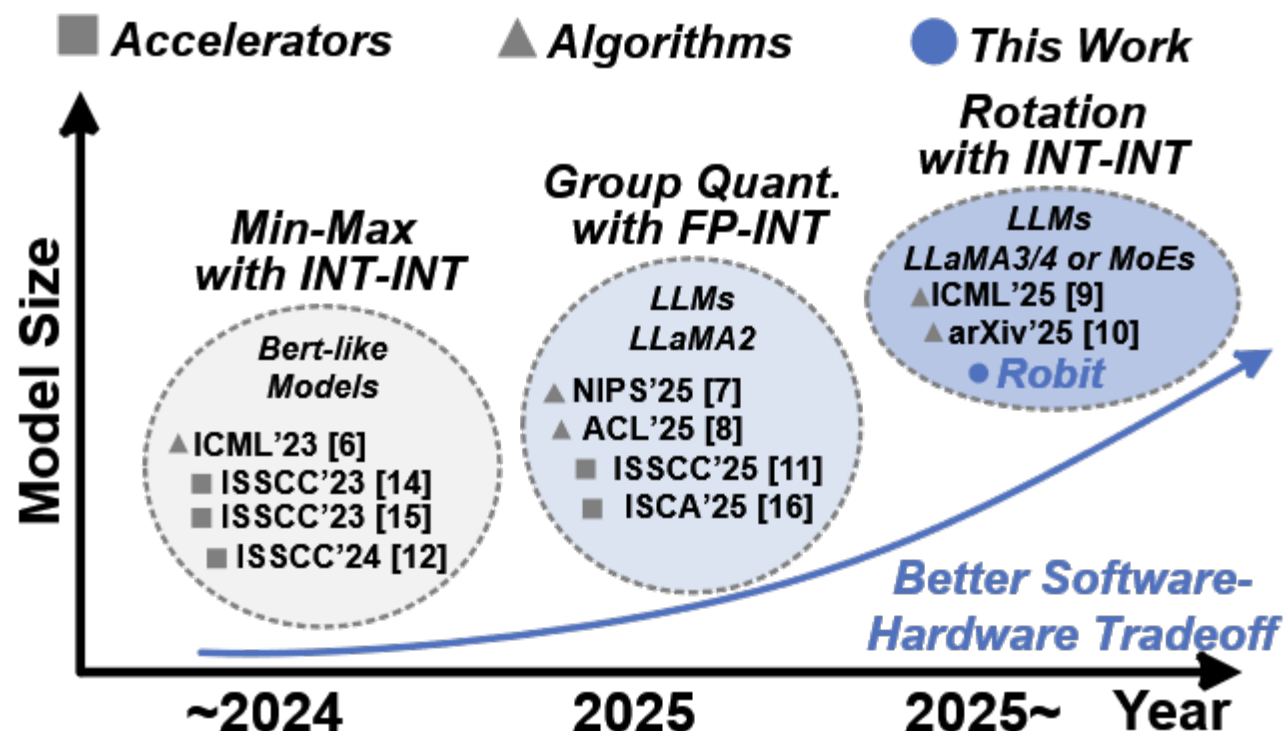
Chatbots



Math

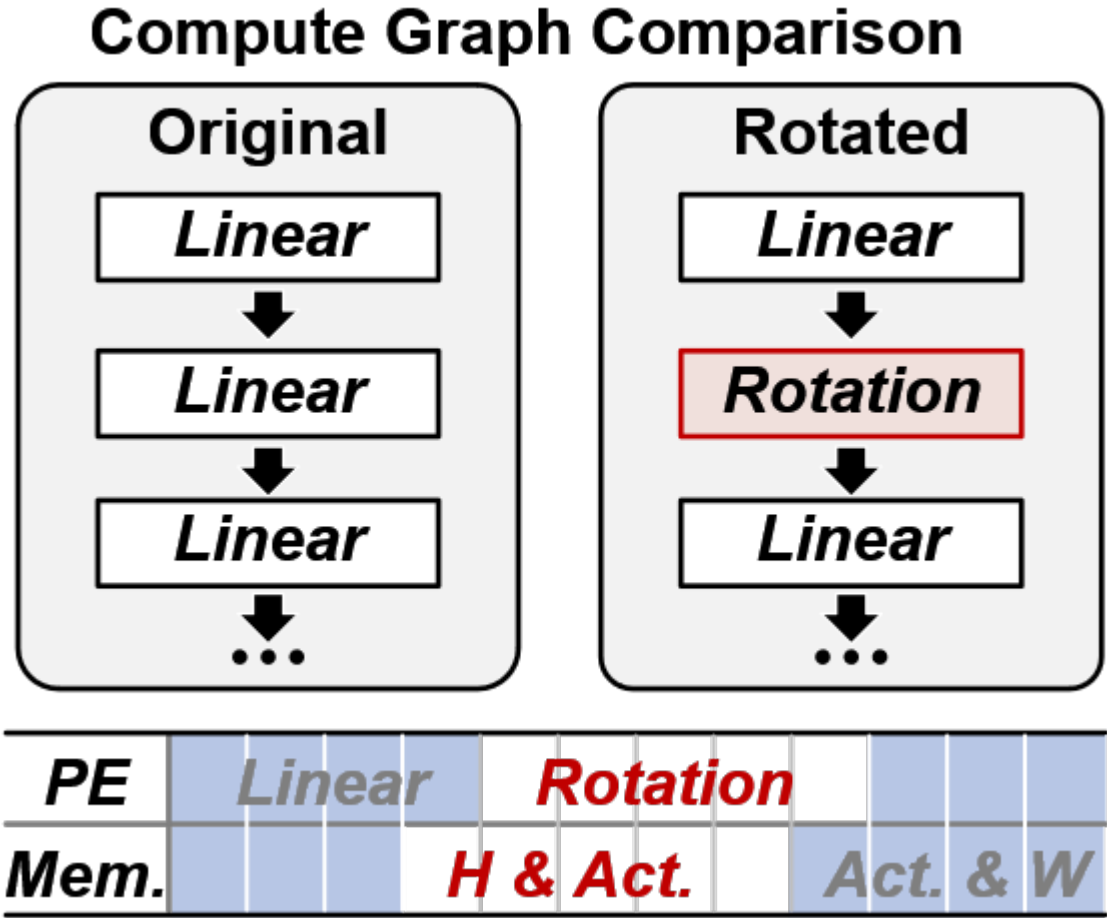


Quantized DNN Deployment Overview

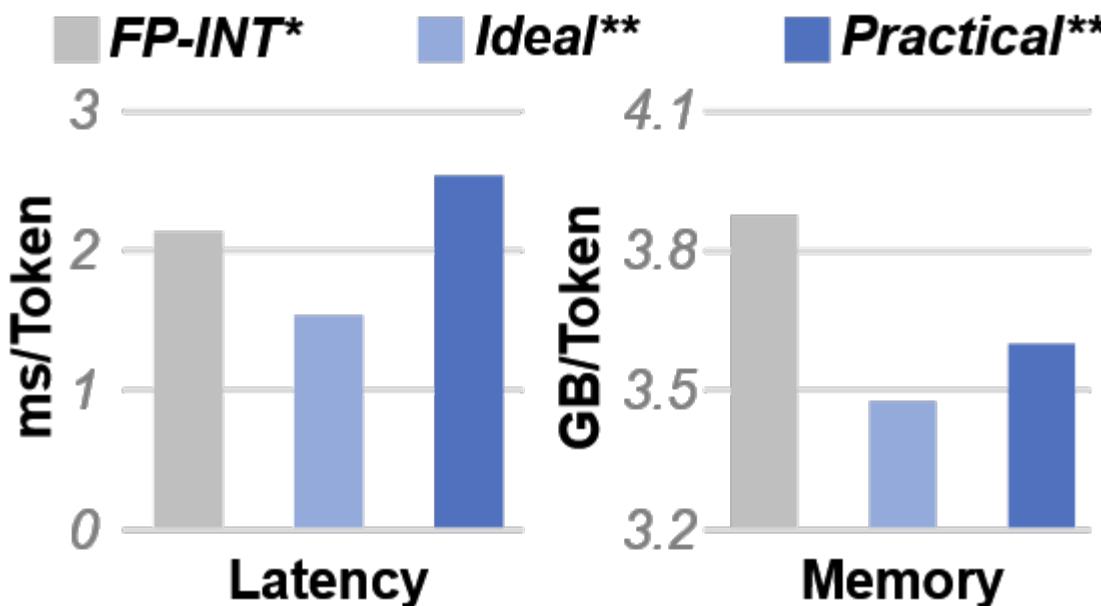


- LLMs have become fundamental to modern social life.
- Rotation-based quantization offers a promising trade-off.

Introduction – Challenges for Rotation LLM



Latency and Memory Access Analysis with LLaMA2-7B@WIKI2

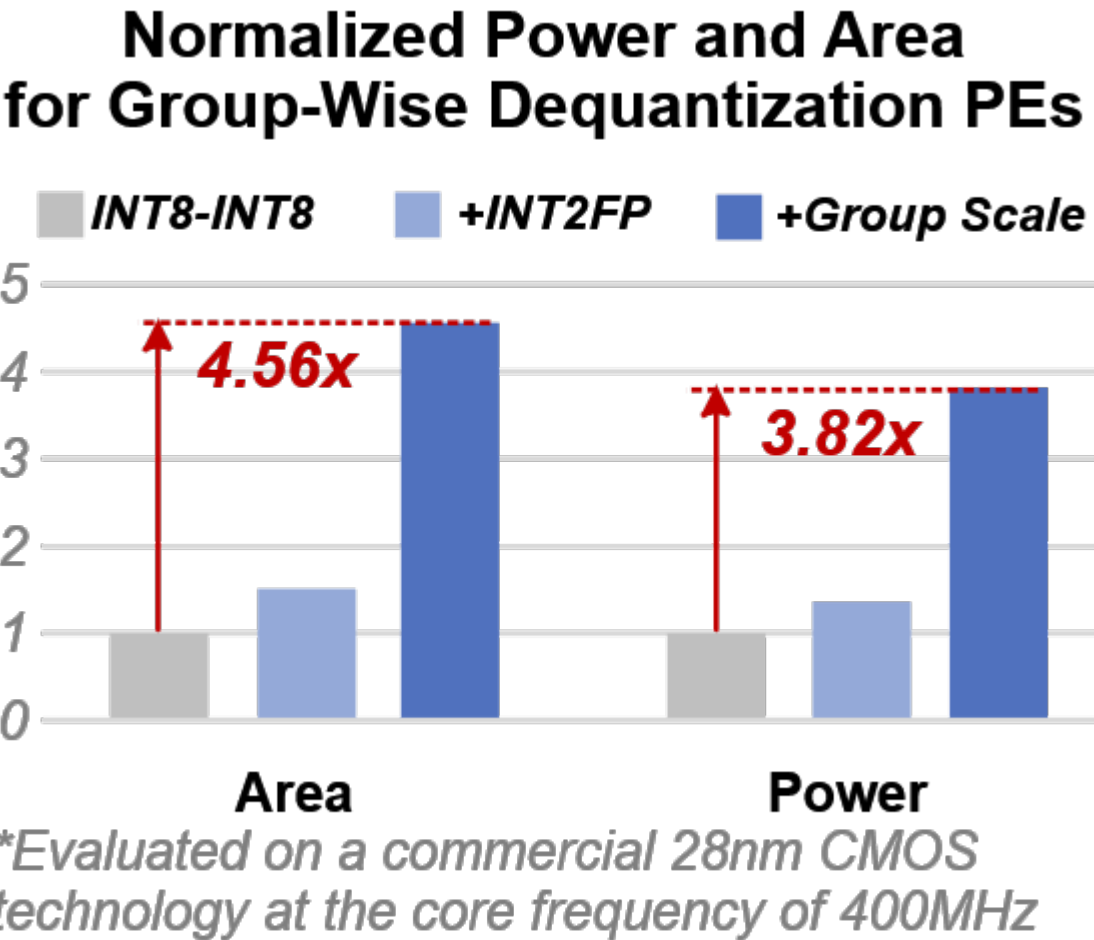
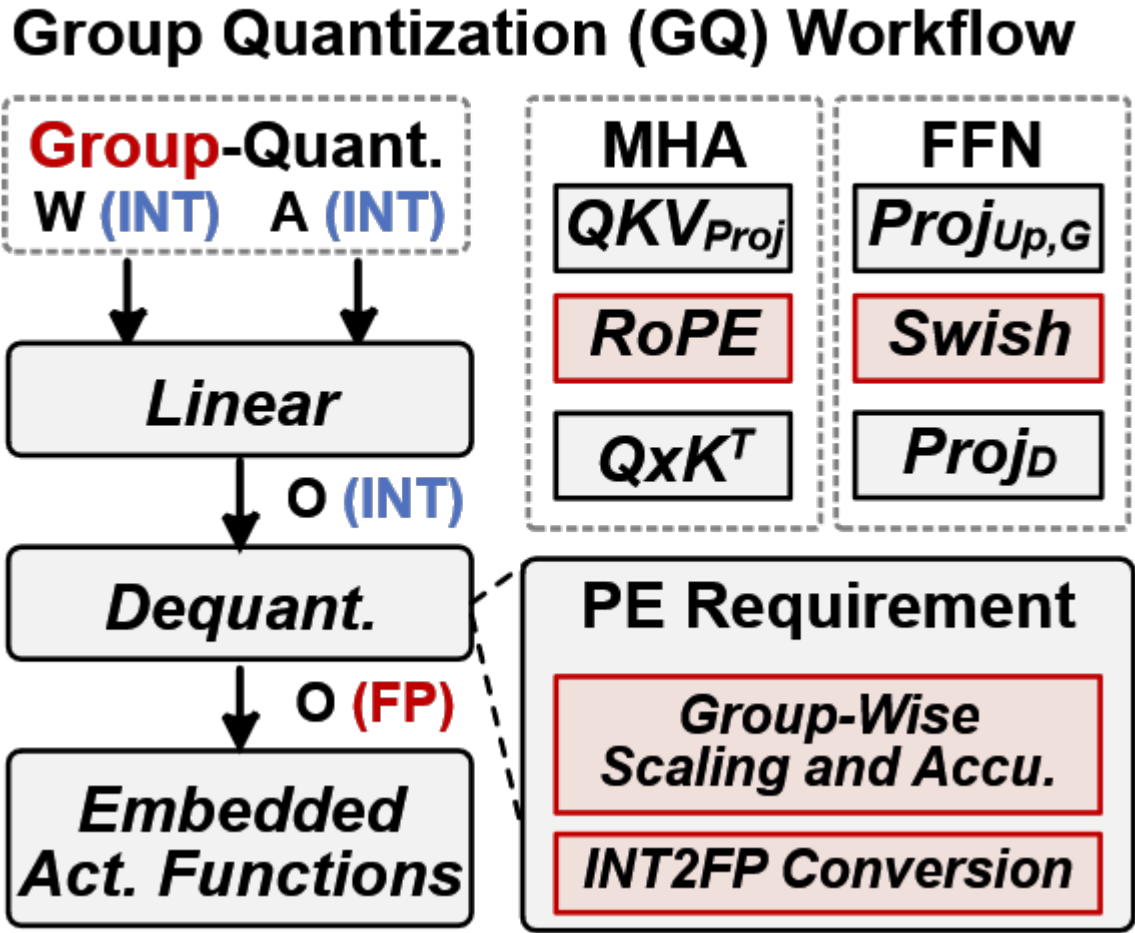


*SOTA LLM Accelerator [1]. ** Evaluated with dense [1] at A4W4KV4. All PPL Loss < 0.5.

■ Challenge 1: Additional Latency and Memory Access for Rotation

[1] Y. Qin et al., ISSCC, pp. 420-422, 2025.

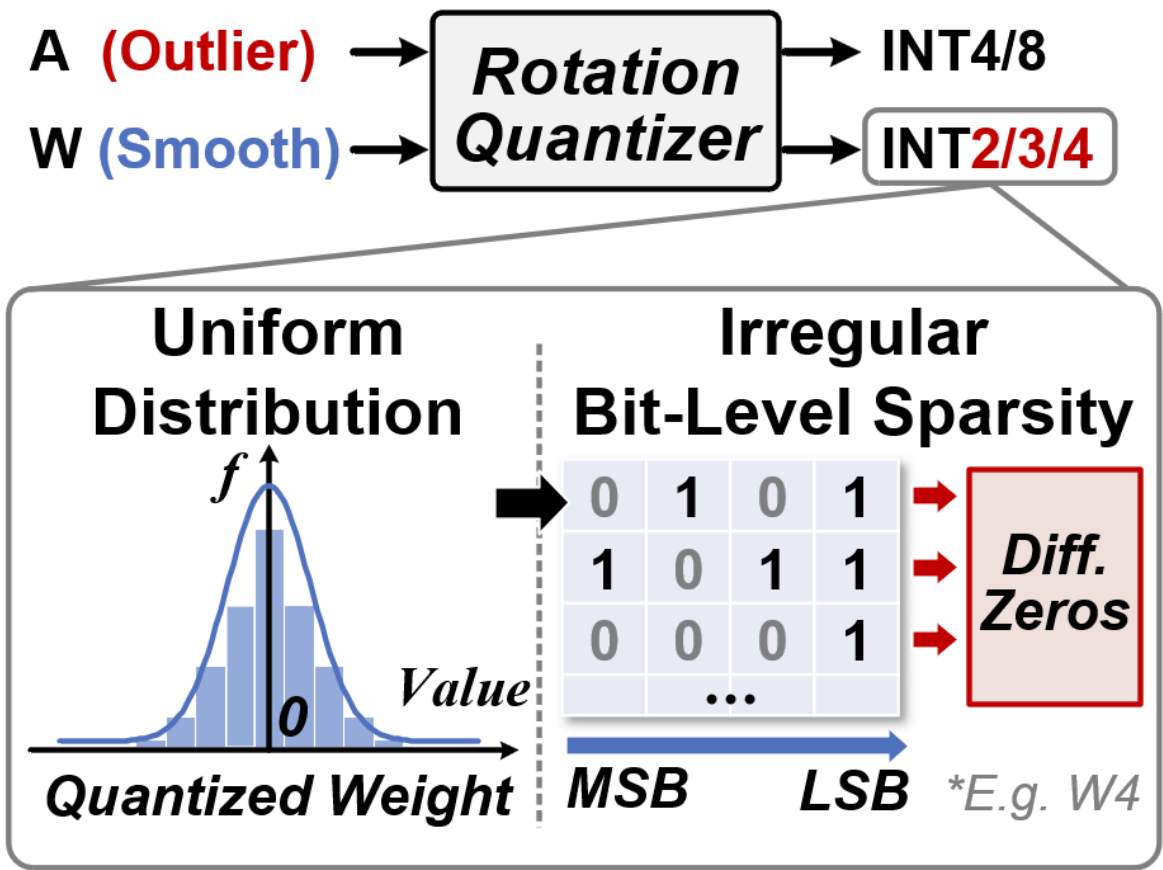
Introduction – Challenges for Rotation LLM



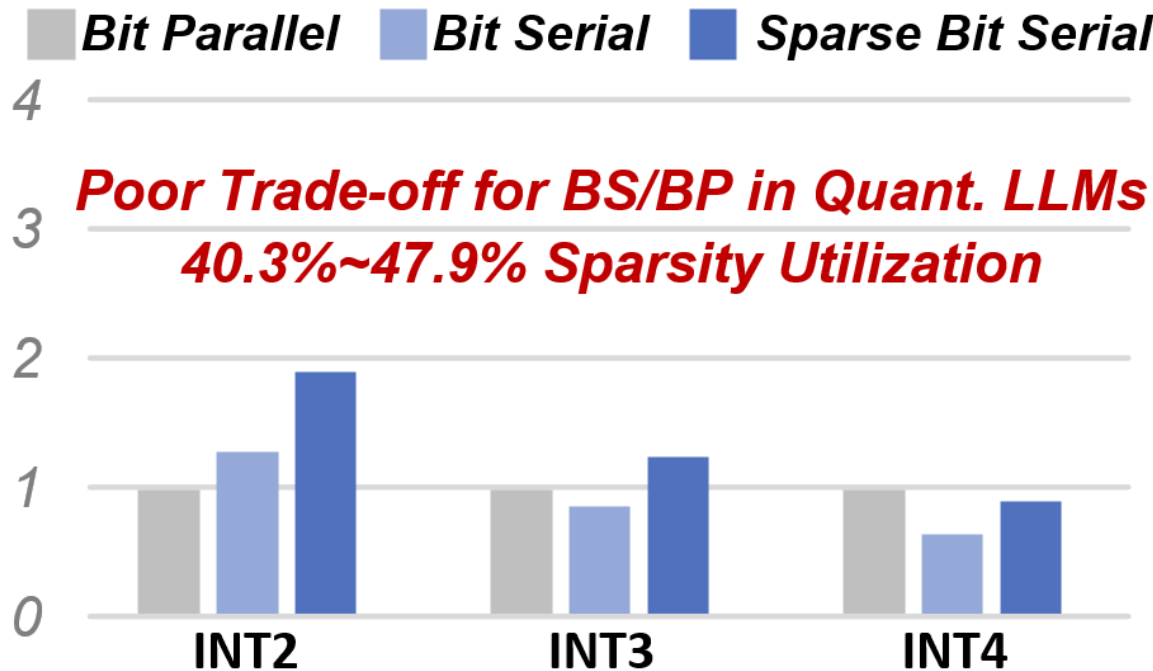
■ Challenge 2: Weak INT Path of GQ and Embedded Act. Functions

Introduction – Challenges for Rotation LLM

Flexible Weight Config. with Sparsity



MAC Energy Efficiency under Variable Weight Bit-Widths



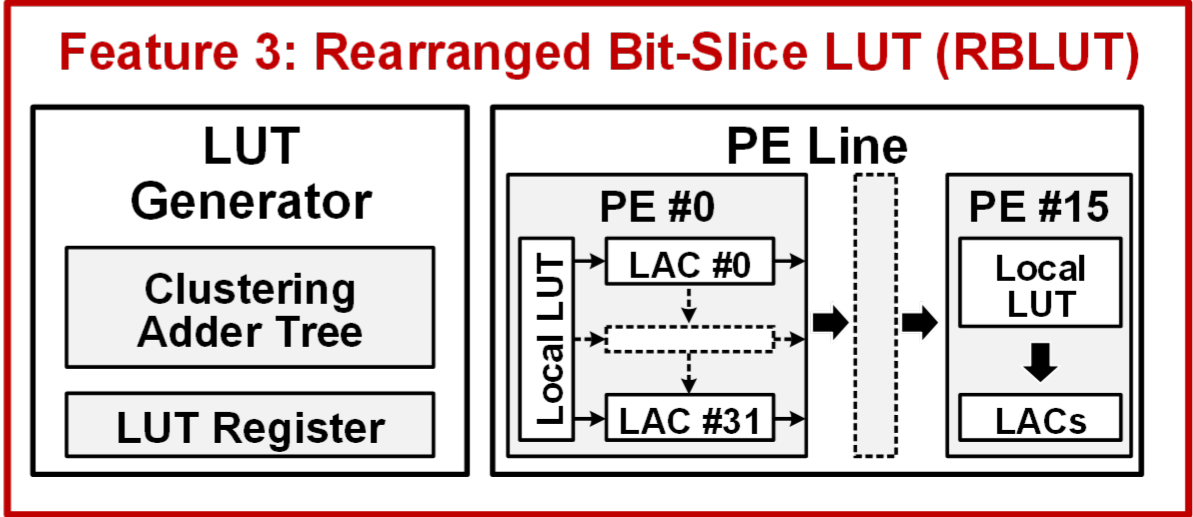
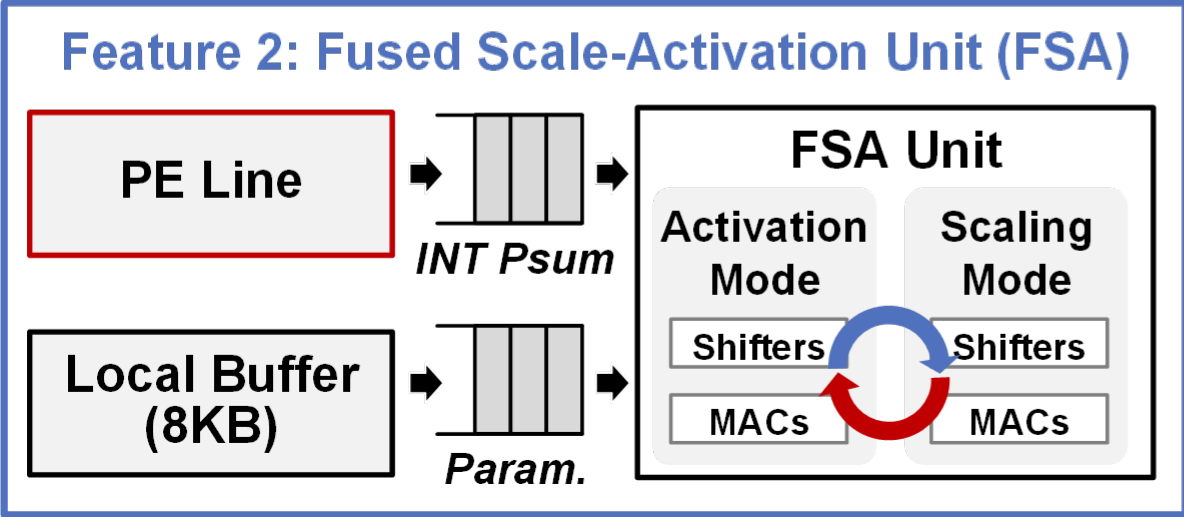
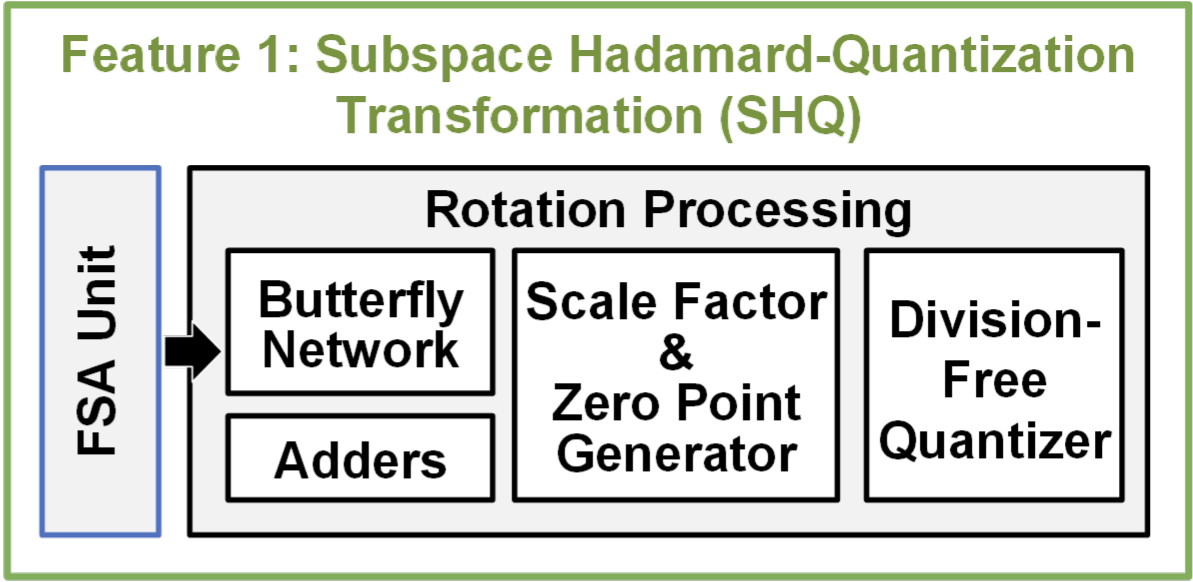
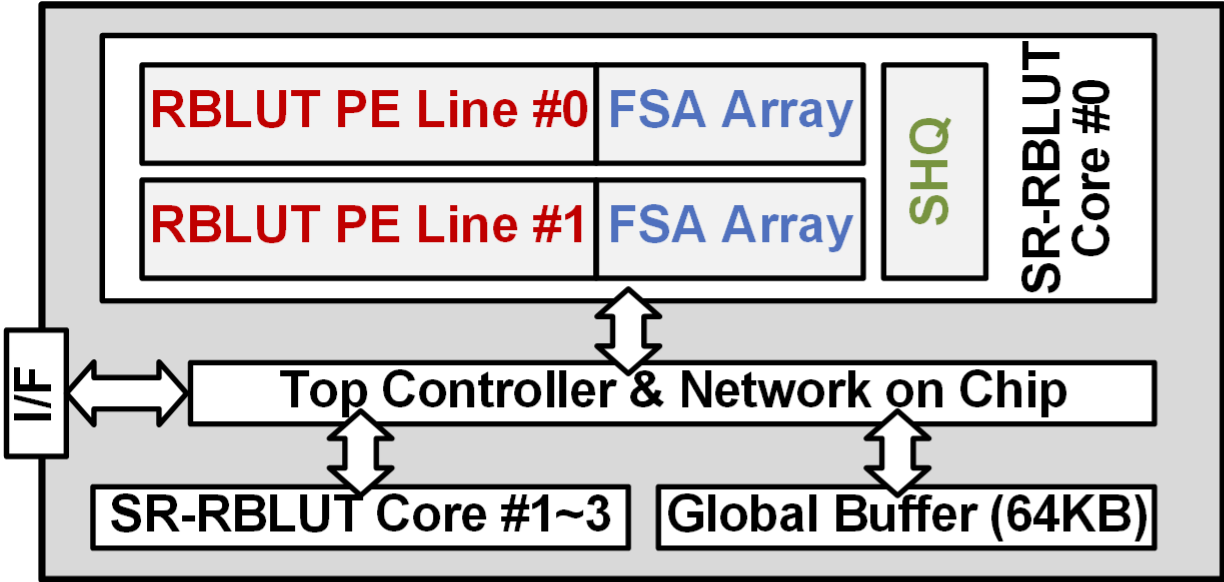
**Bit Parallel (BP) and Bit Serial (BS) MAC unit are normalized to product 8b-4b MAC/cycle.*

■ Challenge 3: Poor MAC Flexibility-Efficiency Trade-off

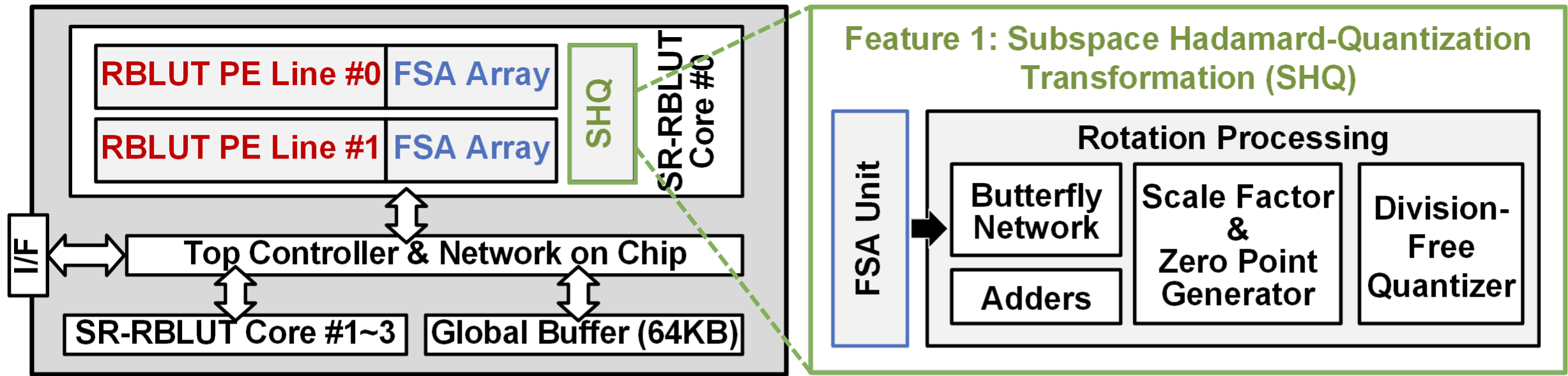
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Overall Architecture



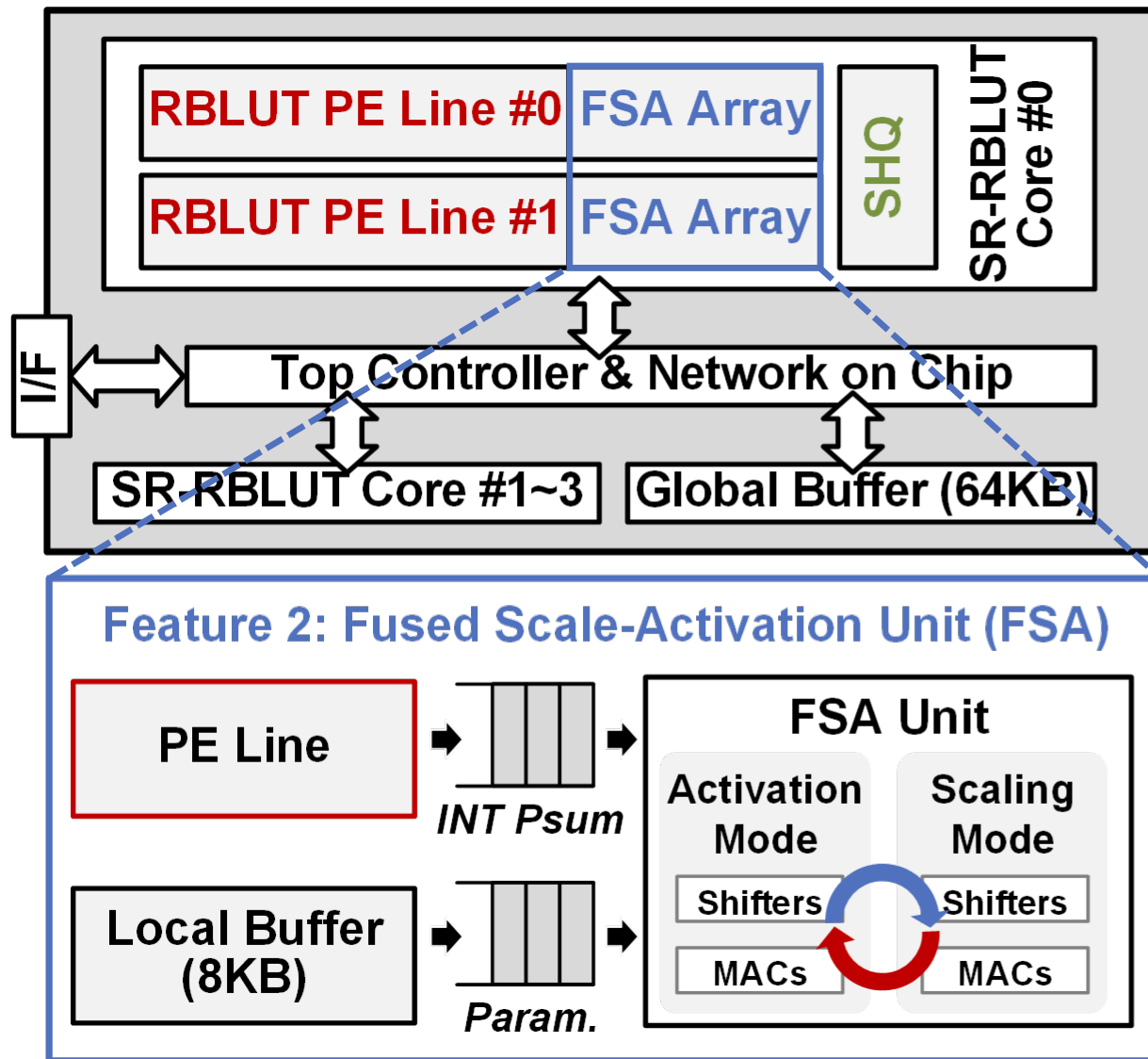
Overall Architecture



■ Subspace Hadamard Quantization Transformation (SHQ)

- Replace the general matrix multiplication (GEMM)
- Fully pipelined with rotation excluded from the critical path

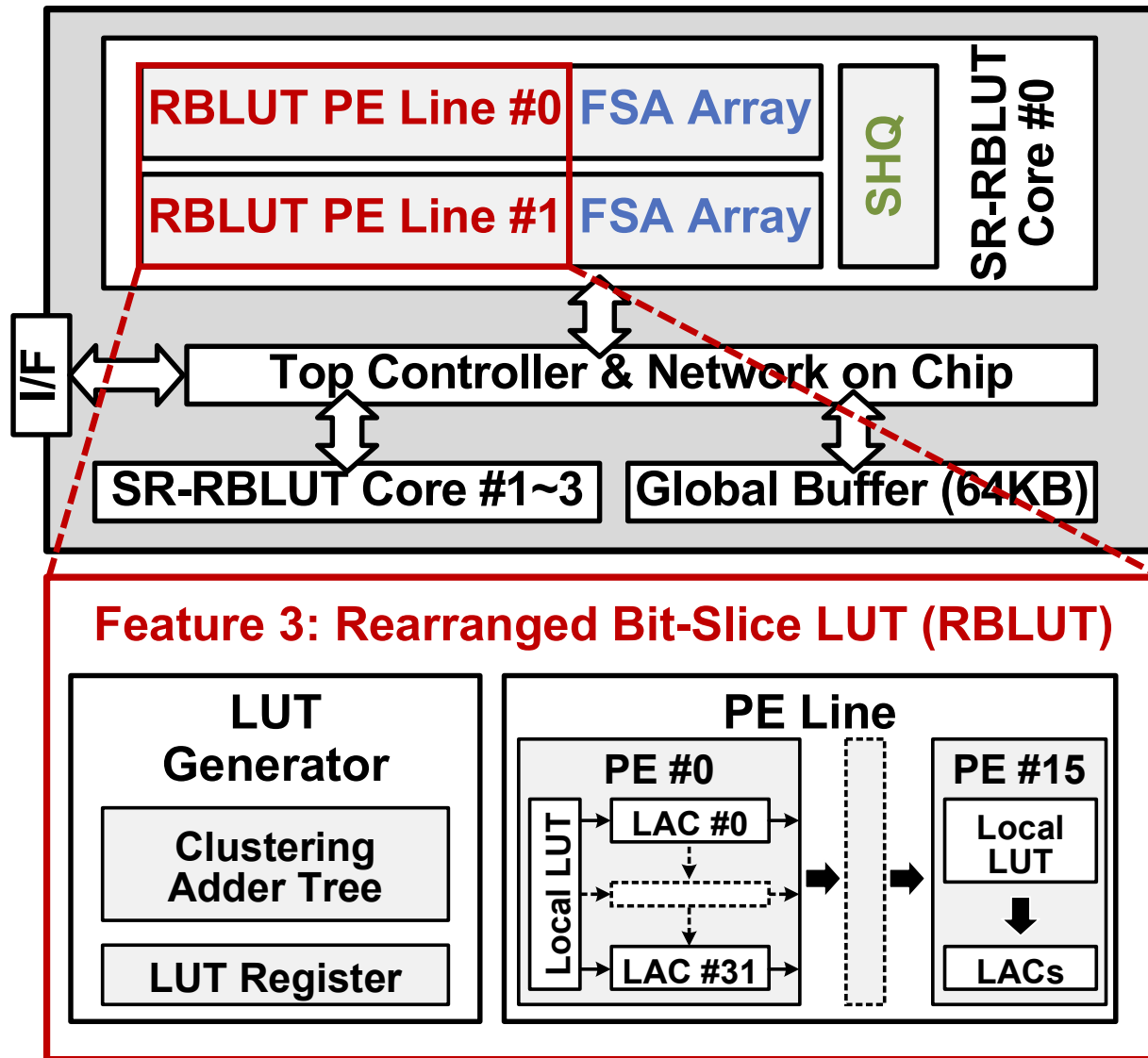
Overall Architecture



■ Fused Scale-Activation Unit (FSA)

- Unify scaling and activation in the INT domain
- Eliminate frequent INT-to-FP conversions

Overall Architecture



■ Rearranged Bit-Slice LUT Engine (RBLUT)

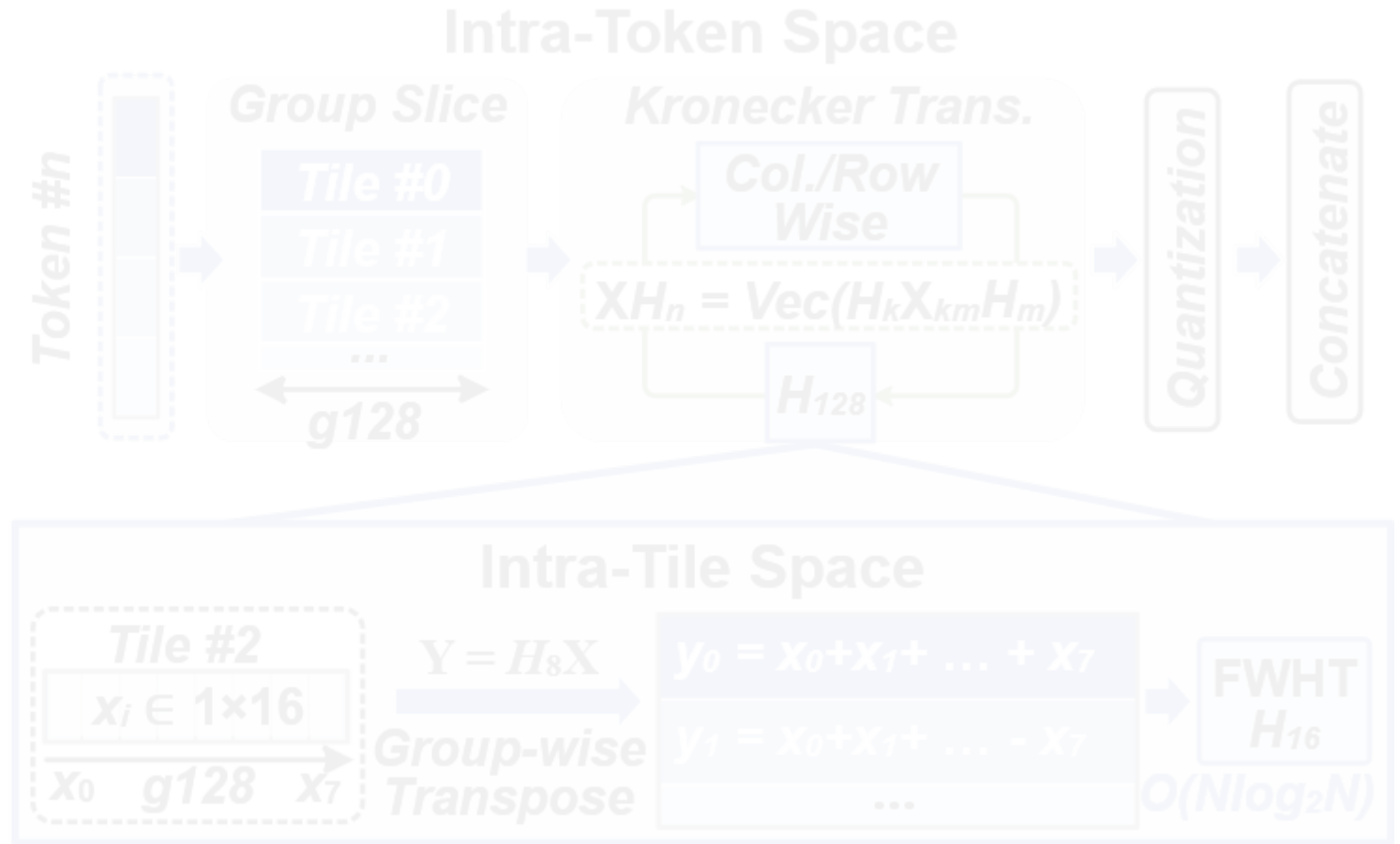
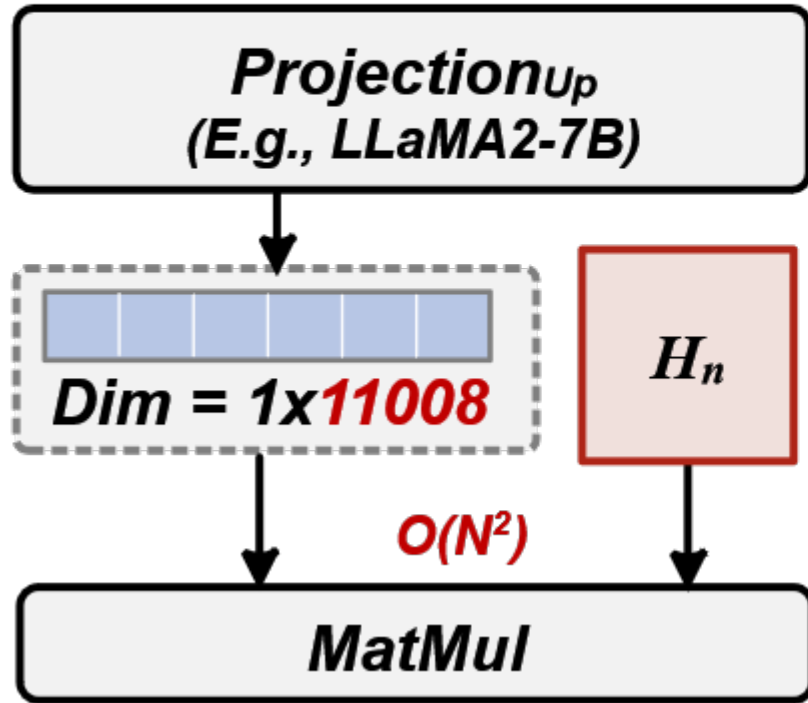
- Trade-off between flexibility and energy efficiency
- Ensure high and stable utilization across all precisions

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Subspace Hadamard-Quantization (SHQ)

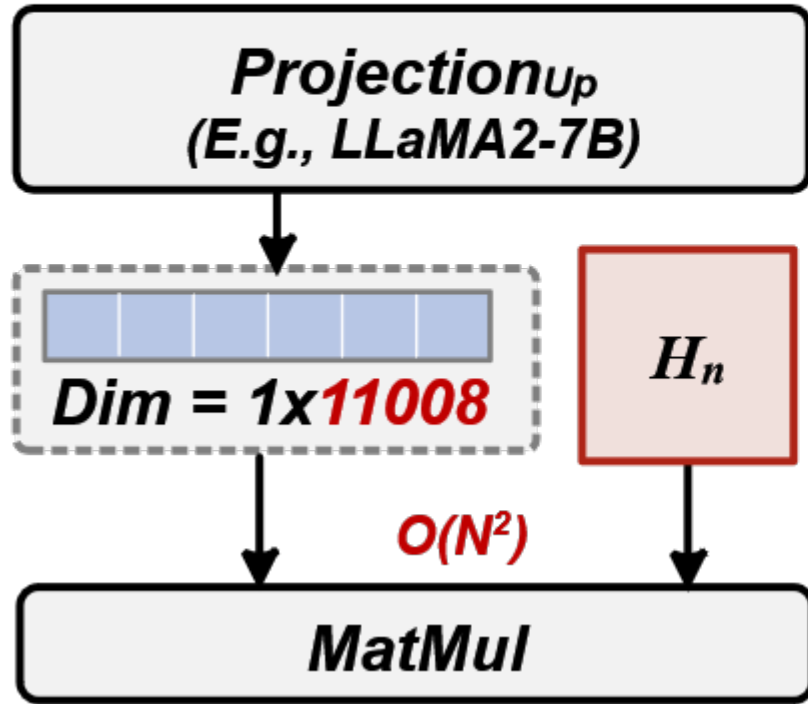
High-dimensional
Hadamard Transform. (HT)



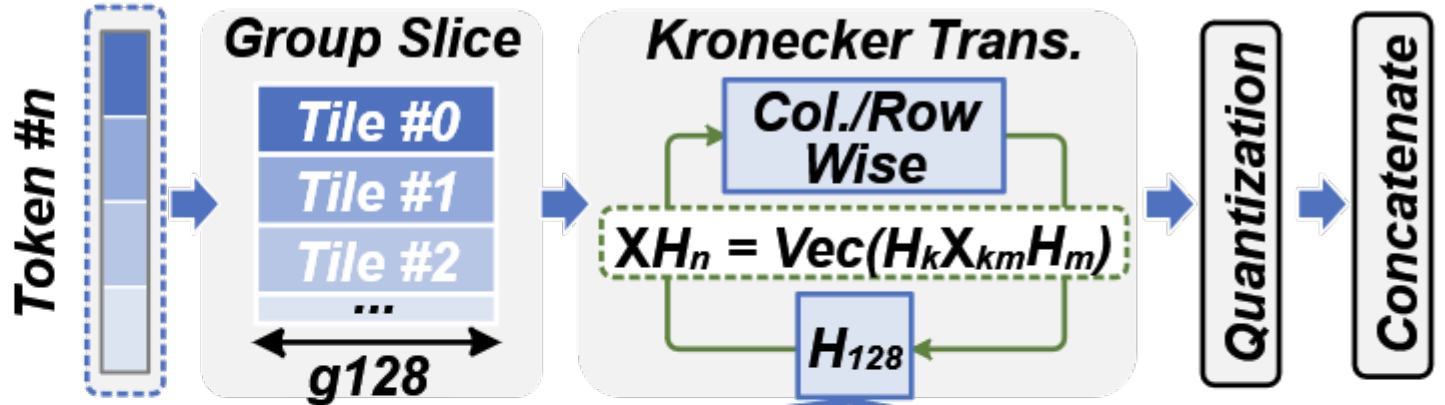
- Kronecker decomposition breaks down the large-scale HT into block operations from Intra-token and Intra-tile spaces.

Subspace Hadamard-Quantization (SHQ)

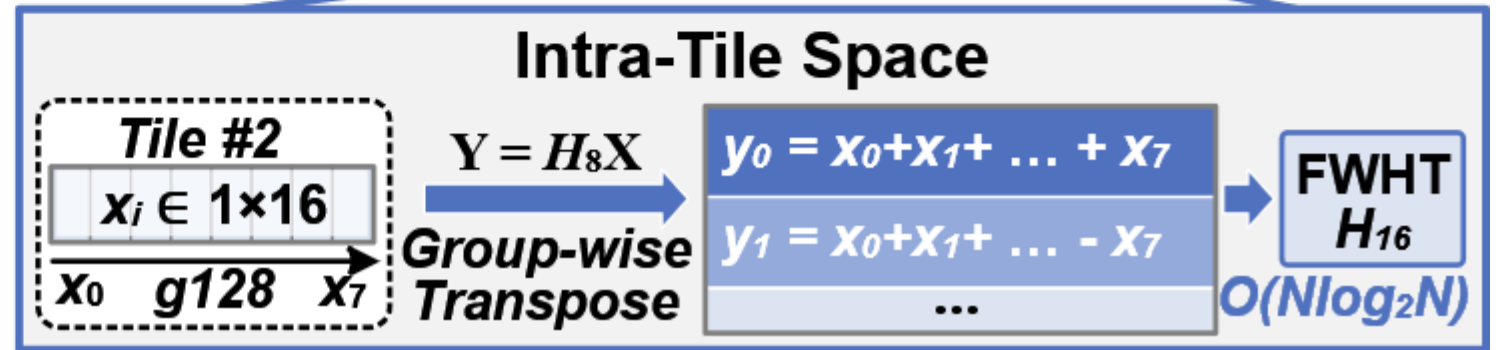
High-dimensional
Hadamard Transform. (HT)



Intra-Token Space

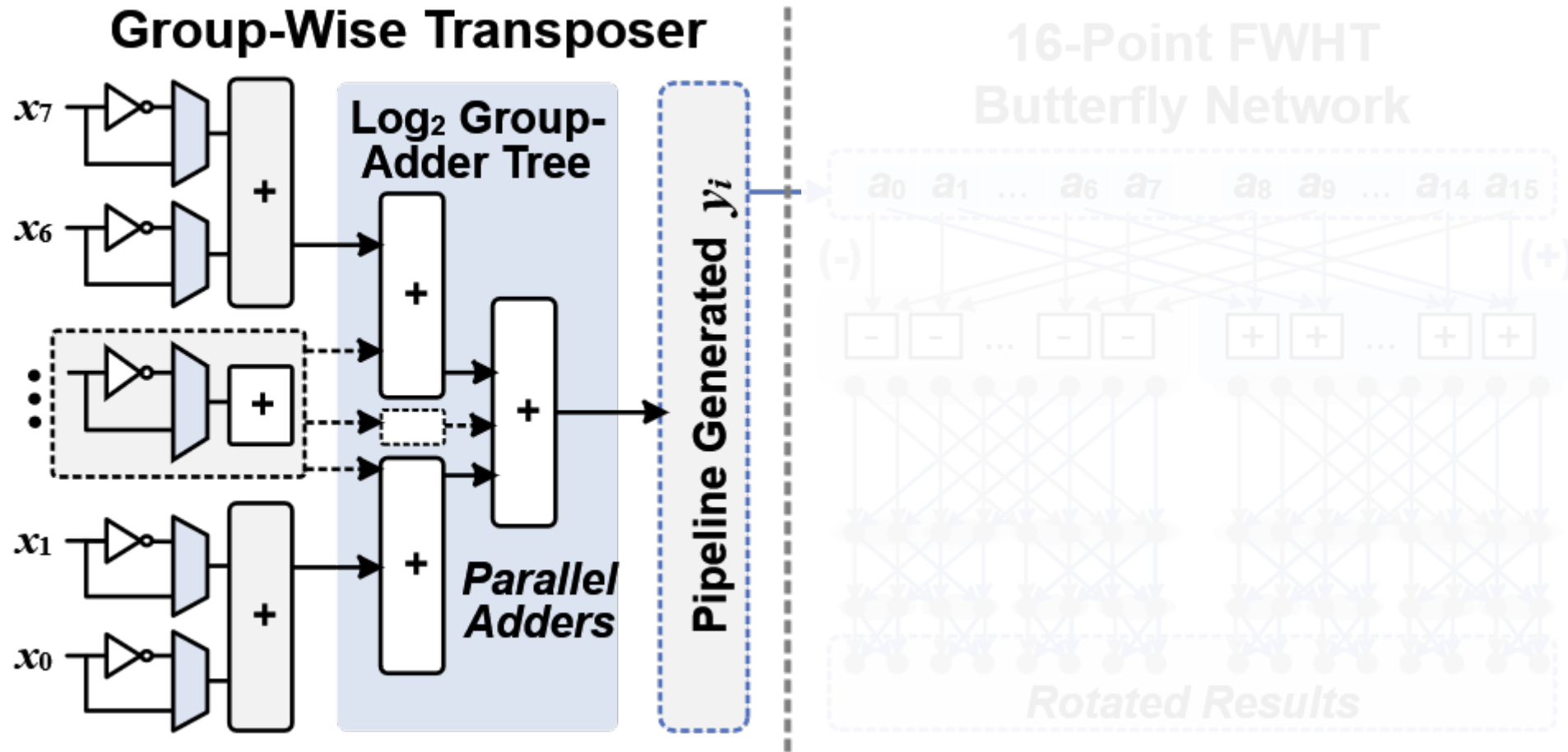


Intra-Tile Space



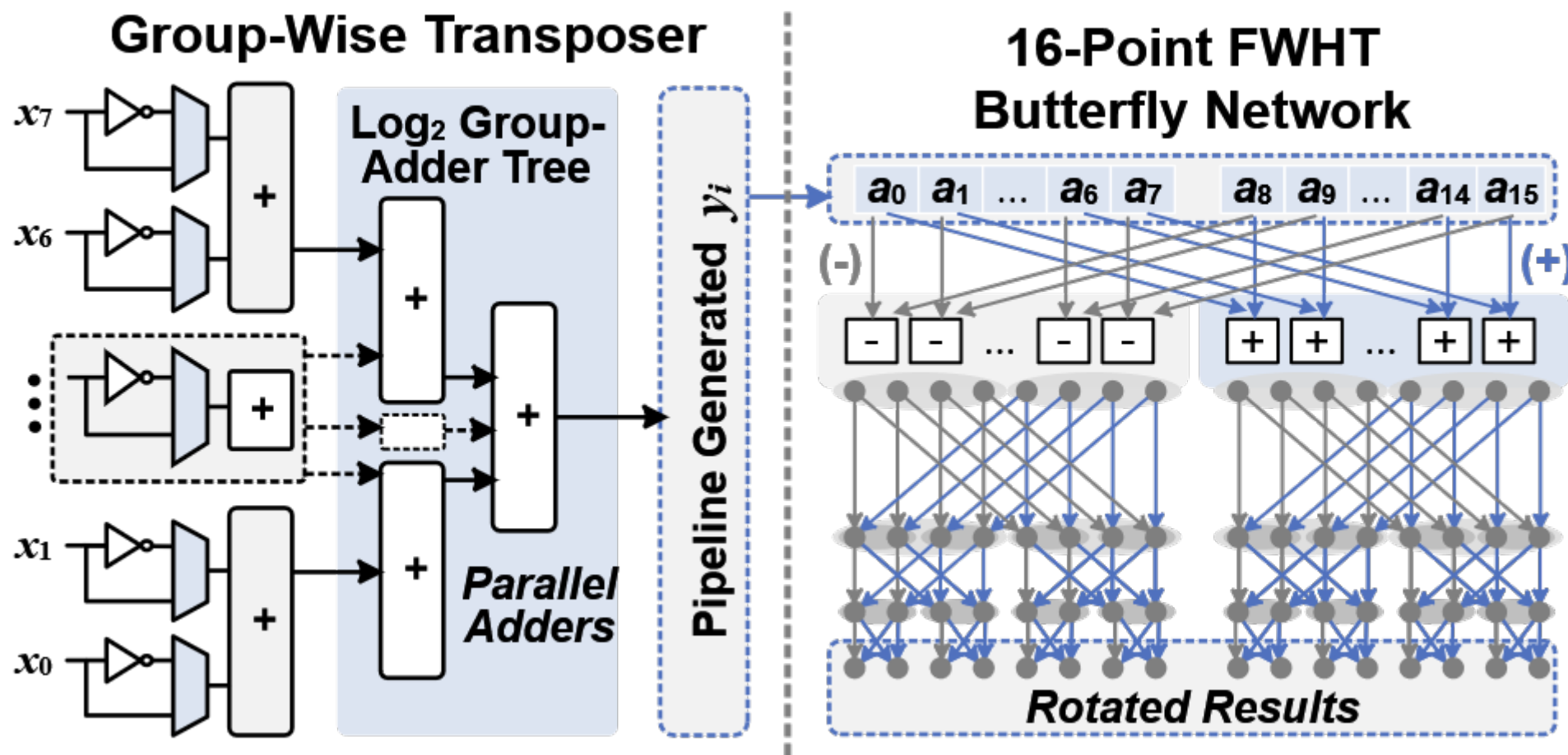
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Subspace Hadamard-Quantization (SHQ)



- Solve linear combination with MUX array and Log₂ adder tree.
- Perform H_{16} FWHT with a pipelined butterfly network.

Subspace Hadamard-Quantization (SHQ)

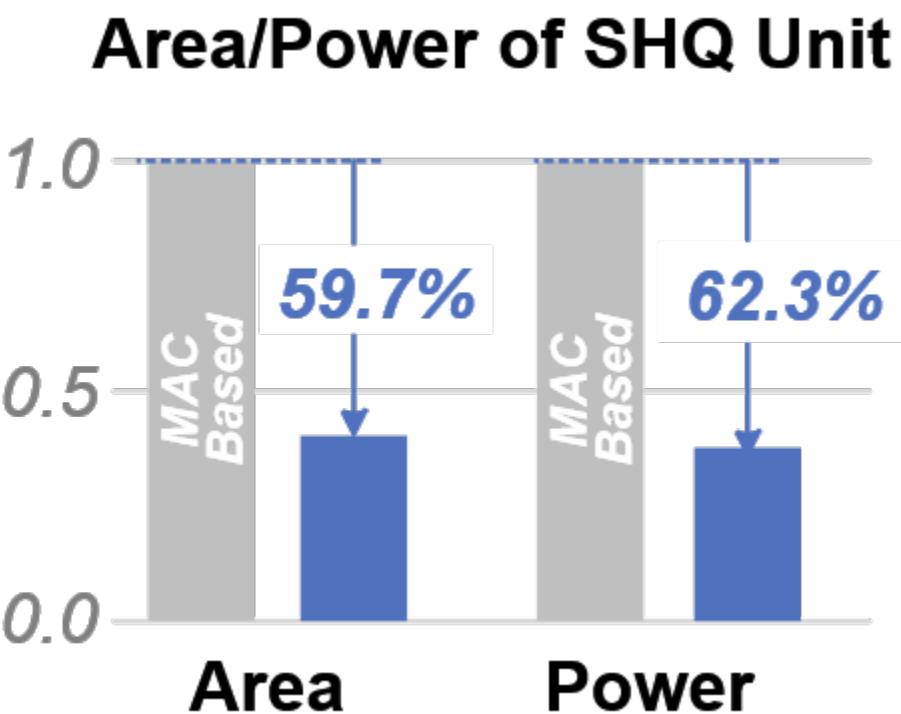


- Solve linear combination with MUX array and Log₂ adder tree.
- Perform H_{16} FWHT with a pipelined butterfly network.

Subspace Hadamard-Quantization (SHQ)

Rotation-based Operation Reduction and Speedup Improvement with SHQ

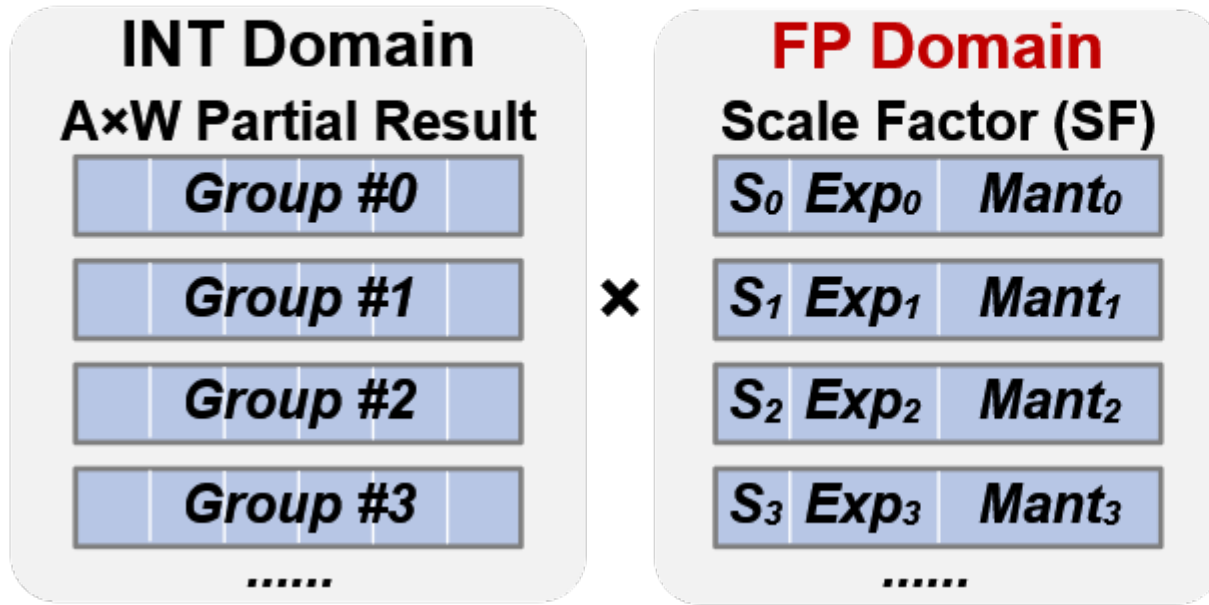
Model	MHA Reduction	FFN Reduction	Block Speedup
LLaMA2-13B	33.14%	47.06%	1.74×
LLaMA2-7B	31.83%	45.39%	1.68×
LLaMA3.2-3B	29.78%	42.13%	1.61×



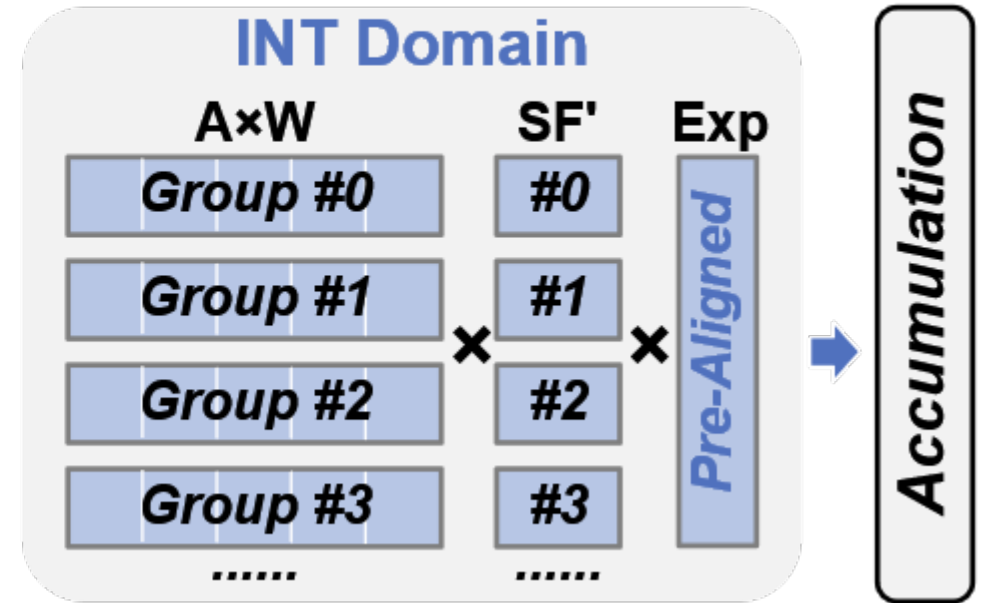
- SHQ achieves up to a **1.74×** rotation-based LLM block speedup.
- SHQ unit reduces area by **59.7%** and power by **62.3%** compared to the MAC-based implementation.

Fused Scale-Activation Unit (FSA)

FP Multiplication and **Frequent Alignment**

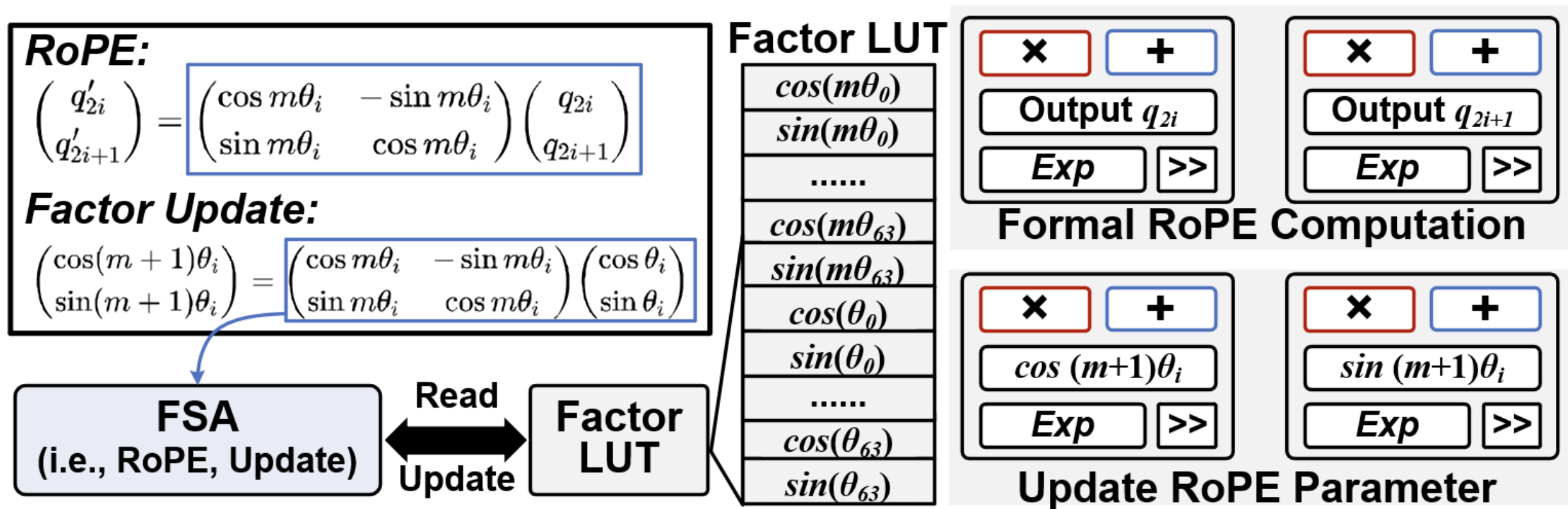


INT Domain and **Alignment-free**



- Per-channel scale factors exhibit very **small exponent spread**.
- FSA pre-aligns scale factors within each channel.
- Group partial sums are reduced in **INT domain**.

Fused Scale-Activation Unit (FSA)

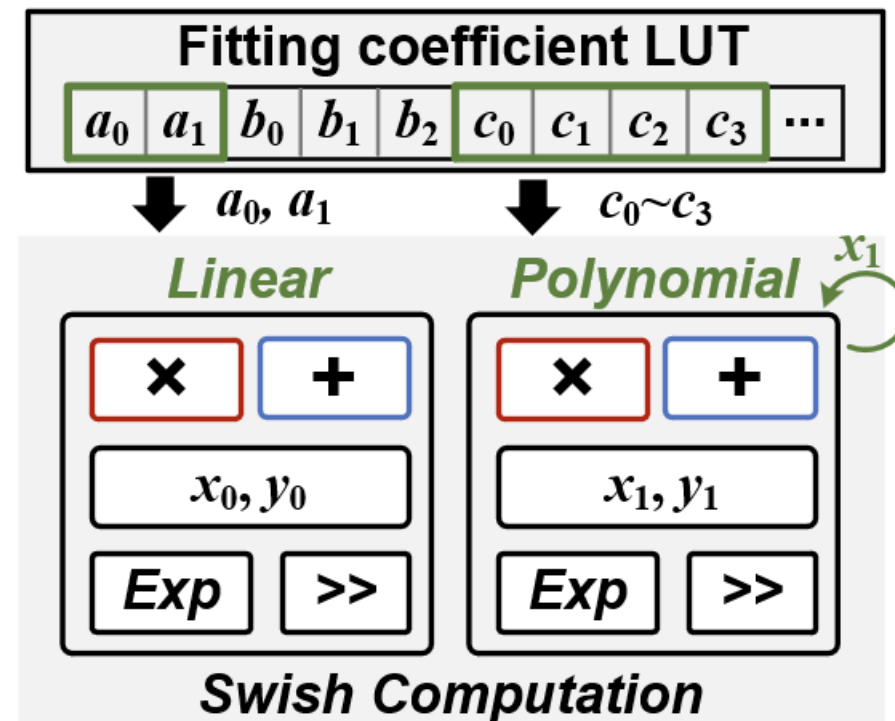
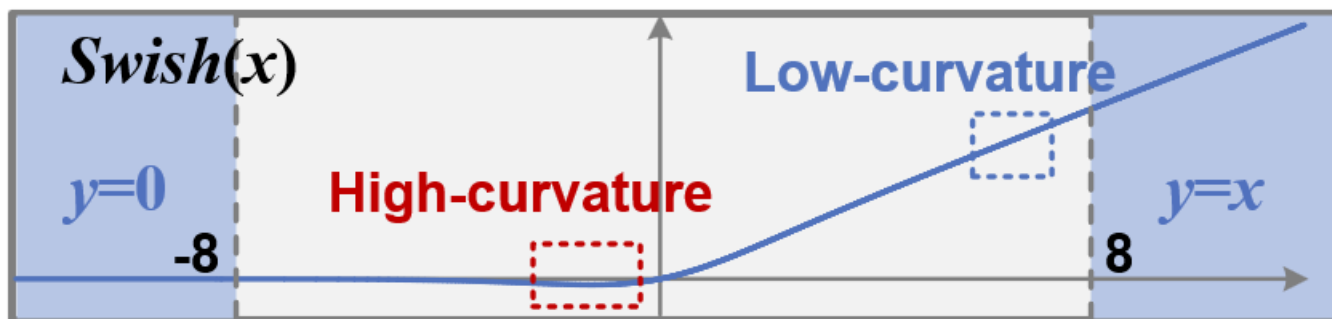


- A small step-angle LUT supplies **pre-aligned** sin/cos values.
- An angle accumulation data path **updates the trigonometric** values in a **MAC** manner.

Fused Scale-Activation Unit (FSA)

For input x , Original Swish: $\frac{x}{1+e^{-x}}$
 Piece-wise polynomial fitting:

$$y = \begin{cases} c_0 + x[c_1 + x(c_2 + c_3x)] \\ b_0 + x(b_1 + xb_2) \\ a_0 + a_1x \end{cases}$$
 Fit High Curvature Region
 Fit Low Curvature Region



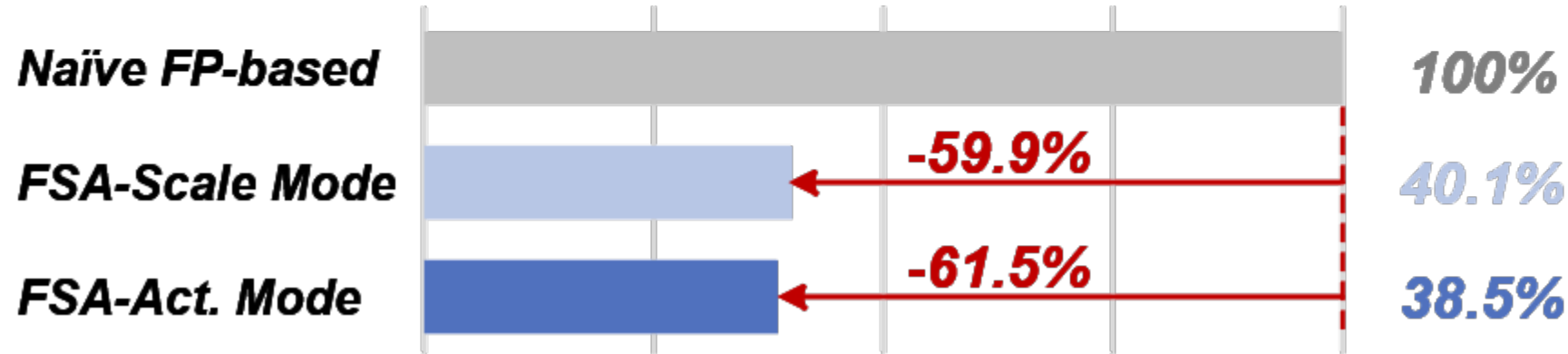
- Approximate Swish with **magnitude-aware** piece-wise polynomial
- Produce powers iteratively with the **same MAC lanes**, using a **Horner-style sequence** method.

Fused Scale-Activation Unit (FSA)

Exponent Difference of Channel-Wise Scale Factors

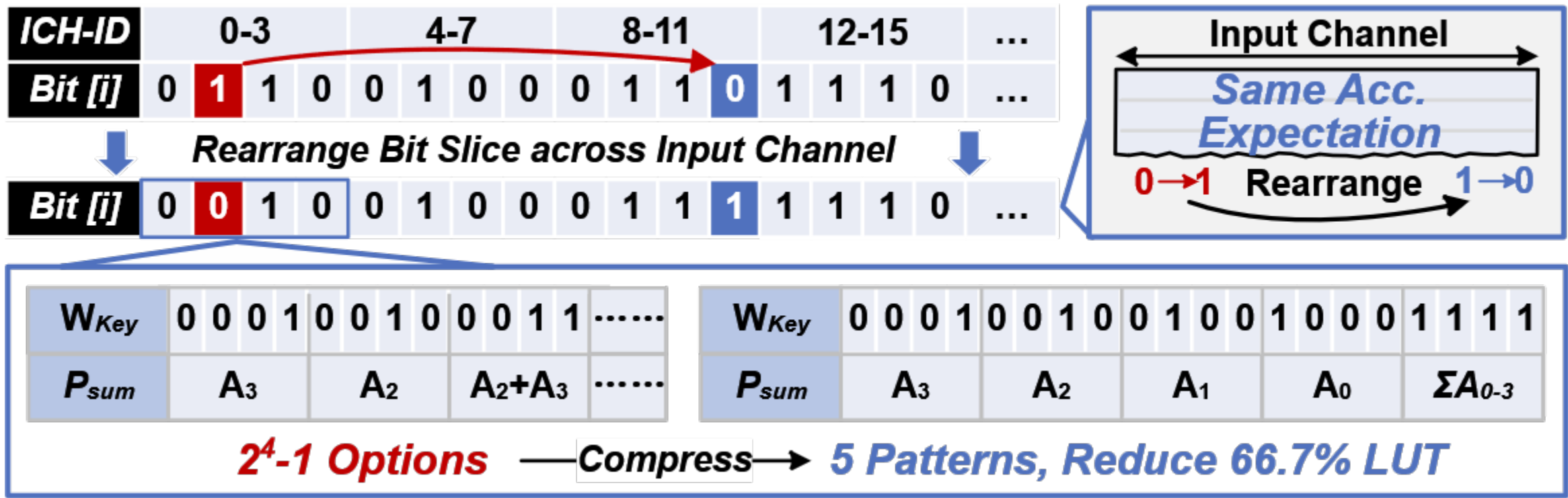
Model	LLaMA2-13B	LLaMA2-7B	LLaMA3.2-3B
Max $\text{Exp}_\Delta \leq 2$	97.26%	97.83%	98.32%

Norm. Energy of Group Scale and Act. Function



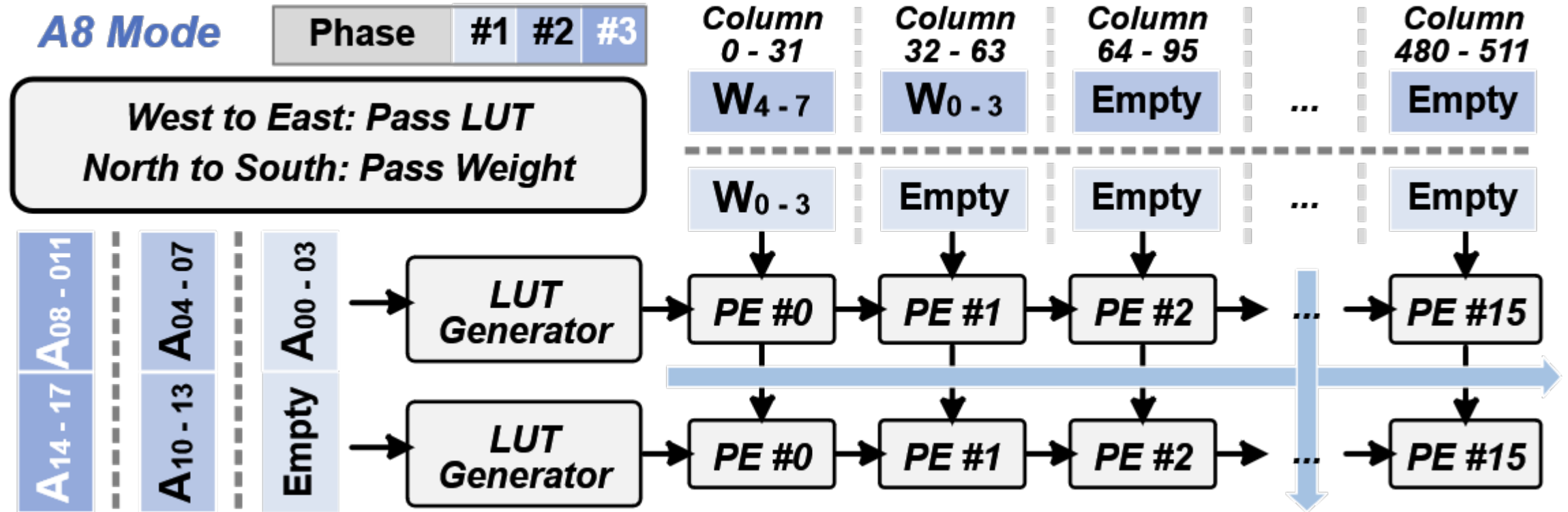
- Compared to a naïve FP-based implementation, FSA achieves:
 - Scaling Mode: 59.9% energy saving.
 - Activation Mode: 61.5% energy saving.

Rearranged Bit-Slice LUT Engine (RBLUT)



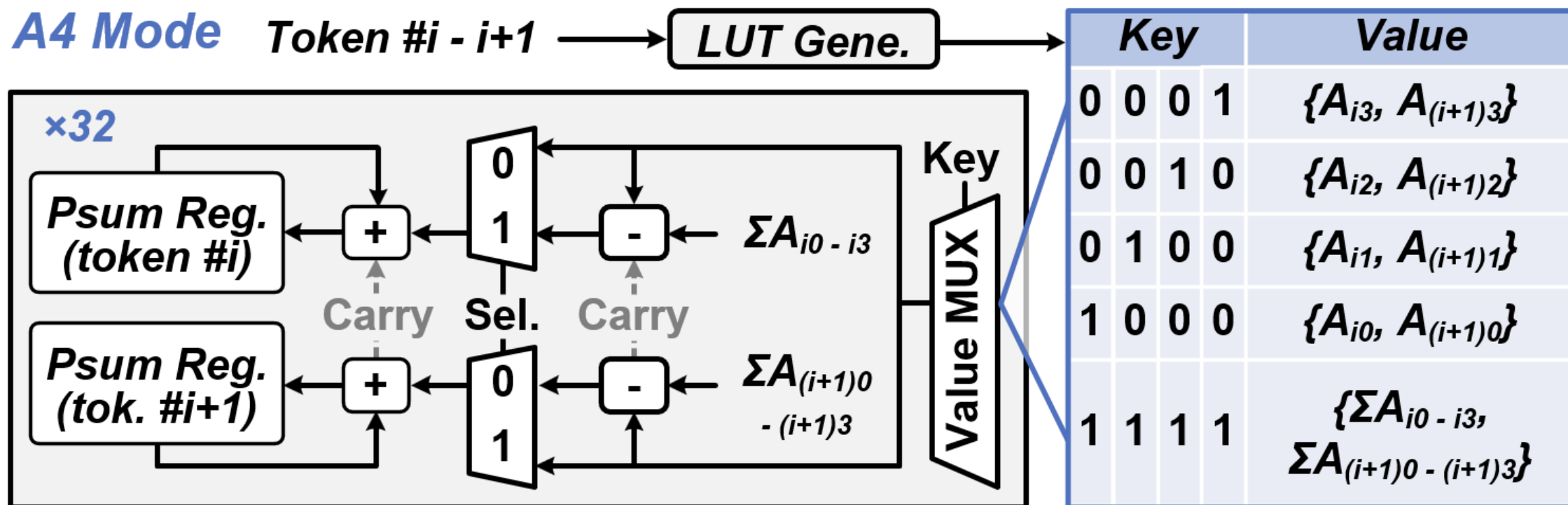
- Reduce LUT size with inherent **bit sparsity** and **Hamming-weight awareness** in LUT-based bit-slice computation paradigm.

Rearranged Bit-Slice LUT Engine (RBLUT)



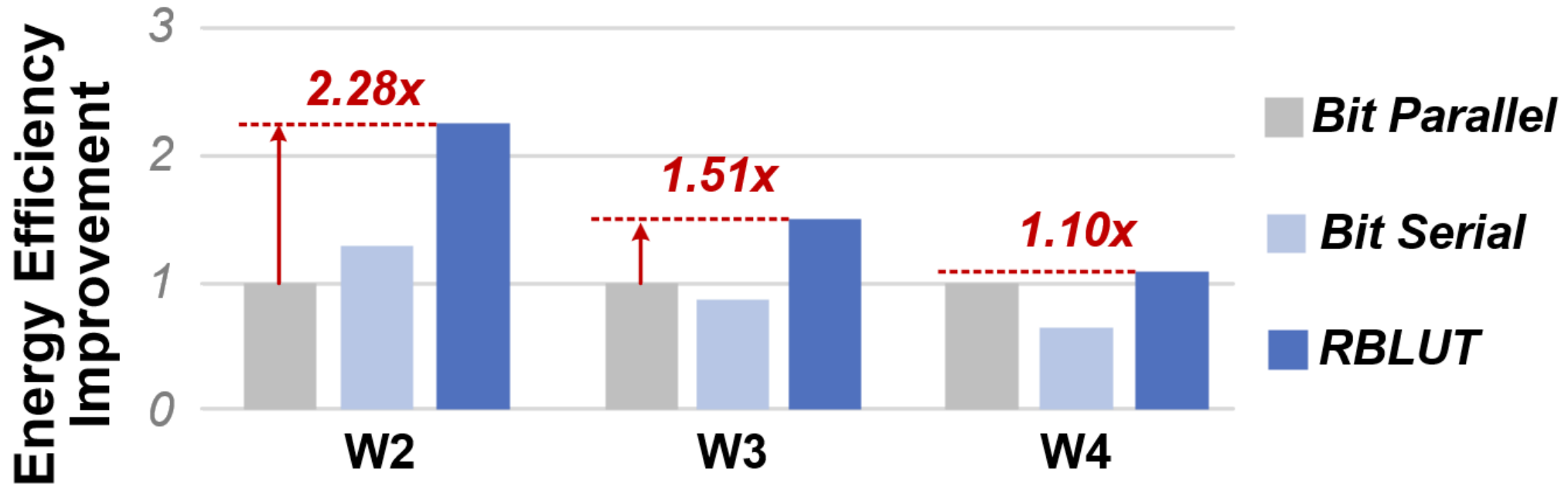
- PE array is organized in a systolic manner. Each PE integrates:
 - A **local LUT** that is forwarded downstream for reuse.
 - 32 **lookup-accumulation (LAC)** units that share the LUT.

Rearranged Bit-Slice LUT Engine (RBLUT)



- PE is reconfigurable, switching between **A8** and **A4** modes:
 - A8 Mode: Carry paths enabled for 8-bit accumulation.
 - A4 Mode: Carry paths disabled to **maximize throughput**.

Rearranged Bit-Slice LUT Engine (RBLUT)



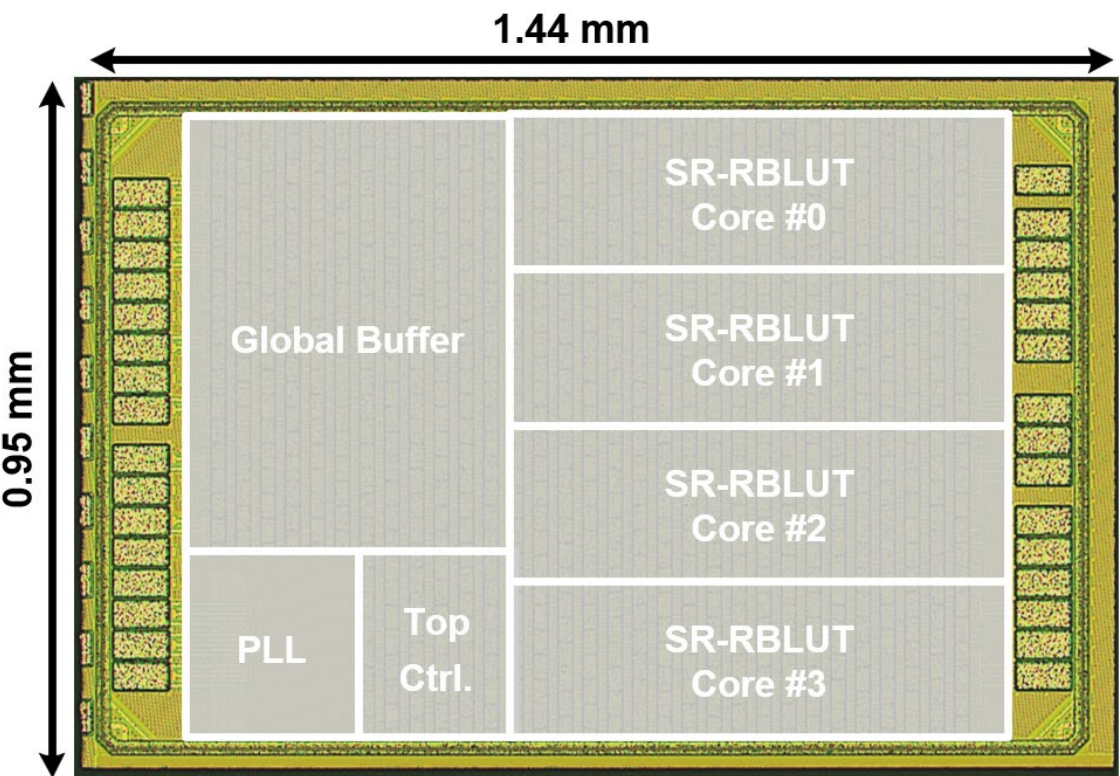
**Bit Parallel, Bit Serial, and RBLUT are normalized to same throughput under 8b-4b workloads. Evaluated with TOPS/W.*

- Compared to bit-parallel and bit-serial designs, RBLUT achieves:
 - W2 Mode: up to **2.28x** energy efficiency improvement.
 - W4 Mode: up to **1.10x** energy efficiency improvement.

Outline

- Introduction & Background
- Overall Architecture
- Key Features
 - Subspace Hadamard-Quantization Transformation (SHQ)
 - Fused Scale-Activation Unit (FSA)
 - Rearranged Bit-Slice LUT Engine (RBLUT)
- **Experimental Results**
- Conclusion

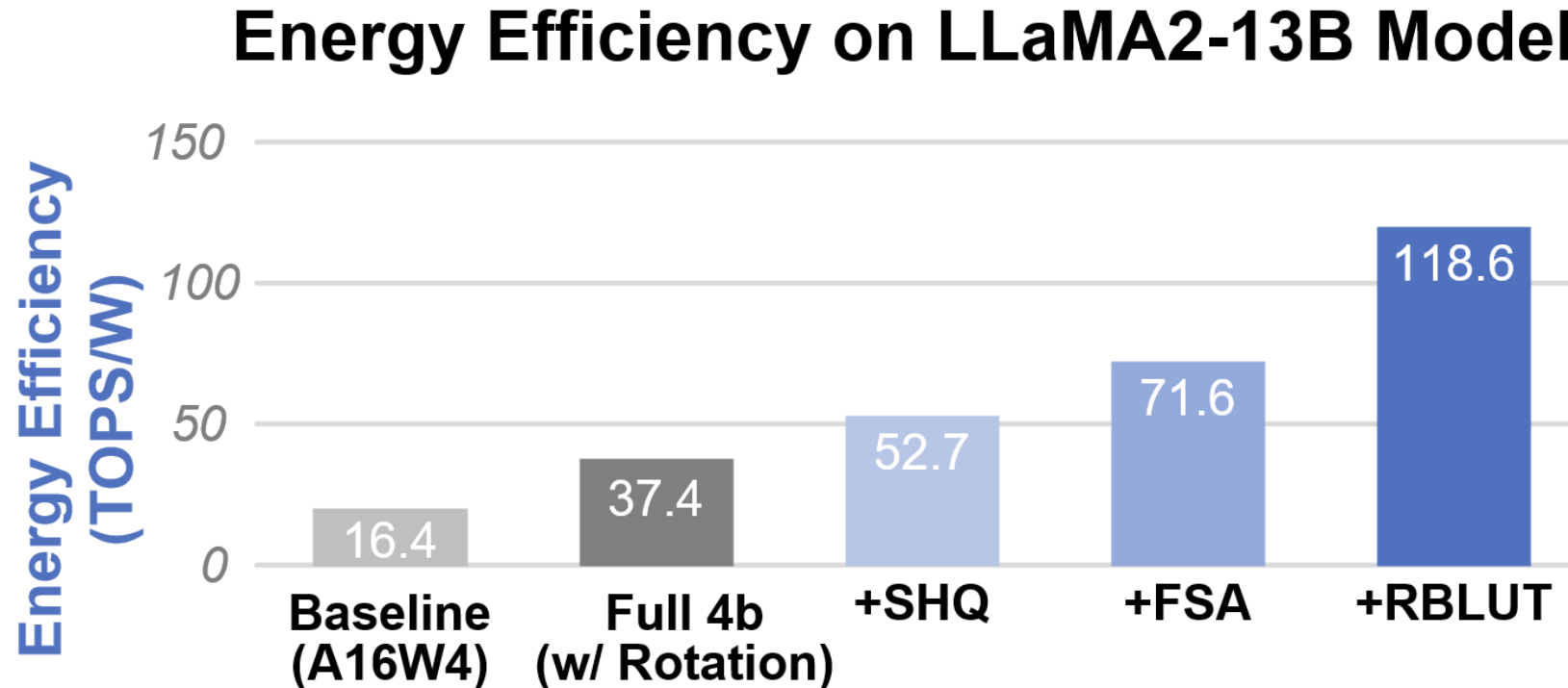
Experimental Results



1) Evaluated at 1.0V, 450MHz core frequency, 1 MAC equivalent to 2 operations. 2) Configured with A8W4KV8 quantization. 3) Configured with A4W2KV4 quantization. 4) External memory access is excluded. 5) Results at the highest energy efficiency point.

Specification		
Technique (nm)	28	
Die Area (mm ²)	1.37	
Supply Voltage (V)	0.6-1.0	
SRAM(KB)	96	
Frequency (MHz)	50-450	
Data Precision	INT8/4 (A) INT2/3/4 (W)	
Power (mW)	11.6-68.4	
Peak Performance (TOPS) ¹⁾	3.87 ²⁾ -15.79 ³⁾	
Area Efficiency (TOPS/mm ²) ¹⁾	2.82 ²⁾ -11.51 ³⁾	
Energy Efficiency (TOPS/W) ⁴⁾	0.69V 188MHz ⁵⁾	82.08 ²⁾ -318.21 ³⁾
	1.0V 450MHz	57.72 ²⁾ -236.20 ³⁾

Experimental Results



- Compared to naïve rotation implementation, our work achieves:
 - An average efficiency of **118.6 TOPS/W** at peak performance.
 - **1.41×**, **1.92×**, and **3.17×** efficiency improvement from 3 features.

Experimental Results

	ISSCC'23 [14]	ISSCC'24 [13]	ISSCC'24 [12]	ISSCC'25 [11]	This work
Workloads	Transformers	Diffusion Models	Language Models	General LLMs	General LLMs
Quantization	General INT	General FP	General INT	Uniform Weight Only	Rotation-based Uniform Act. & W
Technique (nm)	28	28	28	28	28
Die Area (mm ²)	3.93	3.67	20.25	3.52	1.37
Supply Voltage (V)	0.64-1.03	0.6-1.0	0.7-1.1	0.63-1.0	0.6-1.0
Frequency (MHz)	20-320	50-540	50-200	50-460	50-450
Precision	INT8	BF16/FP16	INT8/INT16	BF16 (A) INT8 (W)	INT8/4 (A) INT2/3/4 (W)
Power (mW)	8.27-250.65	8.268-170.0	47.5-469.2	16.95-105.0	11.6-68.4
Performance (TOPS)	0.49-3.33	6.636 (BF16) 4.424 (FP16)	3.41 (INT8)	0.70-3.58	3.87-15.79
Energy Efficiency (TOPS/W)	1.96-53.83	74.34 (BF16) 67.89 (FP16)	22.9-47.8 (INT8)	10.07-88.36	82.08-318.21

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Conclusion

- **An accelerator for Rotation-based dual-quantized LLMs inference.**
- **Subspace Hadamard-Quantization (SHQ)**
 - Improve on-chip rotation operator efficiency
- **Fused Scale-Activation Unit (FSA)**
 - Maintain benefits of dual-quantized LLMs
- **Rearranged Bit-Slice LUT Engine (RBLUT)**
 - Improve flexibility-efficiency trade-off of INT MACs
- **Achieves 51.6 μ J/Token at LLaMA3.2-3B**